Mathematics III

III Semester SEE Question Papers

Course codes: MA231TB, 21MA31C, 18MA31C, 16MA31C

- Feb/Mar 2026 (MA231TB)
- Jan/Feb 2025 (MA231TB)
- Apr 2024 (MA231TB) - Set 1
- Apr 2024 (MA231TB) - Set 2
- Jul 2023 Makeup (21MA31C)
- Apr 2023 (21MA31C)
- Mar 2021 (18MA31C)
- Nov/Dec 2019 (16MA31C)
- Nov/Dec 2019 (18MA31C)
- Fast Track Jul 2019 (16MA31C)
- Nov/Dec 2018 (16MA31C)
- Fast Track Jul 2018 (16MA31C)
- Nov/Dec 2017 (16MA31C)

Question Paper ID: 946564

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: MA231TB

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

Common to AS / CH / IEM / ME

Statistics, Laplace Transform And Numerical Methods

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator & handbook for mathematics is permitted.

Part A

Question No	Question	M	CO	BT
1.1	The second, third and fourth central moments of a distribution are 7.0575, 36.1588, 408.7895. The coefficient of kurtosis is_____.	02	2	2
1.2	Regression lines of y on x and x on y are $4x - 5y + 33 = 0$ and $20x = 9y + 107$ respectively. The correlation coefficient is_____.	02	2	2
1.3	Test the analyticity of the function $f(z) = 2xy + i(x^2 - y^2)$ by using Cauchy-Riemann equations.	02	1	1
1.4	Taylor's series expansion of $sin(z)$ about the point $z = \pi/4$ is _____.	02	2	2
1.5	The Laplace transform of $f(t) = 2^t$ is _____.	02	1	1
1.6	If $f(t) = e^{2t}\,\delta(t-3)$, then $L[f(t)]$ =_____.	02	2	3
1.7	Find $L^{-1}\left[\frac{2s^2-5s+8}{s^3}\right]$.	02	1	1
1.8	$L^{-1}\left[\frac{1}{s(s-\sqrt{2})}\right]$ is _____.	02	2	2
1.9	The partial differential equation $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = 0$ is classified as _____.	02	1	1
1.10	For one dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0,t) = u(5,t) = 0$ and $u(x,0) = x - x^2$, the value of $u(1, 0.5)$ by taking $h = 1$ and $k = 0.25$ is _____.	02	2	2

Part B

<table>
<tr><th>Question No</th><th>Question</th><th>M</th><th>CO</th><th>BT</th></tr>
<tr><td>2a</td><td>The following table gives the results of the measurements of train resistances; x is the velocity in miles per hour, y is the resistance in pounds per ton. By the method of least squares find the best values of a, b, c for the quadratic equation $y = a + bx + cx^2$ which fits the data
<table>
<tr><td>x</td><td>20</td><td>40</td><td>60</td><td>80</td><td>100</td><td>120</td></tr>
<tr><td>y</td><td>5.5</td><td>9.1</td><td>14.9</td><td>22.8</td><td>33.3</td><td>46</td></tr>
</table>
Represent the required values in tabular form.</td><td>06</td><td>2</td><td>2</td></tr>
<tr><td>2b</td><td>Calculate the first four moments of the following distribution about the mean, also find β1 and β2.
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr>
</table></td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>3a</td><td>Determine the analytic function $f(z) = u + iv$ in-terms of x, y. Given that $u = e^{2x}[xcos(2y) - ysin(2y)]$.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>3b</td><td>Find the power series expansion of $f(z) = \frac{z}{(z^2+1)(z^2+4)}$ in the following regions. $(i)|z| < 1, \quad (ii)1 < |z| < 2, \ (iii)|z| > 2.$</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Show that $u(x, y) = e^x(xcosy - ysiny)$ is a harmonic function and find the conjugate harmonic function $v(x, y)$. Also write $f(z)$ in terms of z only.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td>4b</td><td>Evaluate the integral $\int_C \frac{\sin(\pi z^2)+\cos(\pi z^2)}{(z-1)^2(z-2)}\, dz$, where C is the circle $|z| = 3$ using Cauchy's residue theorem.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>5a</td><td>Evaluate: $\mathcal{L}\{e^{t/3}t^{-1/2} + t^2\cos(2t) + 2\}$.</td><td>08</td><td>1</td><td>2</td></tr>
<tr><td>5b</td><td>Find the Laplace transform of the full wave rectifier $f(t) = E\sin(\omega t),\ 0 < t < \frac{\pi}{\omega}$,
$f(t + \pi/\omega) = f(t)$. Also represent this function graphically.</td><td>08</td><td>2</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>Evaluate: $\mathcal{L}\left\{\sinh(2t)\sin(3t) + \frac{1-\cos t}{t}\right\}$.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>6b</td><td>Express the following in terms of unit step function and hence find the Laplace transform of
$f(t) = \begin{cases} 1, & 0 < t \le 1 \\ 2t, & 1 < t \le 2 \\ 3t^2, & t > 2 \end{cases}$</td><td>08</td><td>2</td><td>4</td></tr>
<tr><td>7a</td><td>Show that $L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right] = \frac{1}{2a^3}[\sin at - at\cos at]$.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td>7b</td><td>Solve the initial value problem
$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = 5,\ y(0) = 0,\ y'(0) = 0$, using Laplace transform</td><td>08</td><td>4</td><td>3</td></tr>
</table>

	method.			
	OR			
8a	Evaluate (i) $L^{-1}\left[\frac{s+2}{s^2-4s+13}\right]$ (ii) $L^{-1}\left[\log\left(\frac{s+b}{s+a}\right)\right]$.	08	3	3
8b	Apply convolution theorem to evaluate $L^{-1}\left\{\frac{1}{(s+2)^2(s-2)}\right\}$.	08	4	3
9a	Solve the heat equation $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0,t)=0=u(4,t)$ and $u(x,0)=sin(\frac{\pi x}{4})$ by taking $h=1$ for $0\le t\le 2$ using Bendre-Schmidt method.	08	3	3
9b	Solve the elliptic equation $u_{xx}+u_{yy}=0$ in $0\le x\le 3, 0\le y\le 3$, given that $u(x,0)=0, u(0,y)=0, u(x,3)=x^2$ and $u(3,y)=3y$. Take $h=k=1$ and employ Gauss Seidel method to solve this problem. Carry out three iterations by considering the value up to 4 decimals.	08	3	3
	OR			
10a	Solve $\frac{\partial^2 u}{\partial t^2}=\frac{\partial^2 u}{\partial x^2}$ subject to the boundary conditions $u(0,t)=u(1,t)=0, t\ge 0$ and initial conditions $u(x,0)=sin(\pi x)$, $\frac{\partial u}{\partial t}(x,0)=0, 0\le x\le 1$ taking$h=k=0.2$ for $0\le t\le 1$.	08	4	3
10b	Solve the heat equation $\dfrac{\partial u}{\partial t}=\dfrac{1}{4}\dfrac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,0)=\dfrac{x^2}{2}(1-x^2)$, $0\le x\le 1, u(0,t)=u(1,t)=0, t\ge 0$ using Schmidt method. Carry out computations for two levels, taking $h=0.25$ and $k=0.01$.	08	3	3

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Common to AS / BT / CH / IM / ME

STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. 2nd year Hand book of mathematics to be provided.

PART-A

			M	CO	BT
1	1.1	In a partially destroyed laboratory record of an analysis of a correlation data, the following results were observed: variance of $y = 16$, equations of lines of regression of y on x is $5y = 8x - 17$ and x on y is $5x = 2y + 14$. For the given data, value of correlation coefficient is _____ the variance of x is _____.	02	2	2
	1.2	The first three moments of a distribution about the value 2 of the variables are $1, 16$ and -40. The third moment about the mean is _____.	02	1	1
	1.3	The real and imaginary parts of the complex valued function $f(z) = e^{2z}$ respectively are _____ and _____.	02	1	1
	1.4	The coefficient of z^2 in Maclaurin's series expansion of $f(z) = \log(1 + z)$ is _____.	02	1	1
	1.5	Using Laplace transform the value of the integral $\int_0^\infty \cosh(at)\, dt, (a > 0)$, is _____.	02	2	1
	1.6	$L[t \sin(2t)]$ is _____.	02	2	2
	1.7	The frequency response of a system is found to be $\frac{1}{(9s^2-16)}$, the corresponding time response is _____.	02	1	2
	1.8	If $F(s) = \frac{1}{s^2+4s+2}$, the inverse Laplace transform of $F(s)$ is _____.	02	1	1
	1.9	The partial differential equation $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ is classified as _____.	02	1	1
	1.10	For the one dimensional wave equation $u_{xx} = u_{tt}$ with the boundary conditions $u(0,t) = 0, u(1,t) = 0, u_t(x,0) = 0$ and the initial displacement $u(x,0) = x(1-x)$. The value of $u(0.25, 0.25)$ by taking $h = 0.25$ and $k = 0.25$ is _____.	02	2	2

PART-B

2	a	As part of a research study on software engineer salaries at a tech company have been provided with salary data (in thousands of dollars) for a set of employees. The salaries are categorized into specific ranges, with the corresponding number of employees in each range as follows:						
		Salary Range ($K) | 60 – 70 | 70 – 80 | 80 – 90 | 90 – 100 | 100 – 110 Number of Engineers | 1 | 3 | 7 | 3 | 3							
		Using this data calculate the first three central moments.	06	2	3

	b	Fit a straight line of the form $y = ax + b$ for the following data by the method of least squares. | x | 5 | 10 | 15 | 20 | 25 | | y | 16 | 19 | 23 | 26 | 30 |	05	1	2
	c	Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in Engineering Mathematics (x) and Engineering Graphics (y): | x | 15 | 20 | 28 | 12 | 40 | 60 | 20 | 80 | | y | 40 | 30 | 50 | 30 | 20 | 10 | 30 | 60 |	05	2	2
3	a	Verify that $u = y + e^x \cos y$ is harmonic. If so find v such that $f(z) = u + iv$ is analytic using Milne-Thomson method.	08	2	3
	b	Obtain the Laurent's series expansion for $f(z) = \frac{z}{(z-1)(z-3)}$ in the region: i) $1 < \|z\| < 3$ ii) $\|z\| > 3$	08	2	3
		OR			
4	a	Find the Taylor's series expansion of $f(z) = \frac{1}{z^2+4}$ about $z = -i$.	08	2	3
	b	Evaluate the integral $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$ where C is the circle $\|z\| = 3$ using Cauchy's residue theorem.	08	2	3
5	a	A periodic function $f(t)$ of period $(2\pi)/\omega, \omega > 0$ is defined by, $f(t) = \begin{cases} \sin \omega t, & 0 < t < \pi/\omega \\ 0, & \pi/\omega < t < (2\pi)/\omega \end{cases}$; Show that $L\{f(t)\} = \frac{\omega}{(s^2+\omega^2)(1-e^{-(\pi s)/\omega})}$.	08	3	3
	b	Obtain the Laplace transform of $\frac{\cos(2t) - \cos(3t)}{t} + 3(t^2+1)^2$	08	2	2
		OR			
6	a	Express the following piecewise – continuous function in terms of unit step function and hence find the Laplace transform, $f(t) = \begin{cases} 2t^2, 0 \le t < 3 \\ t + 4, 3 < t < 5 \\ 9, t \ge 5 \end{cases}$	08	3	3
	b	Find $L\{\int_0^t e^{-3t} \sin^2 2t dt + t \cos 3t \sin 2t\}$.	08	2	2
7	a	Employing convolution theorem find inverse Laplace transform of $\frac{s^2}{(s^2+9)^2}$.	08	2	3
	b	The functional form of a signal in frequency domain is given by $\frac{4s+5}{(s-1)^2(s+2)}$. Determine the function in time domain using Laplace transform.	08	2	2
		OR			
8	a	By applying Laplace transform solve the differential equation $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 5e^{2t}$, when $y(0) = 2$ and $y'(0) = 1$.	08	2	3
	b	Find the inverse Laplace transform of the function $\frac{s+2}{s^2-2s+5} + \log\left(\frac{1-s^2}{s^2}\right)$	08	2	2

9	a	Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ subject to the conditions $u(0,t) = 0, u(1,t) = 0$ and $u(x,0) = \sin(\pi x)$for $0 \le t \le 0.1$ by taking $h = 0.2, k = 0.02$. Also find the following values from the table i) u(0.2,0.04) ii) $u(0.4, 0.08)$ iii) $u(0.6, 0.06)$.	08	3	3
	b	The temperature (x, y) in a steady state heat flow in a square plate bounded by $x = 0, y = 0, x = 4, y = 4$ satisfies the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Taking a square mesh of size unity and the boundary values given by $u = 0$ along $x = 0, y = 0, u = x^3$ along $y = 4, u = 16y$ along $x = 4$. Compare two iterations using Gauss Seidel iterative method.	08	3	4
		OR			
10	a	Using Schmidt's formula solve $\frac{\partial^2 u}{\partial x^2} = 2\frac{\partial u}{\partial t}$ with the conditions $u(0,t) = 0, u(12,t) = 9$ for $0 \le t \le 12$ and $u(x,0) = \frac{x}{4}(15 - x)$ for $0 \le x \le 12$ taking $h = k = 3$.	08	3	3
	b	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $\frac{\partial^2 u}{\partial t^2} = 25\frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(0,t) = u(5,t) = 0$ and initial conditions $u(x,0) = \begin{cases} 2x, & 0 \le x \le 2.5 \\ 10 - 2x, & 2.5 \le x \le 5 \end{cases}$ and $\frac{\partial u}{\partial t} = 0$ at $t = 0$. Solve the equation numerically for half period of vibration, taking $h = 1$ and $k = \frac{1}{5}$.	08	3	4

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III Semester B.E. Examinations April – 2024
Common to AS, BT, CH, IEM, ME
Department of Mathematics

STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

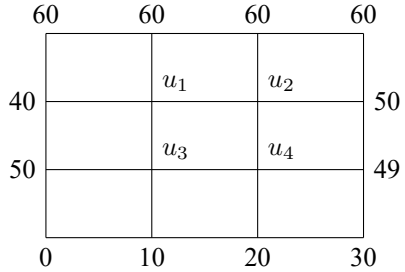
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator and hand book for mathematics is permitted.

PART-A

			M	BT	CO
1	1.1	If the first four moments of a distribution about the value 25 of the variables are $-3, 11, -29$ and 71, then the second moment about the mean is _____.	02	1	2
	1.2	The equations of regression lines are y = 0.5x + 1 and x = 0.4y + 3. The correlation coefficient is _____.	02	2	2
	1.3	The real part of the complex function $f(z) = \log_e z$ in cartesian co-ordinate system is _____.	02	1	1
	1.4	Taylor's series expansion of $\cos z$ about the point $z = \pi/4$ is _____.	02	2	1
	1.5	The Laplace transform of the signal $f(t) = t + \cos 2t$ is _____.	02	1	1
	1.6	Region of convergence of $\mathcal{L}[a^t]$ to hold good is _____.	02	1	1
	1.7	$\mathcal{L}^{-1}\left[\frac{1}{4s-1}\right]$ = _____.	02	1	2
	1.8	If $F(s) = \frac{1}{s^2+4s+4}$, then the inverse Laplace transform of $F(s)$ = _____.	02	2	1
	1.9	The equation $u_{xx} + u_{yy} - 4u_x - 6u_y = 3$ is classified as _____.	02	1	1
	1.10	For the one dimensional wave equation $u_{xx} = u_{tt}$ with the boundary conditions $u(0,t) = 0$, $u(1,t) = 0$, $u_t(x,0) = 0$ and $u(x,0) = x - x^2$, the value of $u(0.25, 0.25)$ by taking h = 0.25 and k = 0.25 is _____.	02	2	2

PART-B

2	a	From the following frequency distribution, compute first three moments about the mean. x: 1, 2, 3, 4, 5, 6 y: 1, 6, 13, 25, 30, 22	05	2	3
	b	A simply supported beam carries a concentrated load P at its mid-point. Corresponding to various values of P, the maximum deflection Y is measured and is given in the following table. P: 100, 120, 140, 160, 180, 200 Y: 0.45, 0.55, 0.60, 0.70, 0.80, 0.85 Fit a law of the form Y = a + bP and hence estimate Y when P is 150.	06	3	2

	c	Calculate the rank correlation coefficient from the following data: x: 81, 78, 73, 73, 69, 68, 62, 58 y: 10, 12, 18, 18, 18, 22, 20, 24	05	3	3
3	a	Determine the analytic function whose real part is $u = e^x(x\cos y - y\sin y)$ by Milne-Thomson method. Also find its harmonic conjugate v.	08	2	2
	b	Obtain the Laurent's series expansion for $f(z) = \dfrac{1}{(z+1)(z+3)}$ in the region (i) $1 < \|z\| < 3$ (ii) $\|z\| > 3$ (iii) $\|z\| < 1$.	08	2	2
		OR			
4	a	Find the analytic function $f(z) = u + iv$ where $u - v = 2xy + x^2 - y^2 + x - y$.	08	2	2
	b	Evaluate the integral $\displaystyle\int_C \frac{e^{2z}}{(z+1)(z-2)^2}\,dz$ where C is the circle $\|z\| = 3$ using Cauchy's residue theorem.	08	2	2
5	a	A periodic function f(t) of period 2a, a > 0 is defined by $f(t) = K$ for $0 \le t \le a$ and $-K$ for $a < t \le 2a$. Show that $\mathcal{L}\{f(t)\} = \dfrac{K}{s}\tanh\left(\dfrac{as}{2}\right)$.	08	2	2
	b	Obtain the Laplace transform of $\dfrac{e^{-t} - e^{7t}}{t} + \left(2\sqrt{t} - \dfrac{3}{\sqrt{t}}\right)^3$.	08	3	3
		OR			
6	a	Express the following in terms of unit step function and hence find the Laplace transform of $f(t) = \begin{cases} 3t^2, & 0 < t < 3 \\ 5t, & t > 3 \end{cases}$	08	2	2
	b	Find $\mathcal{L}\left\{e^{-t}\left(\int_0^t t\sin 4t\,dt\right) + \cos 2t\sin 5t\right\}$.	08	3	3
7	a	The signal in frequency domain is given by $\dfrac{s}{(s-3)(s^2+4)}$. Determine the transformation in time domain.	08	2	3
	b	Employing convolution theorem, find inverse Laplace transform of $\dfrac{1}{(s^2+a^2)^2}$.	08	3	4
		OR			
8	a	Evaluate $\mathcal{L}^{-1}\left[\dfrac{s+1}{s^2-s+1}\right]$.	08	2	3
	b	Solve the differential equation using Laplace transform $\dfrac{d^2y}{dt^2} - 3\dfrac{dy}{dt} + 2y = 12e^{-t}$ when $y(0) = 2$ and $y'(0) = 6$.	08	3	4
9	a	Determine the value of $u(x, y)$ satisfying the Laplace equation $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of a square region with boundary values as shown in Fig 9a. Carry out 3 iterations.  Fig 9a	08	2	4

	b	Employ Bendre-Schmidt method with h = 1 to find the values of $u(x,t)$ up to t = 1 for the equation $2\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $0 < x < 4$, $t > 0$ under the conditions $u(x,0) = x(4-x), 0 < x < 4$; $u(0,t) = u(4,t) = 0, t \geq 0$.	08	3	3
		OR			
10	a	The temperature distribution $u(x,t)$ is modeled as the one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{4}{5}\frac{\partial^2 u}{\partial x^2}$, $0 < x < 100$, $t > 0$ subject to the boundary conditions $u(0,t) = 0 = u(100,t)$, $t \geq 0$ and the initial condition $u(x,0) = x$ for $0 \leq x \leq 50$ and $100 - x$ for $50 \leq x \leq 100$. Determine the temperature distribution $u(x,t)$ by taking h = 20 and k = 10 using Schmidt relation for 5 levels.	08	2	3
	b	Solve $\frac{\partial^2 u}{\partial t^2} = 16\frac{\partial^2 u}{\partial x^2}$ taking h = 1 in five steps given that $u(x,0) = x^2(5-x)$, $u_t(x,0) = 0$, $u(0,t) = u(5,t) = 0$.	08	3	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
III Semester B.E. April – 2024 Examinations
Common to AS, BT, CH, IEM, ME
MODEL QUESTION PAPER (2022 SCHEME)
Department of Mathematics

STATISTICS, LAPLACE TRANSFORMS AND NUMERICAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

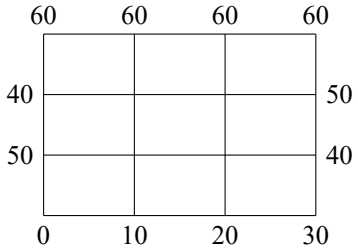
PART-A

			M	BT	CO
1	1.1	A frequency distribution curve is said to be 'platykurtic' if _____.	01	–	–
	1.2	If two variables x and y are perfectly correlated then, coefficient of correlation 'r' is _____.	01	–	–
	1.3	If the equations of regression lines of two variables x and y are x − y = 0, 4x − y = 3 then the means of x and y are _____ and _____.	02	–	–
	1.5	The real and imaginary parts of the function $f(z) = z + e^z$ is _____ and _____.	02	–	–
	1.6	The residue of $f(z) = \dfrac{1}{(z+2)(z+3)}$ at the pole z = 3 is _____.	02	–	–
	1.7	Compute: $\mathcal{L}[e^{3t}\cosh(2t)]$ is _____.	02	–	–
	1.8	The Laplace transform of the signal $f(t) = t\,\delta(t-3)$ is _____.	02	–	–
	1.9	Evaluate: $\mathcal{L}^{-1}\left[\dfrac{3}{s^2-4}\right]$.	02	–	–
	1.10	If $F(s) = \dfrac{1}{s^2+2s+7}$, then $\mathcal{L}^{-1}[F(s)]$ = _____.	02	–	–
	1.11	The forward difference approximation for u_t and central difference approximation u_x are _____ and _____.	02	–	–
	1.12	The 1D wave equation $c^2u_{xx} - u_{tt} = 0$ is classified as _____.	02	–	–

PART-B

2	a	Compute the first four moments about the mean of the frequency distribution given below: x: 0, 1, 2, 3, 4 f: 1, 8, 28, 56, 70	10	–	–
	b	The following table gives ranking of 10 students according to their achievements in both the laboratory and lecture portions of an engineering course. Find the coefficient of rank correlation. Lab (x): 8, 3, 9, 2, 7, 10, 4, 6, 1, 5 Lecture (y): 9, 5, 10, 1, 8, 7, 3, 4, 2, 6	6	–	–
3	a	Show that the function $f(z) = z^2$ is analytic and hence obtain its derivative.	8	–	–
	b	Expand the function $f(z) = \dfrac{z}{(z+1)(z+3)}$ as power series in the region $1 < \lvert z\rvert < 3$.	8	–	–

		OR			
4	a	Construct an analytic function $f(z) = u + iv$, if the real part of the function is $u = x^2 + x - y^2$.	8	–	–
	b	Evaluate $\int_C \frac{3z-4}{z(z-1)(z-2)}\,dz$ by using the residue theorem if C: $\|z\|$ = 3/2.	8	–	–
5	a	For the periodic function $f(t) = \begin{cases} 1, & 0 < t < \frac{a}{2} \\ -1, & \frac{a}{2} < t < a \end{cases}$, where $f(t+a) = f(t)$, find $\mathcal{L}[f(t)]$. Sketch the graph of the function.	8	–	–
	b	Find (i) $\mathcal{L}[te^{-3t} + \sin(t)\sin(3t)]$ (ii) $\mathcal{L}\left[\frac{\sin(2t)}{t}\right]$.	8	–	–
		OR			
6	a	Evaluate: $\mathcal{L}\left[\left(2\sqrt{t} - \frac{1}{3\sqrt{t}}\right)^3 + t\cos(2t)\right]$.	8	–	–
	b	Express the function in terms of unit step function and hence, obtain the Laplace transform of $f(t) = \begin{cases} \cos(t), & 0 < t < \pi \\ 1, & \pi < t \le 2\pi \\ \sin(t), & t \ge 2\pi \end{cases}$	8	–	–
7	a	Find the inverse Laplace transform of the following: (i) $\frac{3s+2}{2s^2-4s+3}$ (ii) $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$.	8	–	–
	b	Employ convolution theorem to find $\mathcal{L}^{-1}\left[\frac{1}{(s+1)s^2}\right]$.	8	–	–
		OR			
8	a	Find the inverse Laplace transform of the following: (i) $\log_e\left(\frac{s+a}{s+b}\right)$ (ii) $\frac{s\,e^{-s/2} + \pi e^{-s}}{s^2+\pi^2}$.	8	–	–
	b	Using Laplace transform solve the problem of undamped forced vibrations of a spring governed by the equation $\frac{d^2x}{dt^2} + \beta^2 x = A\sin\omega t$, $x(0) = x_0$, $x'(0) = v_0$.	8	–	–
9	a	Determine the values of $u(x,t)$ satisfying the parabolic equation $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0,t) = u(8,t) = 0$ and $u(x,0) = 4x - \frac{x^2}{2}$ at the points x = i : i = 0, 1, 2, …, 8 and j = 0, 1, 2, …, 5.	8	–	–
	b	The temperature $u(x,y)$ for a steady state heat flow in a square plate bounded by x = 0, y = 0, x = 4, y = 4 satisfies Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Taking a square mesh of size unity and the boundary values given by u = 0 along x = 0 and y = 0, u = x^3 along y = 4, u = 16y along x = 4. Compute two iterations using Gauss-Seidel iterative method.	8	–	–
		OR			
10	a	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$ with boundary conditions u = 0 at x = 0, t > 0 and u = 0 at x = 4, t > 0 and initial conditions u = x(4 − x) and $\frac{\partial u}{\partial t}$ = 0 at t = 0, 0 ≤ x ≤ 4. Solve the equation numerically for one half period of vibration, taking h = 1 and k = 1/2.	8	–	–

b	Determine the values of $u(x, y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of a square region, with boundary values as shown in the following figure. Carry out three iterations.  	8	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations July-2023 (Makeup)

(Common to CH, ME, IM, CV, BT)

INTEGRAL TRANSFORMS, OPTIMIZATION AND NUMERICAL TECHNIQUES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6,7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.

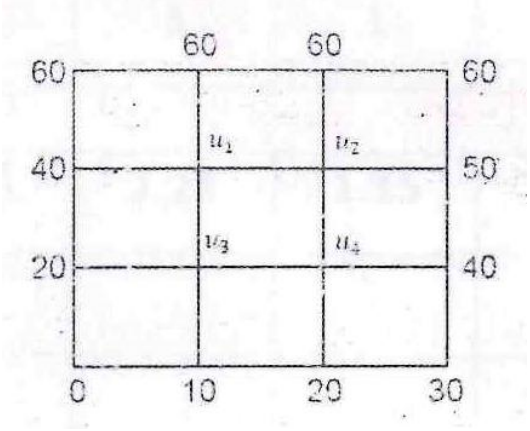
PART-A

1	1.1	If $L[f(t)] = \frac{6s}{(s^2+9)^2}$, then $L[f(2t)]$ = _____.	01
	1.2	Region of convergence of Laplace transform of e^{3t} is _____.	01
	1.3	For the function $f(x) = x^3$ over the interval $(-3, 3)$ with $f(x+6) = f(x)$, then the Fourier coefficient b_1 = _____.	01
	1.4	If $f(x) = \begin{cases} 2-x, & 0 < x < 2 \\ \pi, & 2 < x < 4 \end{cases}$, then the Fourier series at the point $x = 2$ converges to _____.	01
	1.5	$L\left[5t^2 + \sin\left(\frac{t}{2}\right)\right]$ = _____.	02
	1.6	$L^{-1}\left[\frac{1}{3s+2} + \frac{1}{s^4}\right]$ = _____.	02
	1.7	$L^{-1}\left[\frac{1}{s(s+3)}\right]$ = _____.	02
	1.8	If $f(x) = x + 3$ over the interval $(0, 2\pi)$ with $f(x + 2\pi) = f(x)$, then the Fourier coefficient a_0 is _____.	02
	1.9	Obtain all the basic feasible solutions to the system of linear equations: $x_1 + 2x_2 + x_3 = 4$ and $2x_1 + x_2 + 5x_3 = 5$.	02
	1.10	Find the corner points of the feasible region for the objective function $z = 3x + 2y$ subject to the constraints $x + 2y \leq 10$, $3x + y \leq 15$ and $x, y \geq 0$.	02
	1.11	The standard and diagonal five-point formula for $u_{i,j}$ = _____.	02
	1.12	Forward difference approximation for $\frac{\partial u}{\partial x}$ = _____ and backward difference approximation for $\frac{\partial u}{\partial y}$ = _____.	02

PART-B

2	a	i) Find the Laplace transform of $\frac{\cos(2t)-\cos(3t)}{t} + e^{7t}\sin(5t)$. ii) Evaluate $\int_0^\infty t\, e^{-5t}\cos(4t)dt$ using Laplace transform.	08
	b	Show that the Laplace transform of the periodic function $f(t) = \begin{cases} 1+t, & 0 \leq t \leq 1 \\ 3-t, & 1 \leq t < 2 \end{cases}$ with $f(t+2) = f(t)$ is $\frac{1}{s} + \frac{1}{s^2}\tanh\left(\frac{s}{2}\right)$. Also sketch the graph of the function.	08

3	a	Find: i) $L^{-1}\left[\log_e\left(\frac{s+a}{s+b}\right)\right]$ ii) $L^{-1}\left[\frac{s+2}{s^2-4s+13}\right]$	08
	b	Use convolution theorem to find the inverse Laplace transform of $\frac{s}{(s^2+k^2)^2}$ where k is a constant.	08
		OR	
4	a	Find: i) $L^{-1}\left[\frac{4s+5}{(s+1)^2(s+2)}\right]$ ii) $L^{-1}\left[\cot^{-1}\left(\frac{s}{a}\right)\right]$	08
	b	Solve $\frac{d^2y}{dt^2}+4\frac{dy}{dt}+4y=e^{-t}$ with $y=\frac{dy}{dt}=0$ at $t=0$ using Laplace transform.	08
5	a	Find the Fourier series for the function: $f(x)=\begin{cases}\pi+x, & -\pi<x<0\\ \pi-x, & 0<x<\pi\end{cases}$ with $f(x+2\pi)=f(x)$.	08
	b	Obtain the Fourier series for the function: $f(x)=\frac{\pi-x}{2}$ over the interval $(0,2\pi)$ with $f(x+2\pi)=f(x)$. Deduce $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots$	08
		OR	
6	a	Obtain the Fourier series for the function $f(x)=\lvert x\rvert$ over the interval $(-l,l)$ with $f(x+2l)=f(x)$.	08
	b	Expand the function $f(x)=x(\pi-x)$ over the interval $(0,\pi)$ in half range Fourier cosine series.	08
7	a	Maximize $z=3x+5y$ subject to the constraints $3x+2y\le 18$, $x\le 4$, $y\le 6$ and $x,y\ge 0$ using simplex method.	08
	b	Maximize $=3x_1+2x_2+3x_3$ subject to the constraints $2x_1+x_2+x_3\le 2$, $3x_1+4x_2+2x_3\ge 8$, and x_1, x_2, $x_3\ge 0$ using Big M method.	08
		OR	
8	a	A factory uses three types of machines to produce two types of electronic gadgets. The first gadget requires 12, 4 and 2 *hours* respectively on the three types of machines. The second gadget requires 6, 10 and 3 *hours* on the machine respectively. The total available time in hours respectively on the machines is 6000, 4000 and 1800 respectively. If the two types of gadgets respectively fetch a profit of $Rs.400$ and $Rs.1000$, formulate the data as a linear programming problem and find the number of gadgets of each type to be produced for getting maximum profit using graphical method.	08
	b	Maximize $F=6x+10y+2z$ subject to the constraints: $2x+4y+3z\le 40$, $x+y\le 10$, $2y+z\le 12$ and x, y, $z\ge 0$ using simplex method.	08
9	a	Solve the wave equation $u_{tt}=4u_{xx}$ subject to $u(0,t)=0$, $u(4,t)=0$, $u_t(x,0)=0$ and $u(x,0)=x(4-x)$ by taking $h=1$, $k=0.5$ up to four steps.	08
	b	Solve $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$ using Schmidt method with the conditions: $u(x,0)=\sin(\pi x)$, $0\le x\le 1$, $u(0,t)=0$, $u(1,t)=0$. Carryout the computations for two-time levels taking $h=\frac{1}{3}$ and $k=\frac{1}{36}$.	08
		OR	

10	a	Solve $u_{xx} + u_{yy} = 0$ in the following square region with the boundary conditions indicated as shown in the Fig. 10.a  Fig. 10.a	
		Perform five iterations.	08
	b	Solve $\frac{\partial u}{\partial t} = 5\frac{\partial^2 u}{\partial x^2}$ using Schmidt method with the conditions $u(0,t) = 0$, $u(5,t) = 60$, $u(x,0) = \begin{cases} 20x, & 0 \le x \le 3 \\ 60, & 3 \le x \le 5 \end{cases}$	
		Carryout the iterations up to three-time levels with $h = 1$ and $k = 0.1$.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations April-2023

(Common to CH, ME, IM, CV, BT)

INTEGRAL TRANSFORMS, OPTIMIZATION AND NUMERICAL TECHNIQUES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6,7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.

PART-A

1	1.1	Laplace transform of the input signal $t\sin 3t$ is _____.	02
	1.2	If $L[f(t)] = \frac{e^{-1/s}}{s}$ then $L[e^t f(3t)]$ is _____.	02
	1.3	$L^{-1}\left[\frac{1}{2s^2-8}\right] =$ _____.	02
	1.4	If $F(s) = \frac{1}{s^2+8s+16}$, then $L^{-1}[F(s)]$= _____.	02
	1.5	For an even function $f(x)$ defined over the interval $(-2,2)$, the Euler coefficients in the Fourier seris expansion of $f(x)$, a_0 is _____ and a_n is _____.	02
	1.6	For the function $f(x) = \begin{cases} \pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$, $f(x+2\pi) = f(x)$, the value of the function at $x = 0$ is _____.	02
	1.7	Basic solutions of equation $x - 4y = 16$ are _____ & _____ and if $x \geq 0, y \geq 0$, the feasible solution is _____.	02
	1.8	Write the following LPP in the standard form: Maximize $z = x_1 + 2x_2 + 2x_3$ Subject to the constraints $x_1 + x_2 + x_3 \leq 1$, $x_1 + 2x_2 + x_3 \leq 2$, $x_1, x_2, x_3 \geq 0$.	02
	1.9	$u_{xx} + 2u_{yy} + y_{yy} = 0$ is classified as _____ partial differential equation.	02
	1.10	Using the explicit scheme the value of $u(2,1)$ for the one dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subjected to the boundary conditions of $u(0,t) = u(5,t) = 0$ and $u(x,0) = x^2(25 - x^2)$ by taking $h = 1$ and $k = 1/2$ is _____.	02

PART-B

2	a	The saw-tooth wave function in the time domain is defined as $(t) = \frac{at}{T}$, $0 < t < T$. Find the frequency domain signal. (Given $f(t+T) = f(t)$). Represent $f(t)$ graphically.	08
	b	Obtain the Laplace transform of the following functions: i) $\frac{\cos(t)-\cos(2t)}{t}$ ii) $\int_0^t \left(t^{-3/2} + \cos^2(2t)\right)dt$	08
3	a	Obtain the inverse Laplace transform of following functions: i) $\cot^{-1}(s-5)$ ii) $\frac{3s+2}{(s-2)(s+1)}$.	08

	b	The equation of motion for the forced oscillation of a body is $y'' - 2y' + y = e^{2t}$, $y(0) = 0$, $y'(0) = -1$. Using Laplace transform determine the displacement y as a function of t.	08
		OR	
4	a	Use convolution theorem to determine $L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$.	08
	b	Find the inverse Laplace transform of the function $\frac{5s+3}{(s-1)(s^2+2s+5)}$.	08
5	a	Expand $f(x) = (x-1)^2$ as half range Fourier cosine series for $0 \le x \le 1$. Hence deduce $\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$.	08
	b	If $f(x) = \begin{cases} 0, -\pi < x \le 0 \\ \sin x, \ 0 < x < \pi \end{cases}$. Prove that $f(x) = \frac{1}{\pi} + \frac{sinx}{2} - \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2-1}$.	08
		OR	
6	a	Expand the function $f(x) = x(2\pi - x)$ in Fourier series over the interval $(0,2\pi)$. Hence deduce $\frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.	08
	b	Expand $f(x) = \begin{cases} \frac{1}{4} - x, & 0 < x < \frac{1}{2} \\ x - \frac{3}{4}, & \frac{1}{2} < x < 1 \end{cases}$ as the Fourier series of sine terms.	08
7	a	A firm produces an alloy having the following specification: i) Specific Gravity ≤ 0.98 ii) Chromium content ≥ 8% iii) Melting point ≥ 450^0C Raw materials A and B are used to produce the alloy. Raw material A has specific gravity 0.92, chromium content 7% and melting point 440^0C. Raw material B has specific gravity 0.99, chromium content 13% and melting point 490^0C. Raw material A costs $Rs.90$ per kg and Raw material B costs $Rs.280$ per kg. Formulate the problem as a linear programming problem.	06
	b	Using Big-M method to solve the following: Minimize $Z = 2x_2 + x_2$ subject to the constraints $3x_1 + x_2 = 3$, $4x_1 + 3x_2 \ge 6$, $x_1 + 2x_2 \le 3$ and $x_1, x_2 \ge 0$.	10
		OR	
8	a	Solve the following linear programming problem graphically, Minimize $z = 5x + 8y$ subject to the constraints $12x + 2y \ge 42$, $x + 3y \ge 12$, $x + y \le 10$ and $x \ge 0$, $y \ge 0$.	08
	b	Using Simplex method Maximize $z = 4x_1 + 10x_2$ subject to the constraints $2x_1 + x_2 \le 50$, $2x_1 + 5x_2 \le 100$, $2x_1 + 3x_2 \le 90$ and $x_1, x_2 \ge 0$	08

9	a	Solve $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$, subject to the conditions $u(x,0)=0$, $u(x,1)=x$, $0\le x\le 1$, $u(0,y)=0$, $u(1,y)=y$, $0\le y\le 1$ with $h=k=\frac{1}{3}$ using Gauss-Seidel iterative method (Perform one iteration).	08
	b	For the equation $2\frac{\partial^2 u}{\partial x^2}=\frac{\partial u}{\partial t}$, $0<x<4$, $t>0$ under the conditions: i) $u(x,0)=x(4-x), 0<x<4$ ii) $u(0,t)=u(4,t)=0, t\ge 0$ employ Bendre-Schmidt method with $h=1$ to find the values of u upto $t=1$.	08
		OR	
10	a	Solve numerically $\frac{\partial^2 u}{\partial t^2}=0.0625\frac{\partial^2 u}{\partial t^2}$ subject to the conditions $u(0,t)=0=u(5,t)$, $u(x,0)=x^2(x-5)$ and $u_t(x,0)=0$ by taking $h=1$ for $0\le t\le 1$.	08
	b	The temperature distribution $u(x,t)$ is modeled as the one-dimensional heat equation $2\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial t^2}$,$0\le x\le 5$ subject to the boundary conditions $u(0,t)=u(5,t)=0$, and the initial condition $u(x,0)=5x-x^2$. Determine the temperature distribution $u(x,t)$ by taking $\Delta x=h=1$ and $\Delta t=k=\frac{1}{3}$ using Schmidt relation for 5 levels.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations March-2021

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ENGINEERING MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

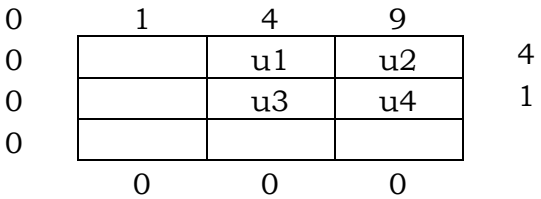
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The necessary condition for $\int_{x_1}^{x_2} f(x,y,y')dx$ to be extremum is _____	01
	1.2	The shortest distance between two points in a plane is a _____.	01
	1.3	If $f(x)\begin{cases}-x, & -\pi < x < 0\\ x, & o < x < \pi\end{cases}$ then $f(x)$ is a _____ function in $(-\pi,\pi)$.	01
	1.4	The smallest period of the function $\sin(\frac{2n\pi x}{k})$ is _____	01
	1.5	If $L\{f(t)\} = F(s)$, then $L\{\frac{f(t)}{t}\}$=_____.	01
	1.6	$L^{-1}\{(s+2)^{-2}\}$ =_____.	01
	1.7	Give an example of tridiagonal matrix of order 4.	01
	1.8	If 'p' is multiplicity of a root then the Newton-Raphson formula to find the root is_____.	01
	1.9	Bendre-Schmidt recurrence relation for one dimensional heat equation is _____.	01
	1.10	The finite difference form of $\frac{\partial^2 u}{\partial x^2}$ is _____.	01
	1.11	The extremal $\int_{x_1}^{x_2}(y' + x^2(y')^2)dx$ is _____.	02
	1.12	For the function $f(x) = \frac{\pi - x}{2}$ in $0 \le x \le 2\pi$, the value of the Fourier co-efficient a_0 is _____.	02
	1.13	$L\{t^5 e^{2t}\}$ =_____.	02
	1.14	For the system of equation $2x + 3y = 7,\ 3x - y = 5$ and co-efficient matrix $A = LU$ where $L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22}\end{bmatrix}$ and $U = \begin{bmatrix}1 & U_{12}\\ 0 & 1\end{bmatrix}$ then find L.	02
	1.15	The diagonal five-point formula is given by u_{ij}=_____.	02

PART-B

2	a	Show that the geodesics on a sphere of radius $'a'$ are its great circles.	08
	b	Find the extremal of the functional $\int_0^1 \{(y')^2 + 12xy\}\,dx, y(0) = 0, y(1) = 1$.	08

3	a	Obtain the complex form of the Fourier Series for the function: $f(x) = \begin{cases} -k & in \quad -\pi < x < 0 \\ k & in \quad 0 < x < \pi \end{cases}$	08
	b	Obtain the half range sine series of $f(x) = \begin{cases} \frac{1}{4} - x & in\ 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & in \quad \frac{1}{2} < x < 1 \end{cases}$	08
		OR	
4	a	Find the half range cosine series of $f(x) = xsinx\ in\ 0 < x < \pi$. Deduce that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \cdots \ldots = \frac{\pi - 2}{4}$.	08
	b	Obtain the Fourier series expansion of the function : $f(x) = \begin{cases} x & in \quad 0 < x < \pi \\ x - 2\pi & in \quad \pi < x < 2\pi \end{cases}$. Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$	08
5	a	Using Convolution theorem, evaluate $L^{-1}\{\frac{s^2}{(s^2+a^2)^2}\}$.	08
	b	A periodic function $f(t)$ of period $2a, a > 0$ is defined by $f(t) = \begin{cases} a, & 0 \le t < a \\ -a, & a < t < 2a \end{cases}$. Show that $L\{f(t)\} = \frac{a}{s}\tanh(\frac{as}{2})$.	08
		OR	
6	a	Find the inverse Laplace transform of : i) $\frac{2s-1}{s^2+2s+17}$ ii) $\frac{s}{(s^2+a^2)^2}$	08
	b	Solve the differential equation $y'' + 4y' + 3y = e^{-t}$ with $y(0) = 1 = y'(0)$, Using Laplace Transform method.	08
7	a	The equation $f(x) = 3x^3 + 4x^2 + 4x + 1 = 0$ has a root in the interval $(-1,0)$. Determine an iteration function $\emptyset(x)$, such that the sequence of iterations obtained from $x_{k+1} = \emptyset(x_k),\ x_0 = -0.5, k = 0,1,2, \ldots$ converges to the root. Perform five iterations to find the root.	08
	b	Solve the following linear system of equations, using LU decomposition method. $x + y + z = 1$ $3x + y - 3z = 5$ $x - 2y - 5z = 10$	08
8	a	Solve $25U_{xx} = U_{tt}$ at the pivotal points given $U(0,t) = 0 = U(5,t);\ U_t(x,0) = 0$ and $U(x,0) = \begin{cases} 20x, & 0 \le x \le 1 \\ 5(5-x), & 1 \le x \le 5 \end{cases}$ by taking h=1. Compute $U(x,t)$ for $0 \le t \le 1$.	08
	b	Evaluate the function $U(x,y)$ satisfying $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points given the values on the boundary as indicated in the Fig8.b: Carry out one iteration.  Fig 8.b	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations Nov/Dec-19

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APPLIED MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	$L[(2-a^t)]$ as a function of $F(s)$ is _____.	01
	1.2	If $L[f(t)] = \frac{5-3s}{s^2+4s+1}$, then $L\left[\frac{1}{2}f\left(\frac{t}{2}\right)\right]$ is _____.	01
	1.3	The inverse Laplace transform of $F(s) = e^{-s/3}$ is _____.	01
	1.4	Laplace transform of a signal is found to be $\frac{1}{\left(s-\frac{1}{2}\right)^4}$, then the signal in the time domain is _____.	01
	1.5	If the function $f(x, y, y')$ is independent of x and y, then the Euler's equation is _____.	01
	1.6	The curve passing through any two points rotated about the horizontal axis, and which has minimum surface area is a _____.	01
	1.7	An homogeneous system of linear equation $AX = 0$ has infinitely many solution if rank of the matrix A is _____.	01
	1.8	If the product of two Eigen values of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is 16, then the third Eigen value is _____.	01
	1.9	If two variables x and y are uncorrelated then the line of regression of y on x is _____.	01
	1.10	The formula for coefficient of correlation 'r' between the variables x and y is _____.	01
	1.11	$L[(t+2)u(t-2)]$ is _____.	02
	1.12	The inverse Laplace transform of the function $\frac{1}{s^2+2s+3}$ is _____.	02
	1.13	The extremal of the functional $I = \int_l^m [(y')^2 + 2\cos x]\, dx$ *is* _____.	02
	1.14	The augmented matrix of the linear system of equation is $\left[\begin{array}{ccc\|c} 1 & -2 & 1 & b \\ 1 & 1 & -2 & c \\ -2 & 1 & 1 & a \end{array}\right]$, the system will have no solution if _____.	02
	1.15	For the data (W_i, V_i), $i = 1,2,\ldots,n$, the normal equations to fit $W = a + bV$, where 'a' and 'b' are constants are _____.	02

PART-B

2	a	i) Find $L[\int_0^t \frac{2\sin t \sin 5t}{t}\, dt]$ ii) Evaluate $\int_0^\infty te^{-2t}sin4tdt$, using Laplace transforms.	08
	b	Given $f(t) = \begin{cases} k, & 0 < t < a/2 \\ -k, & \frac{a}{2} < t < a \end{cases}$, where $f(t+a) = f(t)$, show that $L[f(t)] = \frac{K}{s}tanh\left(\frac{as}{4}\right)$. Represent the function graphically.	08
3	a	Find: i) $L^{-1}\left[\frac{s^2}{(s^2+1)(S^2+4)}\right]$, ii) $L^{-1}[\log\left(\frac{s^2+1}{s(s+1)}\right)]$	08
	b	Solve $y' + y = e^{3t} - 1, y(0) = 0$ using Laplace transform technique.	08
		OR	
4	a	Using convolution theorem, find the inverse Laplace transform of $\frac{1}{(s^2+4)^2}$.	08
	b	Solve by using Laplace transform: $y''(t) + y(t) = F(t)$ where $F(t) = \begin{cases} 4, & 0 < t < 1 \\ 3, & t > 1 \end{cases}$ subject to the conditions $y(0) = 0$ and $y'(0) = 1$.	08
5	a	Find the path on which a particle in the absence of friction, will slide from one point to another in the shortest time under the action of gravity.	08
	b	Find the curves on which the functional $\int_0^1 (y'^2 + 12xy)dx$ with $y(0) = 0$ and $y(1) = 1$ can be extremised.	08
		OR	
6	a	Obtain the geodesic on a right circular cylinder	08
	b	Show that the extremal of $\int_0^{\pi/2}(y^2 - y'^2 - 2ysinx)dx$ under the end conditions $y(0) = y(\frac{\pi}{2}) = 0$ is $y = -\frac{1}{2}xcosx$.	08
7	a	Determine the rank of the matrix $\begin{bmatrix} 4 & 3 & -7 & 1 \\ 8 & -3 & -5 & -2 \\ 2 & 5 & -7 & -3 \\ 6 & 3 & 9 & -4 \end{bmatrix}$ by reducing to Echelon form.	05
	b	Solve: $83x + 11y - 4z = 95$, $3x + 8y + 29z = 71$, $7x + 52y + 13z = 104$, using Gauss Seidel method. Carry out 4 iterations.	05
	c	Use the power method to estimate the dominant Eigen value and its corresponding Eigen vector for the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with $X^0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Carry out 5 iterations.	06

8	a	Fit a curve of the form $y = ae^{bx}$ to the following data: x | 5 | 15 | 20 | 30 | 35 | 40 y | 10 | 14 | 25 | 40 | 50 | 62	08
	b	The following table gives the stopping distance y in meters of a motor bike moving at a speed of x kms/hour when the breaks are applied x | 16 | 24 | 32 | 40 | 48 | 56 y | 0.39 | 0.75 | 1.23 | 1.91 | 2.77 | 3.81 Find the correlation coefficient between the speed and the stopping distance and the equations of regression lines. Hence estimate the maximum speed at which the motor bike could be driven if the stopping distance is not to exceed 5 meters.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Common to AS / CV / BT / CH /ME / IM

ENGINEERING MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

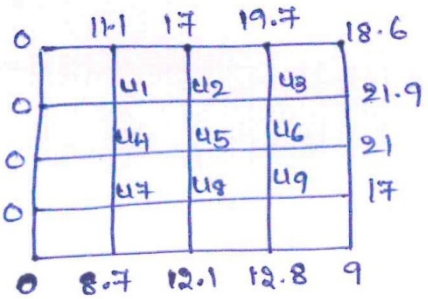
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Write two alternate forms of Euler's equation.	02
	1.2	The shortest distance between two points on the surface of the plane is _____.	01
	1.3	A particle moving in a vertical plane purely under gravity. The path of least time is _____.	01
	1.4	Represent the following function graphically: $f(x)=\begin{cases}-\pi, & -\pi<x<0\\ x, & 0<x<\pi\end{cases}$	01
	1.5	Write the Fourier expansion of $f(x)$ in $(-\pi,\pi)$ when $f(x)$ is even.	01
	1.6	For the function $f(x)=x^2$ in $0<x<2\pi$, the value of the Fourier coefficient a_0 is _____.	02
	1.7	$L\{e^{at}\sin bt\}=$ _____.	01
	1.8	If $L^{-1}\{f(s)\}=F(t)$, then $L^{-1}\{f(s-a)\}=$ _____.	01
	1.9	$L\left\{\frac{\sin t}{t}\right\}=$ _____.	02
	1.10	Write the formula to find multiple root of the equation $f(x)=0$ using modified Newton-Raphson method.	01
	1.11	Write the condition for the convergence of fixed point iteration method.	01
	1.12	If $A=\begin{bmatrix}1 & 2 & 3\\ -1 & 4 & 2\\ 5 & 2 & 1\end{bmatrix}$ is partitioned as, $A=\left[\begin{array}{c:cc} B & \vdots & C\\ \cdots & \vdots & \cdots\\ D & \vdots & E\end{array}\right]$ then $B^{-1}=$ _____.	02
	1.13	The finite difference approximation for u_{xx} is _____.	01
	1.14	Explicit formula for the solution of wave equation is _____.	01
	1.15	The standard five point formula to evaluate $u_{2,2}$ is _____.	02

PART-B

2	a	Determine the Geodesics on a right circular cylinder.	08
	b	Find the extremal of the functional $\int_{x1}^{x2}(y^2+(y')^2+2ye^x)dx.$	08

3	a	Determine the Fourier series expansion of the function $f(x) = 2x - x^2$ in (0,3).	08
	b	Express $f(x) = \frac{\pi^2}{12} - \frac{x^2}{4}$ as a Fourier series in $(-\pi, \pi)$.	08
		OR	
4	a	Find the half range sine series for $f(x) = x(\pi - x)$ in $0 < x < \pi$.	08
	b	Find the Fourier series of $f(x) = \frac{\pi - x}{2}$, $0 < x < 2$.	08
5	a	Find: i) $L\left\{\frac{e^{-at} - e^{-bt}}{t}\right\}$, ii) $L\{\int_0^t e^{-t} \cos t \ dt\}$.	08
	b	Find the Laplace transform of a periodic function $F(t) = \begin{cases} k, & 0 < t < a \\ -k, & a < t < 2a \end{cases}$	08
		OR	
6	a	Using convolution theorem, find $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$.	08
	b	Solve the differential equation by Laplace transform. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = e^{-t}$; $y(0) = 0$, $y'(0) = 1$.	08
7	a	Solve $x^3 = 2x + 5$ for a positive root by iteration method, correct to 4 decimal places.	08
	b	Solve the following system of equations by LU decomposition method: $x + 5y + z = 9$; $2x + y + 3z = 12$; $3x + y + 4z = 16$.	08
8	a	Solve the equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $0 < x < 1$ at $t = 0.002$, $h = 0.1$, $k = 0.001$ and the conditions $u(0,t) = 0$, $u(1,t) = 0$, $u(x,0) = f(x)$ where $f(x) = \begin{cases} 2x, & 0 \le x \le 1/2 \\ 2(1-x), & 1/2 \le x \le 1 \end{cases}$	08
	b	Solve Laplace's equation $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values as shown in the fig 8b. Obtain the values after first iteration of Gauss-Seidal method.  Fig 8b	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Fast Track Examinations July-19

Common to AS / CV / BT / CH /ME / IM

APPLIED MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	If $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\left(\frac{1}{s}\right)$,then $L\left\{\frac{\sin 2t}{2t}\right\}$ is _____.	01
	1.2	If Laplace transform of time function $f(t)$ is $F(s)$ then Laplace transform of $f'(t)$ is _____.	01
	1.3	If the frequency response of the system is $\frac{1}{s^2+3}$, then corresponding time response is _____.	01
	1.4	$L^{-1}\{\frac{1}{\sqrt{s}}\}$ is _____.	01
	1.5	To extremize the functional $\int_{x_1}^{x_2} f(x, y, y')dx$, when f is independent of y. Euler's equation reduces to _____.	01
	1.6	The shortest distance between any two points on a sphere is _____.	01
	1.7	The system of equation $MX = N$ has unique solution if rank of $M =$ _____.	01
	1.8	If $-2, \sqrt{3}, \frac{1}{4}$ are eigen values of matrix P then eigen values of P^2 are _____.	01
	1.9	The range for correlation co-efficient 'r' is _____.	01
	1.10	If the lines of regression are $3x + 2y = 26$ and $6x + y = 31$, then mean values of x and y are respectively _____ and _____.	01
	1.11	$L\{\int_0^t \cosh 2t\, dt\}$ *is* _____.	02
	1.12	The inverse Laplace transform of $\left(\frac{1}{s^2+2s-3}\right)$ is _____.	02
	1.13	The external of the functional $\int_{x_1}^{x_0}\left[\sqrt{x} + (y')^2\right]dx$ is _____.	02
	1.14	The system of equation in echelon form is $\left[\begin{array}{cccc:c} 1 & 2 & 3 & : & 3 \\ 0 & 1 & -2 & : & -3 \\ 0 & 0 & 1 & : & 2 \end{array}\right]$ then using Gauss elimination method, the values of unknown x_1, x_2, x_3 are _____.	02
	1.15	The normal equation to fit a curve $v = au^2 + bu + c$ for (u_i, v_i) where 'a', 'b', and 'c' are constants are _____.	02

PART-B

2	a	Evaluate $L\{te^{-3t}\cos 2t\} + L\left\{\frac{1-e^{at}}{t}\right\} + L\{H(t-4)e^{t-4}\}$.	08
	b	Obtain the Laplace transform of saw-tooth wave function $f(t) = \frac{at}{T}$, $0 < t < T$ where $f(t+T) = f(T)$. Draw the graph of the function.	08
3	a	Using convolution theorem obtain $L^{-1}\left\{\frac{s}{(s^2+4)^2}\right\}$.	08
	b	Solve the problem of undamped forced vibrations of a spring governed by the equation $\frac{d^2x}{dt^2} + \alpha^2 x = A\ \sin\omega t$ given $x = x_0$ and $\frac{dx}{dt} = v_o$ at $t = 0$ using Laplace transforms.	08
		OR	
4	a	The coordinates $(x, y)$of a particle moving along a plane curve at any time t are given by $\frac{dy}{dx} + 2x = sin2t$, $\frac{dx}{dt} - 2y = cos\ 2t, (t > 0)$. Given $x = 1$ and $y = 0$ at $t = 1$, find $x(t)$ and $y(t)$ using Laplace transforms.	08
	b	Find inverse Laplace transform of i) $\frac{1}{s(s+1)(s+2)}$ ii) $Log[\frac{s^2-a^2}{(s-b)^2}]$	08
5	a	Find the extremal of the functional $\int_1^2 \frac{x^3}{(y')^2}dx$, $y(1) = 1, y(2) = 4$.	08
	b	Define geodesics. Find the geodesic on the surface of a plane.	08
		OR	
6	a	Find the curve passing through (x_1, y_1) and (x_2, y_2) which when rotated about the x-axis gives a minimum surface area.	08
	b	Obtain the extremal of $\int_{-1}^{0}[12xy - (y')^2]dx$ with $y(-1) = 1$ and $y(0) = 0$.	08
7	a	Solve the traffic flow problem represented by the following system of equations where unknowns represent the number of cars entering and leaving the junction: $2x_1 + 2x_2 + 4x_3 = 18$ $x_1 + 3x_2 + 2x_3 = 13$ $3x_1 + x_2 + 3x_3 = 14$ Use Gauss Elimination method.	08
	b	Use the power method to estimate the dominant eigen value and its corresponding eigen vector for the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with $X^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Perform 6 iterations.	08
8	a	A simply supported beam carries a concentrated load P at its midpoint. The maximum deflection Y is measured corresponding to various values of P as given below: P 100 120 140 160 180 200 y 0.45 0.55 0.60 0.70 0.80 0.85 Fit a relation of the form $Y = a + bP$.	08
	b	Estimate the coefficients of correlation and lines of regression for the following data: x: 1 2 3 4 5 y: 2 5 3 8 7	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution Affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18

Common to AS / CV / BT / CH /ME / IM

APPLIED MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	$L\left\{t^{-\frac{1}{2}}\right\}$ is _____.	01
	1.2	The laplace transform of $e^{2t}H(t-3)$ is _____.	01
	1.3	If the inverse laplace transform takes a function of frequency, then it yields a function of _____.	01
	1.4	The inverse laplace transform of $\frac{1}{(s+5)^5}$ is _____.	01
	1.5	The Euler-Lagrange equation is _____.	01
	1.6	The geodesics on a plane are _____.	01
	1.7	The system of equations $AX = B$ has infinite solutions if _____.	01
	1.8	The characteristic equation of a square matrix A is _____.	01
	1.9	The range for correlation coefficient r is _____.	01
	1.10	If $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the two regression lines, then mean of x series is _____.	01
	1.11	Given $L\left\{2\sqrt{\frac{t}{\pi}}\right\} = \frac{1}{s^{\frac{3}{2}}}$, then $L\left\{\frac{1}{\sqrt{\pi t}}\right\}$ = _____.	02
	1.12	The inverse laplace transform of $\frac{3s+5\sqrt{2}}{s^2+8}$ is _____.	02
	1.13	The external of the functional $\int_{x_1}^{x_2}(y^2 + x^2y')dx$ is _____	02
	1.14	The reduced echelon form of linear system of equations is $\begin{bmatrix} 2 & 1 & 4 & : & 12 \\ 0 & 1 & -1 & : & 1 \\ 0 & 0 & 3 & : & 3 \end{bmatrix}$, then the values of unknowns x, y, z are _____.	02
	1.15	The normal equations to fit the curve $y = a + bx + cx^2$ are _____.	02

PART-B

2	a	Find the Laplace transform of $\frac{\cos 2t - \cos 3t}{t} + t\sin t$	05
	b	Express the following function in terms of Heaviside function and hence find the Laplace transform: $f(t) = \begin{cases} cost, & 0 < t < \pi \\ sint, & t > \pi \end{cases}$	05
	c	Given $f(t) = \begin{cases} E, & 0 < t < \frac{a}{2} \\ -E, & \frac{a}{2} < t < a \end{cases}$ where $f(t+a) = f(t)$, show that $L\{f(t)\} = \frac{E}{s} \cdot tanh\left(\frac{as}{4}\right)$	06

3	a	Find the inverse Laplace transform of the following: i) $\frac{s+5}{s^2-6s+13}$ ii) $log\left(1-\frac{a^2}{s^2}\right)$	08
	b	Using Laplace transforms, find the solution of the initial value problem: $y'' + 9y = 9\,H(t-3), y(0) = y'(0) = 0.$	08
		OR	
4	a	Find $L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$, using convolution theorem.	08
	b	Solve: $\frac{dx}{dt} - y = e^t, \frac{dy}{dt} + x = \sin t.$ given $x(0) = 1, y(0) = 0,$ using Laplace transform method.	08
5	a	Obtain the necessary condition for the functional $I = \int_{x_1}^{x_2} f(x, y, y')dx$ to be extremum.	06
	b	If a heavy cable hangs freely under gravity between two fixed points then show that the shape of the cable is a catenary.	10
		OR	
6	a	Find the extremal of the functional: $\int_{x_1}^{x_2}[y^2 + (y')^2 + 2ye^x]\,dx$	06
	b	Define gedosics. Show that geodesics on a sphere are minor arcs of great circles.	10
7	a	Test for consistency and solve the following linear system of equations: $5x + y + 3z = 20$ $2x + 5y + 2z = 18$ $3x + 2y + z = 14$	05
	b	Solve the following system of equations by Gauss- Seidal method: $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$	05
	c	obtain the dominant eigen value and the corresponding eigen vector of the matrix: $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by power method taking the initial eigen vector as $[1, 1, 1]$. Perform 6 iterations.	06
8	a	A simply supported beam carries a concentrated load P at its mid point. Corresponding to various values of P, the maximum deflection Y is measured and is given in the following table: P: 50, 70, 100, 120 Y: 12, 15, 21, 25 Find a law of the form $Y = a + bP$.	08
	b	The following data regarding the heights (y) and the weights (x) of 100 college students are given: $\Sigma x = 15000; \Sigma x^2 = 2272500$ $\Sigma y = 6800, \Sigma y^2 = 46.3025, \Sigma xy = 1022250$ Find the correlation coefficient between height and weight and state the equation of regression of height on weight.	08

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R. V. COLLEGE OF ENGINEERING

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III Semester B. E. Fast Track Examinations July-18

Common to AS / CV / BT / CH /ME / IM

APPLIED MATHTHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

9. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
10. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	$L(t^{1/2})$ is _____.	01
	1.2	$L(\delta(t-a)=$ _____.	01
	1.3	If * denotes convolution of two functions $f(t)$ and $g(t)$ then $f(t)*g(t)$ is _____.	01
	1.4	$L^{-1}(\frac{s}{3s^2+6})$ _____.	01
	1.5	When f does not contain x explicitly then Euler's equation is _____.	01
	1.6	The geodesic on a plane is _____.	01
	1.7	The system of equations $AX=B$ is inconsistent if _____.	01
	1.8	In Gauss Jordan method the coefficient matrix must be reduced to _____.	01
	1.9	If regression coefficients of x on y and y on x are -0.8 and -0.5 respectively then correlation coefficient is _____.	01
	1.10	If θ is the angle between the lines of regression them $tan\theta$ = _____.	01
	1.11	Express the following function in terms of unit step function. $f(x)=\begin{cases}\cos t, & 0<t\le\pi\\ \sin t, & t>\pi\end{cases}$	02
	1.12	$L^{-1}(\frac{s}{s^2-2s+1})$ = _____.	02
	1.13	The extremal of the function $\int_{x_2}^{x_2}(y^2-y'^2)dx$ _____.	02
	1.14	If $A=\begin{bmatrix}2&4&3\\0&1&7\\0&0&3\end{bmatrix}$ then eigen values of A^{-1} are _____.	02
	1.15	The normal equations of $y=ax+b$ are _____.	02

PART B

2	a	Probe that the Laplace transform of the following function of period $2a$ $f(x)=\begin{cases}t, & 0<t\le a\\ 2a-t, & a<t<2a\end{cases}$ is $\frac{1}{s^2}tan\,h(\frac{as}{2})$.	08
	b	Find $L\{\int_0^t e^{2t}(\frac{cos\,at-cos\,bt}{t}+t\,sin\,5t)dt\}$	08

3	a	Find: i) $L^{-1}\left(\frac{2}{(S^2+1)(S^2+4)}\right)$ ii) $L^{-1}\left(log\frac{(S+1)(S^2+9)}{(S^2-25)(S-4)}\right)$	08
	b	Using Laplace transforms solve $\frac{d^3y}{dx^3}+2\frac{d^2y}{dx^2}-\frac{dy}{dx}-2y=0$ under the conditions $y(0)=\frac{dy}{dx}(0)=0,\frac{d^2y}{dx^2}(0)=6$.	08
		OR	
4	a	Employing convolution theorem, find Laplace inverse of $\frac{S^2}{(S^2+a^2)^2}$.	08
	b	Solve the following simultaneous equations using Laplace transformations: $\frac{dx}{dt}+y=sin\,t\,,\frac{dy}{dt}+x=cos\,t$, given $x=1,y=0$ when $t=0$.	08
5	a	Derive Euler's equation in the form $\frac{\partial f}{\partial y}-\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)=0$.	08
	b	On which curve the functional $\int_0^{\pi/2}(y'^2-y^2+2xy)dy$ with $y=0,y(\frac{\pi}{2})=0$ be extremised.	08
		OR	
6	a	Find the path in which a particle, in the absence of friction, will slide from one point to another in the shortest time under the action of gravity.	08
	b	Show that the geodesic on the surface of the cylinder is circular helix.	08
7	a	Find the rank of the matrix $\begin{pmatrix}2&-1&-3&-1\\1&2&3&-1\\1&0&1&1\\0&1&-1&-1\end{pmatrix}$	08
	b	Find the dominant eigen value and the corresponding eigen vector of the matrix $\begin{bmatrix}1&3&-1\\3&2&4\\-1&4&10\end{bmatrix}$ with $[1,1,0]^T$ as the initial vector using power method. Perform 6 iterations.	08
8	a	Calculate the coefficient of correlation and obtain the lines of regression for the following data: x: 1 2 3 4 5 6 7 8 9 y: 9 8 10 12 11 13 14 16 15	08
	b	Fit a curve $y=a+bx+cx^2$ to the following data: x: −3 −2 −1 0 1 2 3 y: 4.63 2.11 0.67 0.09 0.63 2.15 4.58	08

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R. V. COLLEGE OF ENGINEERING

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III Semester B. E. Examinations Nov/Dec-17

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APPLIED MATHEMATICS-III

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

7. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
8. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	$L\left[t^{3/2}\right]$ is _____.	01
	1.2	The Laplace transform of the function $t\delta(t-3)$ is _____.	01
	1.3	The inverse Laplace transform of the function $\frac{1}{3s^2+5}$ is _____.	01
	1.4	The frequency response of a system is found to be $\frac{se^{-s}}{s^2-4}$, then the corresponding time response is _____.	01
	1.5	Write any one alternate form of Euler-Lagrange equation for a functional $I=\int_{x_1}^{x_2} f(x,y,y')dx$ to be extremum.	01
	1.6	The path in which a particle will slide from one point to another in the absence of friction in the shortest time under the action of gravity is _____.	01
	1.7	The system of equation $AX=B$ has unique solution if _____.	01
	1.8	If $-3,\sqrt{2}$ are eigen values of matrix M then eigen values of M^2 are _____.	01
	1.9	The range for correlation coefficient r is _____.	01
	1.10	If $y=\frac{1}{2}x+3$ and $x=y+4$ are 2 lines of regression then mean of y series is _____.	01
	1.11	If $L\{f(t)\}=\frac{s^2+1}{s^3+3s+2}$ then $L\{f(t/3)\}$ is _____.	02
	1.12	The inverse Laplace transform of the function $\frac{1}{s(s+2)}$ is _____.	02
	1.13	The extremal of the functional $\int_{x_1}^{x_2}\left[\sqrt{x}+\sqrt{y'}\right]dx$ is _____.	02
	1.14	The reduced echelon form of linear system of equations is $\begin{bmatrix}1 & 2 & -3:-1\\ 0 & 1 & 4:9\\ 0 & 0 & 1:2\end{bmatrix}$, then the values of unknowns x,y,z are _____.	02
	1.15	The normal equations to fit the curve $v=pu^2+qu+r$ are _____ where p,q,r are constants.	02

PART-B

2	a	Find $L\left\{\frac{\sin^2 t}{t} + \int_0^t t\, e^{-t} \sin 4t\, dt\right\}$.	05
	b	Express the following functions in terms of Heaviside function and hence find the Laplace transform. $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$	05
	c	Find the Laplace transform of the saw toothed wave of period T, given by $f(t) = \frac{t}{T}$ for $0 < t < T$.	06
3	a	Find the inverse Laplace transform of the following: i) $\frac{s+2}{s^2-2s+5}$ ii) $\log\left(\frac{1-s^2}{s^2}\right)$	08
	b	Solve the problem of un-damped forced vibrations of a spring governed by the equation $\frac{d^2x}{dt^2} + \beta^2 x = a \sin \omega t\, , x(0) = x_0, x'(0) = v_o$ using Laplace transform method with two cases $\omega = \beta, \omega \neq \beta$.	08
		OR	
4	a	Find $L^{-1}\left\{\frac{s}{(s^2+16)^2}\right\}$ using convolution theorem.	08
	b	Solve the following system of equations: $2\frac{dx}{dt} + \frac{dy}{dt} - x - y = e^{-t}$ $\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = e^{t}$ Given $x(0) = 2, y(0) = 1$ exploring Laplace transform method.	08
5	a	Obtain the necessary condition for the functional $I = \int_{x_1}^{x_2} f(x, y, y')dx$ to be extremum.	06
	b	Show that a heavy cable hangs freely under gravity between two fixed points in the shape of a catenary.	10
		OR	
6	a	Find the extremum of the functional $\int_0^1 [(y')^2 + 12xy]dx$, $y(0) = 0, y(1) = 1$.	06
	b	Define geodesics. Show that geodesics on a sphere are minor arcs of great circles.	10
7	a	Test for consistency and solve the following linear system of equations $4x - 2y + 6z = 8$ $x + y - 3z = -1$ $15x - 3y + 9z = 21$	05
	b	Obtain the solution of following system of equations using Gauss-Seidel method. $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$ Perform four iterations.	05

	c	Using power method estimate the dominant eigen value and corresponding eigen vector for the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ as the initial vector. Perform 6 iterations.	06
8	a	If P is the pull required to lift the weight by means of a pully block, find a linear law if the form $p = a + bw$, connecting p & w using following data $W(lbs)$: 50, 70, 100, 120 $P(lb)$: 12, 15, 21, 25 Compute p when $w = 150lb$.	06
	b	Find the coefficient of correlation between industrial production and export from the following data. Also find the two regression lines. $Production\ (in\ crore\ tons)$: 55, 56, 58, 59, 60, 61, 62 $Exports\ (in\ crore\ tons)$: 35, 38, 38, 39, 44, 43, 45	10

Work Systems Design

III Semester SEE Question Papers

Course codes: IM233AI, 21IM33, 18IM35, 12IM35

- Feb/Mar 2026 (IM233AI)
- Jan/Feb 2025 (IM233AI)
- Oct/Nov 2024 Supplementary (IM233AI)
- Apr/May 2024 (IM233AI)
- Model Question Paper (2022 Scheme) (IM233AI)
- Apr/May 2023 (21IM33)
- Fast Track Jan 2023 (18IM35)
- Mar 2021 (18IM35)
- Fast Track Oct 2020 (18IM35)
- Data tables, Fast Track Oct 2020 (18IM35)
- Nov/Dec 2019 (18IM35)
- Fast Track Jul 2017 (12IM35)
- Nov/Dec 2016 (12IM35)
- Fast Track Jul 2016 (12IM35)

Question Paper ID: 999028

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM233AI

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Work Systems Design

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10

Part A

Question No	Question	M	CO	BT
1.1	Define a Work System as a physical entity.	02	1	1
1.2	Provide the basic formula for Productivity and define its terms.	02	1	1
1.3	Define the Therbligs- Reach and Move.	02	2	1
1.4	State two main categories under which the Principles of Motion Economy are grouped.	02	2	1
1.5	Name the three major types of Allowances (PFD) included in Standard Time.	02	1	1
1.6	How is the total Allowance usually expressed when calculating Standard Time?	02	1	1
1.7	What does Total Productive Maintenance (TPM) focus on?	02	2	1
1.8	What is the Japanese term for Autonomation? Define its core concept.	02	2	1
1.9	Define an Industrial Accident in the context of Occupational Safety.	02	4	1
1.10	Name two potential negative effects of excessive Noise exposure in the workplace.	02	4	1

Part B

Question No	Question	M	CO	BT
2a	Define physical work systems. Describe their components with examples.	08	1	2

<table>
<tr><td>2b</td><td>The standard time for a manual material-handling work cycle is 2.58 min per piece. The PFD allowance factor used to set the standard was 13%. During a particular 8-hour shift of interest, it is known that the worker lost a total of 53 min due to personal time, rest breaks, and delays. On that same day, the worker completed 214 work units. Determine

(i) The number of standard hours accomplished,

(ii) Worker efficiency, and

(iii) The worker’s performance level expressed as a percentage.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>3a</td><td>How does the "USA Principle" of automation apply to Methods Engineering?</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td>3b</td><td>Explain the role of Methods Engineering in improving productivity and work design.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Describe the "Flow Diagram" (Spatial) and its relationship to the Flow Process Chart.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>4b</td><td>Explain how flow process charts are used in methods engineering. Provide an example of how they help improve a manual task.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>5a</td><td>Rank the five primary work measurement methods by accuracy and justify the high ranking of Predetermined Motion Time Systems (PMTS).</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>5b</td><td>Rank the work measurement methods by "Relative Application Speed" (Time to set the standard) and explain why Historical Data is the fastest.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a,b ,d,e and f are operator controlled and external to machine cycle and they are performance rated at 90%.If the machine allowance is 35% and worker PFD allowance is 25%, determine the normal time and standard time for the work cycle.
<table>
<tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>Observed Time(min)</td><td>0.18</td><td>0.30</td><td>0.88</td><td>1.12</td><td>1.55</td><td>1.80</td></tr>
</table></td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>6b</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a,b ,d,e and f are operator controlled and external to machine cycle and they are performance rated at 90%.If the machine allowance is 35% and worker PFD allowance is 25%, determine the normal time and standard time for the work cycle.
<table>
<tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>Observed Time(min)</td><td>0.12</td><td>0.28</td><td>0.78</td><td>1.04</td><td>1.45</td><td>1.80</td></tr>
</table></td><td>08</td><td>3</td><td>3</td></tr>
</table>

7a	Compare and contrast "Waste of Motion" and "Waste of Transportation." Provide one industrial example for each and evaluate how excessive inventory contributes to both.	08	2	4
7b	Define "Muda" and explain the waste of Overproduction. Why is it considered the most serious waste?	08	2	4
	OR			
8a	Why is "Setup Time Reduction" a prerequisite for implementing a Pull system?	08	2	4
8b	Compare and contrast Push vs. Pull systems of production control. Why is Pull preferred in Lean?	08	2	4
9a	Explain the role of Anthropometry in work system design.	08	4	4
9b	Discuss Noise and its control in the workplace.	08	4	4
	OR			
10a	Analyze the role of "Unsafe Acts" as a primary cause of accidents in workplaces.	08	4	4
10b	Develop an "Ergonomic Checklist" for an office worker to self-assess their risk for developing Repetitive Strain Injuries (RSI).	08	4	5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B.E. Regular/Supplementary Examinations Jan/Feb – 2025

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Illustrate the Pyramidal Structure of the work.	02	2	1
	1.2	Distinguish between internal element and external element in cycle time analysis.	02	4	1
	1.3	Enumerate the categories of worker machine systems based on the levels of operator attention required.	02	1	2
	1.4	Differentiate between snap back and cumulative timing methods.	02	4	2
	1.5	What is lean production?	02	2	3
	1.6	Why is an allowance added to the normal time to complete the standard time?	02	2	2
	1.7	Identify the broad categories of factors causing fatigue in workers.	02	2	4
	1.8	List any four techniques of Work Measurement.	02	1	2
	1.9	In the context of lean manufacturing what is production leveling?	02	2	3
	1.10	In lean manufacturing, _____ refers to waste or non-value-adding activities, while _____ refers to overburdening people or equipment.	02	2	3

PART-B

2	a	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	08	4	1
	b	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8-hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8-hour shift that corresponds to the PFD allowance factor of 16%.	08	3	2
3	a	Review the applicability of three categories of Charting / diagrammatic techniques used by industrial engineers for collecting, displaying and analyzing data on a given work system.	08	2	1
	b	Provide an overview on any four techniques of Methods Engineering.	08	2	1
		OR			
4	a	Enumerate and provide a brief note on the principles of motion economy.	09	2	1
	b	Review the phases involved in the systematic approach that can be used to accomplish a methods engineering study.	07	2	1

<table>
<tr><td>5</td><td>a</td><td>What is a Work Element? Mention at least six guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td></td><td>b</td><td>The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element c is a machine controlled and the time is constant. Elements a, b, d, e and f are operator controlled and external to machine cycle and they are performance rated at 90%. If the machine allowance is 12% and worker PFD allowance is 8%, determine the normal time and standard time for the work cycle.
<table>
<tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>Observed Time (min)</td><td>0.12</td><td>0.28</td><td>0.78</td><td>1.04</td><td>1.45</td><td>1.80</td></tr>
</table></td><td>08</td><td>3</td><td>2</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>6</td><td>a</td><td>What is meant by PMTS? Distinguish between First level PMTS and Higher level PMTS on the basis of at least five characteristics.</td><td>08</td><td>2</td><td>2</td></tr>
<tr><td></td><td>b</td><td>Analyse the causes for variability in the observed work element times that occur from one work cycle to another work cycle.</td><td>08</td><td>4</td><td>2</td></tr>
<tr><td>7</td><td>a</td><td>Enumerate the seven wastes as advocated by Taiichi Ohno in Lean Production philosophy. Explain any four in detail.</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td></td><td>b</td><td>Distinguish between Push System of production control and Pull System of production control. Support your answer with an appropriate illustration.</td><td>06</td><td>4</td><td>3</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>8</td><td>a</td><td>In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td></td><td>b</td><td>Explain the role of Total Productive Maintenance (TPM) in automation with examples.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>9</td><td>a</td><td>Employees in an office environment report eye strain, fatigue, and discomfort. Analyse the physical work environment and recommend improvements.</td><td>08</td><td>3</td><td>4</td></tr>
<tr><td></td><td>b</td><td>What is ergonomics and explain its purpose in production systems. Discuss how ergonomics and human factors are connected to engineering.</td><td>08</td><td>2</td><td>4</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>10</td><td>a</td><td>Highlight on the concept of anthropometry and its importance in designing for the human body, with examples.</td><td>08</td><td>2</td><td>4</td></tr>
<tr><td></td><td>b</td><td>Describe how mechanics can be applied to the human body to enhance safety and efficiency in the workplace.</td><td>08</td><td>2</td><td>4</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Supplementary Examinations Oct/Nov – 2024

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Illustrate pyramidal structure of work.	02	2	1
	1.2	What is meant by the term productivity?	02	2	1
	1.3	List the symbols used in traditional industrial engineering charting techniques.	02	1	2
	1.4	List any two objectives of methods engineering.	02	1	2
	1.5	Mention the elements of standard time for manual work.	02	1	3
	1.6	Expand the acronym MOST.	02	1	3
	1.7	What is meant by Autonomation?	02	2	1
	1.8	Mention the main activities involved in 5s.	02	1	2
	1.9	What is meant by the term Anthropometry?	02	1	4
	1.10	Distinguish between the terms: ‘Accident’ and ‘Hazard’.	02	2	4

PART-B

2	a	During the base year in a mill 326000 tons of steel were produced using 203,000 labor hours. In the next year, the output was 341,000 tonns using 246,000 labor hours. Determine the (i) Labor productivity ratio for the base year, (ii) Labor productivity ratio for the second year, (iii) Labor productivity index for the second year.	10	3	1
	b	Categorize the excess non-productive activities. Write a brief note on each of the category.	06	3	1
3	a	Enumerate the steps in systematic approach of methods engineering application for an improvement.	10	2	2
	b	Evaluation of alternative improvement proposals is done considering two categories of features and specifications. Write a note on them.	06	2	2
		OR			
4	a	Review the basic data collection tools and analysis techniques used for problem solving.	10	4	2
	b	Illustrate and elaborate on automation migration strategy.	06	3	2
5	a	Classify the methods to determine time standards. Support your explanation with an illustration for better clarity.	10	3	3
	b	Illustrate/sketch the model indicating the factors that must be standardized before a standard time can be set for a task.	06	3	3
		OR			
6	a	Enumerate the steps involved in direct time study procedure to arrive at standard time for a job/activity.	–	–	–

<table>
<tr><td></td><td>b</td><td>A regular cycle of work consists of three elements a, b, c. Element ‘d’ is irregular element performed every five cycles. Using the following details; compute normal time and standard time for the work cycle, using an allowance factor of 15%.
<table>
<tr><td>Work element</td><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>Observed time (min)</td><td>0.56</td><td>0.25</td><td>0.50</td><td>1.10</td></tr>
<tr><td>Performance Rating (%)</td><td>100</td><td>80</td><td>110</td><td>100</td></tr>
</table></td><td>–</td><td>–</td><td>–</td></tr>
<tr><td>7</td><td>a</td><td>Summarize the seven forms of ‘waste’ that needs to be eliminated as advocated by Taiichi Ohno to make the production system lean.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>What is meant by the term ‘Kaizen’? How is Kaizen implemented for achieving the goal of continuous improvement in lean systems?</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>8</td><td>a</td><td>Distinguish between ‘push’ and ‘pull’ systems of production control. Support your explanation with an illustrative example.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>Illustrate the structure of a lean production system and highlight the key elements.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td>9</td><td>a</td><td>Sketch the block diagram model of the interactions in a human-machine system. Provide a brief note on different elements in the system.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>Write a note on the primary casual factors for the occurrence of industrial accidents.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>10</td><td>a</td><td>Review the three broad topical areas of Ergonomics as a field of study.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>Distinguish between ‘Single incident traumas’ and ‘Cumulative trauma disorders’ in the context of Ergonomics.</td><td>–</td><td>–</td><td>–</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Examinations April/May – 2024

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Illustrate the pyramidal structure of work.	02	2	1
	1.2	Define the term "Productivity".	02	1	1
	1.3	Distinguish between "Methods Design" and "Methods Analysis".	02	2	2
	1.4	One of the approaches for dealing with the automation projects in methods engineering is USA principle. Expand the acronym 'USA'.	02	1	2
	1.5	What is meant by application speed ratio in the context of work measurement?	02	2	3
	1.6	Expand the acronym 'MOST'.	02	1	3
	1.7	What is the principal objective of JIT?	02	2	2
	1.8	What does the term 'Jidoka' refer to?	02	1	2
	1.9	List the four primary variables that influence the climate in the work place environment.	02	1	4
	1.10	Distinguish between the terms, "industrial accident" and "hazard".	02	2	4

PART-B

2	a	Distinguish between the terms, 'basic productive work content' and 'excess non-productive work content.' Also illustrate with two examples for each category of excess non-productive activities that add to total task time.	08	3	1
	b	A work group of 5 workers in a certain month produced 500 units of output working 8 hrs/day for 22 days in a month. Suppose in the next month, the same work group produced 600 units but were only 20 work days in the month. Using labor productivity measure, determine the productivity index using the prior month as a base.	08	4	1
3	a	Enumerate the steps in a systematic approach that can be used to accomplish a methods engineering study.	08	2	2
	b	Summarize the features and applicability of the following tools used by industrial engineers for basic data collection and analysis as part of problem solving process. (i) Defect concentration diagram (ii) Scatter diagram	08	3	2
		OR			
4	a	Enumerate any eight principles of motion economy related to the use of human body.	–	–	–
	b	Illustrate the automation migration strategy and explain the three phases included in the typical strategy followed by manufacturing organizations.	–	–	–

<table>
<tr><td>5</td><td>a</td><td>Depict a model indicating the factors that must be standardized before a standard time can be set for a task. Also explain the elements of the model.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>The observed element times and performance ratings collected in a direct time study are indicated in the table below. The snapback/flyback timing method was used. The PFD allowance is 14%. All the elements are regular elements in the work cycle. Determine: (i) Normal time and (ii) The standard time for the cycle.
<table>
<tr><td>Work Element</td><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>Observed time (min)</td><td>0.22</td><td>0.41</td><td>0.30</td><td>0.37</td></tr>
<tr><td>Performance Rating (%)</td><td>90</td><td>120</td><td>100</td><td>90</td></tr>
</table></td><td>–</td><td>–</td><td>–</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>6</td><td>a</td><td>Examine the three sequence (activity sequence) models in Basic MOST and indicate the standard sequence of model parameters for each.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements a, b, c and d are performed by the operator and element ‘m’ is a machine element; Element ‘b’ is an internal element performed by the operator simultaneously with element ‘m’; and element ‘d’ is an irregular element performed once every 15 cycles. Observed times and performance ratings are given in the table. The PFD allowance factor is 15%, and the machine allowance is 20%. Determine: (i) Normal time and (ii) Standard time for the work cycle.
<table>
<tr><td>Worker element</td><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>Observed time (min) – manual</td><td>0.22</td><td>0.65</td><td>0.47</td><td>0.75</td></tr>
<tr><td>Performance rating (%)</td><td>100</td><td>80</td><td>100</td><td>100</td></tr>
<tr><td>Machine element</td><td></td><td>m</td><td></td><td></td></tr>
<tr><td>Observed time (min) – machine</td><td>idle</td><td>1.56</td><td>idle</td><td>idle</td></tr>
</table></td><td>–</td><td>–</td><td>–</td></tr>
<tr><td>7</td><td>a</td><td>Illustrate the structure of a lean production system and explain in detail any two ingredients of the system.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>Examine the three prerequisites for successful implementation of Just-in-Time production system.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>8</td><td>a</td><td>Examine any four aspects influencing the stable and reliable production operations in a manufacturing environment.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td></td><td>b</td><td>Motivated workers participation in continuous improvement is critical for achieving the objective of lean production system. In the above context, discuss the relevance of visual management and 5s.</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td>9</td><td>a</td><td>Depict the model of interactions in a typical human machine system. Provide an overview of the human components of interest in the system.</td><td>08</td><td>3</td><td>4</td></tr>
<tr><td></td><td>b</td><td>Work is performed in an environment that includes both physical and social aspects. Classify the broad categories into which physical environment can be segregated from ergonomics perspective. Elaborate on any two categories in detail.</td><td>08</td><td>4</td><td>4</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>10</td><td>a</td><td>Elucidate the three factors commonly identified in the ergonomics literature as the primary reasons for occurrence of industrial accidents.</td><td>08</td><td>2</td><td>4</td></tr>
<tr><td></td><td>b</td><td>Describe any four common cumulative trauma and other occupational disorders in a chosen sector of your interest.</td><td>08</td><td>4</td><td>4</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
III Semester B.E. Examinations April – 2024
MODEL QUESTION PAPER (2022 SCHEME)
Industrial Engineering and Management
WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Distinguish between the terms: Production and Productivity.	02	–	–
	1.2	Classify the types of Worker-Machine Systems.	02	–	–
	1.3	Categorize the features and specifications to be considered in selecting among alternative improvement proposals in Methods Engineering.	02	–	–
	1.4	Expand the acronym USA in the context of methods engineering and automation.	02	–	–
	1.5	What is meant by the term Application Speed Ratio?	02	–	–
	1.6	Mention the prerequisites of Standard Time.	02	–	–
	1.7	What is meant by Lean Production System?	02	–	–
	1.8	Distinguish between Pull System and Push System of Production Control.	02	–	–
	1.9	List the three categories of Human-Machine Systems.	02	–	–
	1.10	What is meant by Anthropometry?	02	–	–
		PART-B			
2	a	Distinguish between basic productive work environment and Excess non productive work environment. Also classify the excess non productive activities into three categories. Support your explanation with examples for each category.	08	–	–
	b	The normal time for a repetitive task that produces two work units per cycle is 3.0 minutes. The plant uses a PFD allowance factor of 15%. Determine (a) the standard time per piece and (b) how many work units are produced in an 8 hour shift at standard performance.	08	–	–
3	a	Review any two categories of specialized analysis techniques used in Methods Engineering that are exclusive to operations analysis and industrial engineering.	08	–	–
	b	Identify and explain the types of symbols used in a flow process chart.	08	–	–
		OR			
4	a	Describe any two basic data collection and analysis techniques used in Methods Engineering.	08	–	–
	b	Enumerate the principles of motion economy related to the design of tooling and equipment.	08	–	–
5	a	Discuss the characteristics typical of industrial situations in which time standards would be beneficial?	08	–	–

	b	The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 minutes. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers, which took 1.60 minutes, rated at 120%. The PFD allowance for this class of work is 15%. Determine (a) normal time and standard time (b) worker's efficiency if the worker produces 123 work units during an 8 hour shift.	08	–	–
		OR			
6	a	Summarize different varieties of Predetermined Motion Time Systems.	08	–	–
	b	Review different versions of MOST work measurement packages.	08	–	–
7	a	Provide a historical note on the Lean Production Philosophy.	08	–	–
	b	With the help of an illustration, explain the working of a kanban system between work stations.	08	–	–
		OR			
8	a	Explain any two aspects of Autonomation with suitable examples.	08	–	–
	b	What is Standardized Work Procedure? Explain the components of the same for a given task.	08	–	–
9	a	What is human machine system? Review the two main components of the same.	08	–	–
	b	Discuss the components of Physical work environment.	08	–	–
		OR			
10	a	What is an industrial accident? Analyze the causes for the same from ergonomics perspective.	08	–	–
	b	Describe any four common occupational diseases along with necessary causes.	08	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations April/May-2023

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	The unit for the standard time determination is _____.	01
	1.2	_____ allowance is provided for the operator to attend to personal needs and recover from fatigue caused by doing his work.	01
	1.3	In _____ timing, the hands of the stop watch are returned to zero at the end of each element and are allowed to start immediately.	01
	1.4	*MOST* is abbreviated as _____.	01
	1.5	To decide on the appropriate incentive plan in an organization, the most commonly used measure is _____.	01
	1.6	A work system in which an automated lathe controlled for most of the work cycle by the part program but requiring the effort of worker to unload the finished part is _____.	01
	1.7	The technique used to symbolize and summarize existing sequence of operations is called _____.	01
	1.8	Automation with human touch is termed as _____.	01
	1.9	Physical hazards in the workplace refer to _____.	01
	1.10	The design and study of work environment to address physical and psychological demands of individuals is called _____.	01
	1.11	A time study was conducted on a job, average time of the element is $0.098 min$ and the rating is 80%. Calculate normal time and standard time considering 12% allowances.	02
	1.12	Define predetermined motion time system.	02
	1.13	List the techniques of work measurement (any two).	02
	1.14	How is lean manufacturing achieved by an Industry?	02
	1.15	What are the broad two components of environment in the context and scope of ergonomics analysis?	02

PART-B

2	a	"The management of an enterprise is responsible for seeing that the enterprise resources are combined in the best possible way to achieve the highest productivity". Comment.	06
	b	With a block diagram, classify the powered machinery in worker machine systems.	10
3	a	What are the objectives of using charts and diagrams to conduct work study? Explain the five types symbols used in the flow process chart with example.	06

	b	List the automation strategies that are employed to improve productivity in manufacturing operations.	10
		OR	
4	a	List the different recording techniques used in method study. Explain any two in detail.	08
	b	How many therbligs are available for recording fundamental motions? List any five symbols related to eye and hand motions with an appropriate example.	08
5	a	What is allowance? Elaborate on *PFD* allowances in time study.	08
	b	Explain the five steps in the direct time study procedure.	08
		OR	
6	a	A job has been sub-divided into five elements. The time for each element and respective rating are given below.	

Element Number	*Observed time*	*Rating factor %*
1	0.7	80
2	0.8	100
3	1.3	120
4	0.5	90
5	1.2	100

		Calculate the normal time and standard time for each element and for the job if the allowance is 15%.	10
	b	Define *MTM*. Explain the uses of *MTM* technique.	06
7	a	With a suitable illustration, bring out the distinguishing features of Pull System and Push system.	06
	b	List the advantages and disadvantages in implementing a lean production system for an automobile industry.	10
		OR	
8	a	What does the Japanese word "Kaizen" mean? How does it improve the productivity of an organization?	10
	b	List the seven forms of waste identified by the Taiichi Ohno.	06
9	a	How can workers best contribute to a positive safety culture in the workplace?	06
	b	What are the principal components of the machine in the human machine system, as indicated in the diagram of the system?	10
		OR	
10	a	What is cognitive ergonomics? Explain the components of human cognitive process.	08
	b	State the difference between occupational safety and occupational health.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations January-2023

WORK SYSTEM DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Enumerate the components of work system defined as a field of professional practice.	02
	1.2	During the base year in a steel mill, 5,96,000 *tons* of steel were produced using 3,43,000 *labor hours*. Determine the labor productivity ratio.	02
	1.3	What is a swim-lane chart?	02
	1.4	State any four principles related to the use of human body in principles of motion economy.	02
	1.5	Mention the 4 work measurement techniques used in practice.	02
	1.6	_____ is the ratio of the time required to determine a time standard relative to the standard time itself.	01
	1.7	In a direct time study done for an irregular element performed every 5 cycles, the observed time was recorded as 1.10min and performance rating was recorded as 90%. Determine the normal time for the element for each cycle.	01
	1.8	_____ is an expert system that uses its knowledge and logic to solve a specific work measurement problem.	01
	1.9	What are the characteristics of work situations for which work sampling is most suited? (Mention any 2).	01
	1.10	Distinguish between learning curve and learning rate.	02
	1.11	What is “pokayoke”? Who developed it?	02
	1.12	State the reason why “overproduction” is considered as waste in Lean Production System.	02

PART-B

2	a	A given task performed by a worker can be considered to consist of the basic productive work content and excess non-productive activities. Discuss these in detail with a block diagram.	08
	b	A work group of 6 workers in a certain month produced 600 units of output working 8hours/day for 24days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same group produced 750units working with the same condition, determine the productivity index using the prior month as a base.	08

3	a	Describe the six steps of the systematic approach in methods engineering.	10
	b	Provide an overview of the recording tools used for methods study.	06
		OR	
4	a	Summarize the objectives (any four) of using charts and diagrams to study the work in methods study.	08
	b	Illustrate the various principles of motion economy related to workplace arrangement.	08
5	a	What is a work element? Mention atleast 6 guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.	10
	b	Describe the 3 activity sequence models in basic MOST.	06
		OR	
6	a	A job has been sub-divided into 5 elements. The time for each element and respective rating are given below.	

Element number	Observation time (min)	Rating factor (%)
1	0.7	80
2	0.8	100
3	1.3	120
4	0.5	90
5	1.2	100

		Calculate the normal time and standard time for the job if the allowance is 15%.	10
	b	State the characteristics of a well-implemented performance rating system.	06
7	a	Categorize the work elements considered in a standard data system. Deliberate on any 1 category in detail with examples.	08
	b	List the various factors affecting the learning rate. Explain any two in detail.	08
8	a	With a suitable illustration, bring out the distinguishing features of pull system and push system.	08
	b	Analyze the relevance of artificial intelligence based expert systems in computerized work measurement systems for determining time standards.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations March-2021

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	List any two functions of knowledge workers.	02
	1.2	Represent a work system by means of a process.	02
	1.3	Mention any two charting and diagramming techniques.	02
	1.4	List any two phases included in the typical automation migration strategy..	02
	1.5	Indicate the factors that must be standardized before a standard time can be set for a task.	02
	1.6	Give any two examples of Nonwork related Interruptions	02
	1.7	Mention general characteristics of work situations that are particularly well-suited for work sampling.	02
	1.8	Distinguish between learning curve and learning rate.	02
	1.9	List two principal functions of an ideal computerized work measurement system.	02
	1.10	Why do you consider overproduction as a waste in lean production system?	02

PART-B

2	a	Write a note on the three categories of excess Nonproductive activities, as they are defined in the course?	06
	b	A work group of 5 workers in a certain month produced 500 units of output working 8hours/day for 22 Days in the month. i) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? ii) Suppose that in the next month, the same work group produced 600 units but there were only20 work days in a month, using the same productivity measures as before, determine the productivity index using prior month as a base.	10
3	a	Describe the steps involved in the systematic approach to problem solving in Methods Engineering.	10
	b	Draw a cause and effect diagram for scoring less marks in a particular course of your curriculum.	06
		OR	

<table>
<tr><td>4</td><td>a</td><td>Review any four data collection and analysis techniques used in problem solving process.</td><td>10</td></tr>
<tr><td></td><td>b</td><td>Briefly explain five classifications of Hand and Arm motions.</td><td>06</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>5</td><td>a</td><td>Identify the factors causing Fatigue in workers in a typical manufacturing industry.</td><td>06</td></tr>
<tr><td></td><td>b</td><td>A job has been sub-divided into five elements. The time for each element and respective rating are given below:
<table>
<tr><th>Element No:</th><th>Observation time (min)</th><th>Rating factor (%)</th></tr>
<tr><td>1</td><td>0.7</td><td>80</td></tr>
<tr><td>2</td><td>0.8</td><td>100</td></tr>
<tr><td>3</td><td>1.3</td><td>120</td></tr>
<tr><td>4</td><td>0.5</td><td>90</td></tr>
<tr><td>5</td><td>1.2</td><td>100</td></tr>
</table>
Calculate the normal time and standard time for the job if the allowance is 15%.</td><td>10</td></tr>
<tr><td></td><td></td><td>OR</td><td></td></tr>
<tr><td>6</td><td>a</td><td>Describe the three activity sequence models in Basic MOST.</td><td>06</td></tr>
<tr><td></td><td>b</td><td>List and explain steps involved in Direct time study.</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>7</td><td>a</td><td>Critically evaluate the issues in implementing work sampling methodology to determine the time standards.</td><td>08</td></tr>
<tr><td></td><td>b</td><td>Describe the recommended steps in developing a standard Data System.</td><td>08</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>8</td><td>a</td><td>Identify three ingredients that are required in a computerized work measurement system that is fully automated using expert system technology.</td><td>06</td></tr>
<tr><td></td><td>b</td><td>With the help of suitable examples explain different types of wastes observed in a typical manufacturing industry.</td><td>10</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

Autonomous Institution affiliated to VTU

III Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Define productivity. Write the formula to find labour productivity.	02
	1.2	List the 3 types of works syetms.	02
	1.3	Name the six general categories of causes that affect performance of most production or service processes [$5MS$ & $1P$].	02
	1.4	Draw the fishbone diagram/cause and effect diagram for poor solder joint.	02
	1.5	List the three work measurement techniques classified under engineered standards.	02
	1.6	Distinguish between continuous and snap back timing method.	02
	1.7	List any two work situations where standard data system is most suited.	02
	1.8	What are the characteristics of work situations for which work sampling is most suited?	02
	1.9	Name the two pillars of the Toyota production system.	02
	1.10	What is cycle time?	02

PART-B

2	a	With suitable examples, explain the three categories of excess nonproductive activities that add to total task time.	08
	b	In the operation of a worker machine system, the work cycle consists of several manual work elements and one machine element performed under semiautomatic control. One workpiece is produced each cycle. The manual work elements total a normal time of $1min$ and the semiautomatic machine cycle is a constant $2.0\,min$. The PFD allowance factor A_{pfd} is 15%. Determine the standard time using i) $A_m = 0$ and ii) $A_m = 30\%$. Where A_m is machine allowance.	08

3 a List and explain the six steps of the systematic approach in methods engineering. 08

b Name the three types of process charts and identify the application area for each. 08

OR

4 a State and explain any four principles related to the use of the human body in motion economy and work design. 08

b Explain the three phases in the automation migration strategy. 08

5 a A direct time study was taken on a manual work cycle using the snap back timing method. The regular work cycle consisted of 3 elements identified as a, b, c in the table. Element d is an irregular element performed every $5\ cycles$. Observed time and performance ratings of the elements are given. Determine:

i) The normal times

ii) The standard time for the work cycle using an allowance factor of 15%.

$Work\ element$	a	b	c	d
$Observed\ time\ (min)$	$0.56min$	0.25	0.5	1.10
$performance\ rating\ (\%)$	100	80	110	100

08

b What are the steps in applying a predetermined motion time system? 08

OR

6 a The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements a, b, c & d are performed by the operator and element m is a machine semiautomatic cycle. Element b is an internal element performed simultaneously with element m and element d is an irregular element performed once every $15\ cycles$. Observed times and performance ratings given in the table. The PFD allowance factor is 15% and the machine allowance is 20%. Determine:

i) Normal time

ii) Standard time for the work cycle.

$Work\ element$	a	b	c	d
$Observed\ time, manual\ (min)$	0.22	0.65	0.47	0.75
$performance\ rating\ (\%)$	100	80	100	100
$Machine\ element$		m		
$Observed\ time, machine\ (min)$	$idle$	1.56	$idle$	$idle$

10

b With a neat block diagram, illustrate how the work measurement techniques correspond to different levels in the general task hierarchy. 06

7	a	Explain different work element classification encountered in standard data systems for work measurement.	08
	b	Identify any 4 advantages and disadvantages of work sampling.	08
8	a	With the help of a block diagram, illustrate the role of personal computers and personal digital assistants in the conduction of direct time study.	08
	b	List and explain the seven forms of waste in production as identified by Taichi Ohno.	08

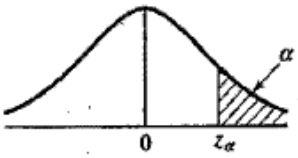
Table A.1 Random Digits

94737	08225	35614	24826	88319	05595	58701	57365	74759
87259	85982	13296	89326	74863	99986	68558	06391	50248
63856	14016	18527	11634	96908	52146	53496	51730	03500
66612	54714	46783	61934	30258	61674	07471	67566	31635
30712	58582	05704	23172	86689	94834	99057	55832	21012
69607	24145	43886	86477	05317	30445	33456	34029	09603
37792	27282	94107	41967	21425	04743	42822	28111	09757
01488	56680	73847	64930	11108	44834	45390	86043	23973
66248	97697	38244	50918	55441	51217	54786	04940	50807
51453	03462	61157	65366	61130	26204	15016	85665	97714
92168	82530	19271	86999	96499	12765	20926	25282	39119
36463	07331	54590	00546	03337	41583	46439	40173	46455
47097	78780	04210	87084	44484	75377	57753	41415	09890
80400	45972	44111	99708	45935	03694	81421	60170	58457
94554	13863	88239	91624	00022	40471	78462	96265	55360
31567	53597	08490	73544	72573	30961	12282	97033	13676
07821	24759	47266	21747	72496	77755	50391	59554	31177
09056	10709	69314	11449	40531	02917	95878	74587	60906
19922	37025	80731	26179	16039	01518	82697	73227	13160
29923	02570	80164	36108	73689	26342	35712	49137	13482
29602	29464	99219	20308	82109	03898	82072	85199	13103
94135	94661	87724	88187	62191	70607	63099	40494	49069
87926	34092	34334	55064	43152	01610	03126	47312	59578
85039	19212	59160	83537	54414	19856	90527	21756	64783
66070	38480	74636	45095	86576	79337	39578	40851	53503
78166	82521	79261	12570	10930	47564	77869	16480	43972
94672	07912	26153	10531	12715	63142	88937	94466	31388
56406	70023	27734	22254	27685	67518	63966	33203	70803
67726	57805	94264	77009	08682	18784	47554	59869	66320
07516	45979	76735	46509	17696	67177	92600	55572	17245
43070	22671	00152	81326	89428	16368	57659	79424	57604
36917	60370	80812	87225	02850	47118	23790	55043	75117
03919	82922	02312	31106	44335	05573	17470	25900	91080
46724	22558	64303	78804	05762	70650	56117	06707	90035
16108	61281	86823	20286	14025	24909	38391	12183	89393
74541	75808	89669	87680	72758	60851	55292	95663	88326
82919	31285	01850	72550	42986	57518	01159	01786	98145
31388	26809	77258	99360	92362	21979	41319	75739	98082
17190	75522	15687	07161	99745	48767	03121	20046	28013
00466	88068	68631	98745	97810	35886	14497	90230	69264

Table A.2 Random Normal Numbers

0.23	−0.17	0.43	2.18	2.13	0.49	2.72	−0.18	0.42
0.24	−1.17	0.02	0.67	−0.59	−0.13	−0.15	−0.46	1.64
−1.16	−0.17	0.36	−1.26	0.91	0.71	−1.00	−1.09	−0.02
−0.02	−0.19	−0.04	1.92	0.71	−0.90	−0.21	−1.40	−0.38
0.39	0.55	0.13	2.55	−0.33	−0.05	−0.34	−1.95	−0.44
0.64	−0.36	0.98	−0.21	−0.52	−0.02	−0.15	−0.43	0.62
−1.90	0.48	−0.54	0.60	−0.35	−1.29	−0.57	0.23	1.41
−1.04	−0.70	−1.69	1.76	0.47	−0.52	−0.73	0.94	−1.63
−.78	0.11	−0.91	−1.13	0.07	0.45	−0.94	1.42	0.75
0.68	1.77	−0.82	−1.68	−2.60	1.59	−0.72	−0.80	0.61
−0.02	0.92	1.76	−0.66	0.18	−1.32	1.26	0.61	0.83
−0.47	1.04	0.83	−2.05	1.00	−0.70	1.12	0.82	0.08
−0.40	1.40	1.20	0.00	0.21	−2.13	−0.22	1.79	0.87
−0.75	0.09	−1.50	0.14	−2.99	−0.41	−0.99	−0.70	0.51
−0.66	−1.97	0.15	−1.16	−0.60	0.50	1.36	1.94	0.11
−0.44	−0.09	−0.59	1.37	0.18	1.44	−0.80	2.11	−1.37
1.41	−2.71	−0.67	1.83	0.97	0.06	−0.28	0.04	−0.21
1.21	−0.52	−0.20	−0.88	−0.78	0.84	−1.08	−0.25	0.17
0.07	0.66	−0.51	−0.04	−0.84	0.04	1.60	−0.92	1.14
−0.08	0.79	−0.09	−1.12	−1.13	0.77	0.40	0.69	−0.12
0.53	−0.36	−2.64	0.22	−0.78	1.92	−0.26	1.04	−1.61
−1.56	1.82	−1.03	1.14	−0.12	−0.78	−0.12	1.42	−0.52
0.03	−1.29	−0.33	2.60	−0.64	1.19	−0.13	0.91	0.78
1.49	1.55	−0.79	1.37	0.97	0.17	0.58	1.43	−1.29
−1.19	1.35	0.16	1.06	−0.17	0.32	−0.28	0.68	0.54
−1.19	−1.03	−0.12	1.07	0.87	−1.40	−0.24	−0.81	0.31
0.11	−1.95	−0.44	−0.39	−0.15	−1.20	−1.98	0.32	2.91
−1.86	0.06	0.19	−1.29	0.33	1.51	−0.36	−0.80	−0.99
0.16	0.28	0.60	−0.78	0.67	0.13	−0.47	−0.18	−0.89
1.21	−1.19	−0.60	−1.22	0.07	−1.13	1.45	0.94	0.54
−0.82	0.54	−0.98	−0.13	1.52	0.77	0.95	−0.84	2.40
0.75	−0.80	−0.28	1.77	−0.16	−0.33	2.43	−1.11	1.63
0.42	0.31	1.56	0.56	0.64	−0.78	0.04	1.34	−0.01
−1.50	−1.78	−0.59	0.16	0.36	1.89	−1.19	0.53	−0.97
−0.89	0.08	0.95	−0.73	1.25	−1.04	−0.47	−0.68	−0.87
0.19	0.85	1.68	−0.57	0.37	−0.48	−0.17	2.36	−0.53
0.49	0.32	−2.08	−1.02	2.59	−0.53	0.15	0.11	0.05
−1.44	0.07	−0.22	−0.93	−1.40	0.54	−1.28	−0.15	0.67
−0.21	−0.48	1.21	0.67	−1.10	−0.75	−0.37	0.68	−0.02
−0.65	−0.12	0.94	−0.44	−1.21	−0.06	−1.28	−1.51	1.39
0.24	−0.83	1.55	0.33	−0.59	−1.24	0.70	0.01	0.15
−0.73	1.24	0.40	−0.61	0.68	0.69	0.07	−0.23	−0.66
−1.93	0.75	−0.32	0.95	1.35	1.51	−0.88	0.10	−1.19
0.08	0.16	0.38	−0.96	1.99	−0.20	0.98	0.16	0.26
−0.47	−1.25	0.32	0.51	−1.04	0.97	2.60	−0.08	1.19

Table A.3 Cumulative Normal Distribution



$$\phi(z_\alpha) = \int_{-\infty}^{z_\alpha} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1 - \alpha$$

z_α	0.00	0.01	0.02	0.03	0.04	z_α
0.0	0.500 00	0.503 99	0.507 98	0.511 97	0.515 95	0.0
0.1	0.539 83	0.543 79	0.547 76	0.551 72	0.555 67	0.1
0.2	0.579 26	0.583 17	0.587 06	0.590 95	0.594 83	0.2
0.3	0.617 91	0.621 72	0.625 51	0.629 30	0.633 07	0.3
0.4	0.655 42	0.659 10	0.662 76	0.666 40	0.670 03	0.4
0.5	0.691 46	0.694 97	0.698 47	0.701 94	0.705 40	0.5
0.6	0.725 75	0.729 07	0.732 37	0.735 65	0.738 91	0.6
0.7	0.758 03	0.761 15	0.764 24	0.767 30	0.770 35	0.7
0.8	0.788 14	0.791 03	0.793 89	0.796 73	0.799 54	0.8
0.9	0.815 94	0.818 59	0.821 21	0.823 81	0.826 39	0.9
1.0	0.841 34	0.843 75	0.846 13	0.848 49	0.850 83	1.0
1.1	0.864 33	0.866 50	0.868 64	0.870 76	0.872 85	1.1
1.2	0.884 93	0.886 86	0.888 77	0.890 65	0.892 51	1.2
1.3	0.903 20	0.904 90	0.906 58	0.908 24	0.909 88	1.3
1.4	0.919 24	0.920 73	0.922 19	0.923 64	0.925 06	1.4
1.5	0.933 19	0.934 48	0.935 74	0.936 99	0.938 22	1.5
1.6	0.945 20	0.946 30	0.947 38	0.948 45	0.949 50	1.6
1.7	0.955 43	0.956 37	0.957 28	0.958 18	0.959 07	1.7
1.8	0.964 07	0.964 85	0.965 62	0.966 37	0.967 11	1.8
1.9	0.971 28	0.971 93	0.972 57	0.973 20	0.973 81	1.9
2.0	0.977 25	0.977 78	0.978 31	0.978 82	0.979 32	2.0
2.1	0.982 14	0.982 57	0.983 00	0.983 41	0.983 82	2.1
2.2	0.986 10	0.986 45	0.986 79	0.987 13	0.987 45	2.2
2.3	0.989 28	0.989 56	0.989 83	0.990 10	0.990 36	2.3
2.4	0.991 80	0.992 02	0.992 24	0.992 45	0.992 66	2.4
2.5	0.993 79	0.993 96	0.994 13	0.994 30	0.994 46	2.5
2.6	0.995 34	0.995 47	0.995 60	0.995 73	0.995 85	2.6
2.7	0.996 53	0.996 64	0.996 74	0.996 83	0.996 93	2.7
2.8	0.997 44	0.997 52	0.997 60	0.997 67	0.997 74	2.8
2.9	0.998 13	0.998 19	0.998 25	0.998 31	0.998 36	2.9
3.0	0.998 65	0.998 69	0.998 74	0.998 78	0.998 82	3.0
3.1	0.999 03	0.999 06	0.999 10	0.999 13	0.999 16	3.1
3.2	0.999 31	0.999 34	0.999 36	0.999 38	0.999 40	3.2
3.3	0.999 52	0.999 53	0.999 55	0.999 57	0.999 58	3.3
3.4	0.999 66	0.999 68	0.999 69	0.999 70	0.999 71	3.4
3.5	0.999 77	0.999 78	0.999 78	0.999 79	0.999 80	3.5
3.6	0.999 84	0.999 85	0.999 85	0.999 86	0.999 86	3.6
3.7	0.999 89	0.999 90	0.999 90	0.999 90	0.999 91	3.7
3.8	0999 93	0.999 93	0.999 93	0.999 94	0.999 94	3.8
3.9	0.999 95	0.999 95	0.999 96	0.999 96	0.999 96	3.9

(continued overleaf)

Table A.3 *(continued)*

z_α	0.05	0.06	0.07	0.08	0.09	z_α
0.0	0.519 94	0.523 92	0.527 90	0.531 88	0.535 86	0.0
0.1	0.559 62	0.563 56	0.567 49	0.571 42	0.575 34	0.1
0.2	0.598 71	0.602 57	0.606 42	0.610 26	0.614 09	0.2
0.3	0.636 83	0.640 58	0.644 31	0.648 03	0.651 73	0.3
0.4	0.673 64	0.677 24	0.680 82	0.684 38	0.687 93	0.4
0.5	0.708 84	0.712 26	0.715 66	0.719 04	0.722 40	0.5
0.6	0.742 15	0.745 37	0.748 57	0.751 75	0.754 90	0.6
0.7	0.773 37	0.776 37	0.779 35	0.782 30	0.785 23	0.7
0.8	0.802 34	0.805 10	0.807 85	0.810 57	0.813 27	0.8
0.9	0.824 94	0.831 47	0.833 97	0.836 46	0.838 91	0.9
1.0	0.853 14	0.855 43	0.857 69	0.859 93	0.862 14	1.0
1.1	0.874 93	0.876 97	0.879 00	0.881 00	0.882 97	1.1
1.2	0.894 35	0.896 16	0.897 96	0.899 73	0.901 47	1.2
1.3	0.911 49	0.913 08	0.914 65	0.916 21	0.917 73	1.3
1.4	0.926 47	0.927 85	0.929 22	0.930 56	0.931 89	1.4
1.5	0.939 43	0.940 62	0.941 79	0.942 95	0.944 08	1.5
1.6	0.950 53	0.951 54	0.952 54	0.953 52	0.954 48	1.6
1.7	0.959 94	0.960 80	0.961 64	0.962 46	0.963 27	1.7
1.8	0.967 84	0.968 56	0.969 26	0.969 95	0.970 62	1.8
1.9	0.974 41	0.975 00	0.975 58	0.976 15	0.976 70	1.9
2.0	0.979 82	0.980 30	0.980 77	0.981 24	0.981 69	2.0
2.1	0.984 22	0.984 61	0.985 00	0.985 37	0.985 74	2.1
2.2	0.987 78	0.988 09	0.988 40	0.988 70	0.988 99	2.2
2.3	0.990 61	0.990 86	0.991 11	0.991 34	0.991 58	2.3
2.4	0.992 86	0.993 05	0.993 24	0.993 43	0.993 61	2.4
2.5	0.994 61	0.994 77	0.994 92	0.995 06	0.995 20	2.5
2.6	0.995 98	0.996 09	0.996 21	0.996 32	0.996 43	2.6
2.7	0.997 02	0.997 11	0.997 20	0.997 28	0.997 36	2.7
2.8	0.997 81	0.997 88	0.997 95	0.998 01	0.998 07	2.8
2.9	0.998 41	0.998 46	0.998 51	0.998 56	0.998 61	2.9
3.0	0.998 86	0.998 89	0.998 93	0.998 97	0.999 00	3.0
3.1	0.999 18	0.999 21	0.999 24	0.999 26	0.999 29	3.1
3.2	0.999 42	0.999 44	0.999 46	0.999 48	0.999 50	3.2
3.3	0.999 60	0.999 61	0.999 62	0.999 64	0.999 65	3.3
3.4	0.999 72	0.999 73	0.999 74	0.999 75	0.999 76	3.4
3.5	0.999 81	0.999 81	0.999 82	0.999 83	0.999 83	3.5
3.6	0.999 87	0.999 87	0.999 88	0.999 88	0.999 89	3.6
3.7	0.999 91	0.999 92	0.999 92	0.999 92	0.999 92	3.7
3.8	0.999 94	0.999 94	0.999 95	0.999 95	0.999 95	3.8
3.9	0.999 96	0.999 96	0.999 96	0.999 97	0.999 97	3.9

Source: W. W. Hines and D. C. Montgomery, *Probability and Statistics in Engineering and Management Science*, 2d ed., © 1980, pp. 592–3. Reprinted by permission of John Wiley & Sons, Inc., New York.

Table A.4 Cumulative Poisson Distribution

	α = Mean											
x	.01	.05	.1	.2	.3	.4	.5	.6	.7	.8	.9	x
0	.990	.951	.905	.819	.741	.670	.607	.549	.497	.449	.407	0
1	1.000	.999	.995	.982	.963	.938	.910	.878	.844	.809	.772	1
2		1.000	1.000	.999	.996	.992	.986	.977	.966	.953	.937	2
3				1.000	1.000	.999	.998	.997	.994	.991	.987	3
4						1.000	1.000	1.000	.999	.999	.998	4
5									1.000	1.000	1.000	5

	α = Mean											
x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	x
0	.368	.333	.301	.273	.247	.223	.202	.183	.165	.150	.135	0
1	.736	.699	.663	.627	.592	.558	.525	.493	.463	.434	.406	1
2	.920	.900	.879	.857	.833	.809	.783	.757	.731	.704	.677	2
3	.981	.974	.966	.957	.946	.934	.921	.907	.891	.875	.857	3
4	.996	.995	.992	.989	.986	.981	.976	.970	.964	.956	.947	4
5	.999	.999	.998	.998	.997	.996	.994	.992	.990	.987	.983	5
6	1.000	1.000	1.000	1.000	.999	.999	.999	.998	.997	.997	.995	6
7					1.000	1.000	1.000	1.000	.999	.999	.999	7
8									1.000	1.000	1.000	8

	α = Mean											
x	2.2	2.4	2.6	2.8	3.0	3.5	4.0	4.5	5.0	5.5	6.0	x
0	.111	.091	.074	.061	.050	.030	.018	.011	.007	.004	.002	0
1	.355	.308	.267	.231	.199	.136	.092	.061	.040	.027	.017	1
2	.623	.570	.518	.469	.423	.321	.238	.174	.125	.088	.062	2
3	.819	.779	.736	.692	.647	.537	.433	.342	.265	.202	.151	3
4	.928	.904	.877	.848	.815	.725	.629	.532	.440	.358	.285	4
5	.975	.964	.951	.935	.916	.858	.785	.703	.616	.529	.446	5
6	.993	.988	.983	.976	.966	.935	.889	.831	.762	.686	.606	6
7	.998	.997	.995	.992	.988	.973	.949	.913	.867	.809	.744	7
8	1.000	.999	.999	.998	.996	.990	.979	.960	.932	.894	.847	8
9		1.000	1.000	.999	.999	.997	.992	.983	.968	.946	.916	9
10				1.000	1.000	.999	.997	.993	.986	.975	.957	10
11						1.000	.999	.998	.995	.989	.980	11
12							1.000	.999	.998	.996	.991	12
13								1.000	.999	.998	.996	13
14									1.000	.999	.999	14
15										1.000	.999	15
16											1.000	16

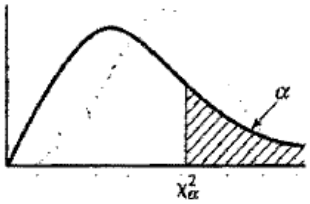
(continued overleaf)

Table A.4 *(continued)*

	α = Mean										
x	6.5	7.0	7.5	8.0	9.0	10.0	12.0	14.0	16.0	18.0	20.0	x
0	.002	.001	.001									0
1	.011	.007	.005	.003	.001							1
2	.043	.030	.020	.014	.006	.003	.001					2
3	.112	.082	.059	.042	.021	.010	.002					3
4	.224	.173	.132	.100	.055	.029	.008	.002				4
5	.369	.301	.241	.191	.116	.067	.020	.006	.001			5
6	.527	.450	.378	.313	.207	.130	.046	.014	.004	.001		6
7	.673	.599	.525	.453	.324	.220	.090	.032	.010	.003	.001	7
8	.792	.729	.662	.593	.456	.333	.155	.062	.022	.007	.002	8
9	.877	.830	.776	.717	.587	.458	.242	.109	.043	.015	.005	9
10	.933	.901	.862	.816	.706	.583	.347	.176	.077	.030	.011	10
11	.966	.947	.921	.888	.803	.697	.462	.260	.127	.055	.021	11
12	.984	.973	.957	.936	.876	.792	.576	.358	.193	.092	.039	12
13	.993	.987	.978	.966	.926	.864	.682	.464	.275	.143	.066	13
14	.997	.994	.990	.983	.959	.917	.772	.570	.368	.208	.105	14
15	.999	.998	.995	.992	.978	.951	.844	.669	.467	.287	.157	15
16	1.000	.999	.998	.996	.989	.973	.899	.756	.566	.375	.221	16
17		1.000	.999	.998	.995	.986	.937	.827	.659	.469	.297	17
18			1.000	.999	.998	.993	.963	.883	.742	.562	.381	18
19				1.000	.999	.997	.979	.923	.812	.651	.470	19
20					1.000	.998	.988	.952	.868	.731	.559	20
21						.999	.994	.971	.911	.799	.644	21
22						1.000	.997	.983	.942	.855	.721	22
23							.999	.991	.963	.899	.787	23
24							.999	.995	.978	.932	.843	24
25							1.000	.997	.987	.955	.888	25
26								.999	.993	.972	.922	26
27								.999	.996	.983	.948	27
28								1.000	.998	.990	.966	28
29									.999	.994	.978	29
30									.999	.997	.987	30
31									1.000	.998	.992	31
32										.999	.995	32
33										1.000	.997	33
34											.999	34
35											.999	35
36											1.000	36

Source: J. Banks and R. G. Heikes, *Handbook of Tables and Graphs for the Industrial Engineer and Manager*, © 1984, pp. 34–35. Reprinted by permission of John Wiley and Sons, Inc., New York.

Table A.6 Percentage Points of The Chi-Square Distribution with ν Degrees of Freedom



ν	$\chi^2_{0.005}$	$\chi^2_{0.01}$	$\chi^2_{0.025}$	$\chi^2_{0.05}$	$\chi^2_{0.10}$
1	7.88	6.63	5.02	3.84	2.71
2	10.60	9.21	7.38	5.99	4.61
3	12.84	11.34	9.35	7.81	6.25
4	14.96	13.28	11.14	9.49	7.78
5	16.7	15.1	12.8	11.1	9.2
6	18.5	16.8	14.4	12.6	10.6
7	20.3	18.5	16.0	14.1	12.0
8	22.0	20.1	17.5	15.5	13.4
9	23.6	21.7	19.0	16.9	14.7
10	25.2	23.2	20.5	18.3	16.0
11	26.8	24.7	21.9	19.7	17.3
12	28.3	26.2	23.3	21.0	18.5
13	29.8	27.7	24.7	22.4	19.8
14	31.3	29.1	26.1	23.7	21.1
15	32.8	30.6	27.5	25.0	22.3
16	34.3	32.0	28.8	26.3	23.5
17	35.7	33.4	30.2	27.6	24.8
18	37.2	34.8	31.5	28.9	26.0
19	38.6	36.2	32.9	30.1	27.2
20	40.0	37.6	34.2	31.4	28.4
21	41.4	38.9	35.5	32.7	29.6
22	42.8	40.3	36.8	33.9	30.8
23	44.2	41.6	38.1	35.2	32.0
24	45.6	43.0	39.4	36.4	33.2
25	49.6	44.3	40.6	37.7	34.4
26	48.3	45.6	41.9	38.9	35.6
27	49.6	47.0	43.2	40.1	36.7
28	51.0	48.3	44.5	41.3	37.9
29	52.3	49.6	45.7	42.6	39.1
30	53.7	50.9	47.0	43.8	40.3
40	66.8	63.7	59.3	55.8	51.8
50	79.5	76.2	71.4	67.5	63.2
60	92.0	88.4	83.3	79.1	74.4
70	104.2	100.4	95.0	90.5	85.5
80	116.3	112.3	106.6	101.9	96.6
90	128.3	124.1	118.1	113.1	107.6
100	140.2	135.8	129.6	124.3	118.5

Source: Robert E. Shannon, *Systems Simulation: The Art and Science*, © 1975, p. 372. Reprinted by permission of Prentice Hall, Upper Saddle River, NJ.

Table A.8 Kolmogorov–Smirnov Critical Values

Degrees of Freedom (N)	$D_{0.10}$	$D_{0.05}$	$D_{0.01}$
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
5	0.510	0.565	0.669
6	0.470	0.521	0.618
7	0.438	0.486	0.577
8	0.411	0.457	0.543
9	0.388	0.432	0.514
10	0.368	0.410	0.490
11	0.352	0.391	0.468
12	0.338	0.375	0.450
13	0.325	0.361	0.433
14	0.314	0.349	0.418
15	0.304	0.338	0.404
16	0.295	0.328	0.392
17	0.286	0.318	0.381
18	0.278	0.309	0.371
19	0.272	0.301	0.363
20	0.264	0.294	0.356
25	0.24	0.27	0.32
30	0.22	0.24	0.29
35	0.21	0.23	0.27
Over 35	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

Source: F. J. Massey, "The Kolmogorov–Smirnov Test for Goodness of Fit," *The Journal of the American Statistical Association*, Vol. 46. © 1951, p. 70. Adapted with permission of the American Statistical Association.

Table A.9 Maximum Likelihood Estimates of the Gamma Distribution

$1/M$	β	$1/M$	β	$1/M$	β
0.020	0.0187	2.700	1.494	10.300	5.311
0.030	0.0275	2.800	1.545	10.600	5.461
0.040	0.0360	2.900	1.596	10.900	5.611
0.050	0.0442	3.000	1.646	11.200	5.761
0.060	0.0523	3.200	1.748	11.500	5.911
0.070	0.0602	3.400	1.849	11.800	6.061
0.080	0.0679	3.600	1.950	12.100	6.211
0.090	0.0756	3.800	2.051	12.400	6.362
0.100	0.0831	4.000	2.151	12.700	6.512
0.200	0.1532	4.200	2.252	13.000	6.662
0.300	0.2178	4.400	2.353	13.300	6.812
0.400	0.2790	4.600	2.453	13.600	6.962
0.500	0.3381	4.800	2.554	13.900	7.112
0.600	0.3955	5.000	2.654	14.200	7.262
0.700	0.4517	5.200	2.755	14.500	7.412
0.800	0.5070	5.400	2.855	14.800	7.562
0.900	0.5615	5.600	2.956	15.100	7.712
1.000	0.6155	5.800	3.056	15.400	7.862
1.100	0.6690	6.000	3.156	15.700	8.013
1.200	0.7220	6.200	3.257	16.000	8.163
1.300	0.7748	6.400	3.357	16.300	8.313
1.400	0.8272	6.600	3.457	16.600	8.463
1.500	0.8794	6.800	3.558	16.900	8.613
1.600	0.9314	7.000	3.658	17.200	8.763
1.700	0.9832	7.300	3.808	17.500	8.913
1.800	1.034	7.600	3.958	17.800	9.063
1.900	1.086	7.900	4.109	18.100	9.213
2.000	1.137	8.200	4.259	18.400	9.363
2.100	1.188	8.500	4.409	18.700	9.513
2.200	1.240	8.800	4.560	19.000	9.663
2.300	1.291	9.100	4.710	19.300	9.813
2.400	1.342	9.400	4.860	19.600	9.963
2.500	1.393	9.700	5.010	20.000	10.16
2.600	1.444	10.000	5.160		

Source: S. C. Choi and R. Wette, "Maximum Likelihood Estimates of the Gamma Distribution and Their Bias," *Technometrics*, Vol. 11, No. 4, Nov. © 1969, pp. 688–9. Adapted with permission of the American Statistical Association.

Table A.7 Percentage Points of The F Distribution with $\alpha = 0.05$

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
	Degrees of Freedom for the Numerator (ν_1)																		
Degrees of Freedom for Denominator (ν_2)																			
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.55	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Source: W. W. Hines and D. C. Montgomery, *Probability and Statistics in Engineering and Management Science*, 2d ed., © 1980, p. 599. Reprinted by permission of John Wiley & Sons, Inc., New York.

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

WORK SYSTEMS DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Illustrate the pyramidal structure of work.	02
	1.2	Distinguish between the terms repetitive and non-repetitive tasks in the context of manual work.	02
	1.3	There are two types of activities recorded in an operation chart. List the same.	02
	1.4	'*USA*' stands for three steps in the analysis and design procedure in automation projects. Expand the three letters in the acronym.	02
	1.5	Standards set by certain work measurement techniques are termed as Engineered standards. List any two of those work measurement techniques.	02
	1.6	List the four prerequisites for valid time standards.	02
	1.7	What is meant by the term 'Application Speed Ratio' in work measurement?	02
	1.8	Categorize the following into suitable type of work element in time studies: a) Push two buttons to initiate press stroke. b) Fetching a kit of components for assembly.	02
	1.9	What is meant by the term 'remission' in the context of learning curves?	02
	1.10	Standing on the foundations of Taichi Ohno's lean production system are two pillars. What are they?	02

PART-B

2	a	In terms of human participation, what are the three basic categories of work systems? Explain the characteristics of each of the category along with appropriate illustrations for each.	06
	b	A work group of 10 workers in a certain month produced 7,200 units of output working 8hrs/day for 22 days in a month. Determine labor productivity ratio using: i) Units of output per worker hour ii) Units of output per work month iii) Suppose that in the next month, the same workgroup produced 6800 units but there were only 20 work days in the month. For each productivity measure as in (i) and (ii), determine the productivity index for the next month using the prior month as a base.	10

3 a Distinguish between the techniques "Methods analysis" and "Methods Design" in terms of their focus of study. 06

b Describe the steps involved in the systematic approach to problem solving in Methods Engineering. 10

OR

4 a Evaluate the relevance of 'Process maps' as a diagrammatic technique in the operations analysis study. 06

b List the principles of motion economy that are related to the use of the human body. Elaborate on any four of the listed. 10

5 a Represent the classification of methods to determine the time standards with a block diagram. Write a note on Engineered standards. 06

b The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements a, b, c and d are performed by the operator; and element 'm' is performed by machine semi-automatic cycle. Element 'b' is an internal element performed simultaneously with element 'm' and element 'd' is an irregular element performed once every 15 cycles. Observed timings and performance ratings are given in the table below. The PFD allowance factor is 15% and the machine allowance is 20%. Determine:

i) Normal time and

ii) Standard time for the work cycle.

$Worker\ Element$	a	b	c	d
$Observed\ time, Manual(min's)$	0.22	0.65	0.47	0.75
$Performance\ rating$	100%	80%	100%	100%
$Machine\ element$		m		
$Observed\ time, Machine(min's)$	$Idle$	1.56	$Idle$	$Idle$

10

OR

6 a Review the steps; the analyst must accomplish to apply a predetermined motion time system to a task. 06

b The snap back timing method was used to obtain average times for work elements in one work cycle. The times are given in the table below. Element 'd' is a machine controlled element and the time is constant. Elements a, b, c, e and f are operator controlled and were performance rated at 80%, however elements 'e' and 'f' are performed during the machine controlled element 'd'. The machine allowance is zero and the PFD allowance is 14%. Determine:

i) Normal time for the cycle

ii) The standard time for the cycle.

$Element$	a	b	c	d	e	F
$Observed\ time(min's)$	0.24	0.30	0.17	0.76	0.26	0.14

10

7 a List the factors affecting the Learning curve. How does product complexity affect the learning curve? 06

	b	A total of 500 observations were made at random times during a one-week period (a total of 40 hrs) in the automatic lathe section. The subjects consisted of 10 identical machines. In each observation, the activity of the machine was identified and classified into one of the three categories: i) Being setup ii) Running production iii) Idle. The number of observations in each category is summarized in the table below. Estimate how many hours per week an average machine spent in each of the three categories. *Category* | *No. of observations* *Being setup* | 75 *Running production* | 300 *Machine idle* | 125 Also determine the 95% confidence internal for the proportion of time spent setting up the machines. Suppose that a total of 1572 work units were completed by the 10 machines and that a total of 23 setups were accomplished during the 5 day period; determine i) Average task time per work unit during production. ii) Average set up time.	10
8	a	What is the principle objective of *JIT* production system? Three pre requisites must be in place for *JIT* to operate successfully. What are they? Elaborate on any one of them.	06
	b	Illustrate a typical structure of a lean production system. List the ingredients of the system and explain the three key aspects of Autonomation.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Classify the following into those which can be considered as inputs and those which can be considered as outputs: i) Number of casting machined. ii) Number of people employed in a canteen.	02
	1.2	List any two factors contributing to the excess work content in typical work system.	02
	1.3	Justify the principle behind method study preceding work measurement in establishing time standards.	02
	1.4	What factors should an industrial engineer consider while selecting a job for methods improvement?	02
	1.5	Indicate the appropriate method study symbol for the following activities: i) A chemical reaction taking place. ii) An accounts assistant verifying a bill.	02
	1.6	Identify the suitable work measurement technique to measure the following activities: i) Overhauling and winding an electric motor. ii) Assemble a rope link.	02
	1.7	Which timing method do you suggest in the following context and Why? To time very short cycle jobs of less than 0.05 minutes.	02
	1.8	How does a time study practitioner level out the effect of variations in the operative's pace of performance?	02
	1.9	Westing house system of rating is based on four criteria. List them.	02
	1.10	What are the two categories of Fatigue allowance intended to provide a workman to recover from fatigue?	02

PART-B

2	a	Illustrate the role of work study in improving the productivity of an organization.	08
	b	"Increased production does not necessarily mean increased productivity". Elaborate the above statement with an appropriate illustration.	08
		OR	

3	a	Briefly explain the contributions made by F.B.Gilbreth and F.W. Taylor to the development of work study as a tool for enhancing productivity in organizations.	08
	b	Provide a review on the methods to increase productivity in organizations.	08
4	a	Describe the steps involved in carrying out a method study.	08
	b	Following is the list of activities for a furniture manufacturing company at Bangalore. Wood cut in various sizes arrive from saw mill. These are moved to stores & stored. Wooden planks of required number are taken to seasoning plant, where seasoning of timber take place (Time approximately 1 month). Depending on the customer order the seasoned wood is transported to the workshop. Components are manufactured using various machines (circular saw, trimmer, thickness etc) components are taken to assembly shop, where assembly takes place. Later inspection of final assembly as done at the inspection department. Inspected assembly is moved to sanding and prefinal finish in the second floor. These are taken to trial booth where primary coating is done. Spray polish is carried out at the spray booth. At the finish department hand polishing and touching is done and inspection is carried out. From there the assembly is moved to upholstery department for upholstery work. And finally furniture is transported to finish goods storage in the first floor there it is inspected and stored. Construct an outline process chart with the use of appropriate symbols for representation.	08
		OR	
5	a	Distinguish between flow diagram and string diagram from the perspective of their application.	08
	b	The operator is engaged on a facing and centering operation on machine as under: Pick up jobs and position in between hydraulic jaws of the vice: $0.15\,min$. Depress "on switch" to secure part: $0.05\,min$. Depress "cycle on" switch to start automatic cycle of m/c: $0.05\,min$. facing and centering of the work piece by the machine on its automatic cycle: $1.60\,min$. Await while jaws of the vice openout to release job: $0.05\,min$. Collect faced and centred job from hydraulic vice: $0.05\,min$. Keep the job aside: $0.05\,min$. Prepare a multiple activity chart for facing and centering operation. Determine cycle time and % utilization of operator and machine.	08
6	a	Analyze the principles of motion economy as applied to use of Human body and arrangement of Work Place.	08
	b	Draw an appropriate chart for recording the activity of changing refill in a ball point pen. Justify the selection of the recording technique.	08
		OR	

7	a	Distinguish between Cyclegraph and Chronocycle graph.	08
	b	Prepare a table indicating the questions to be asked and answered during critical examination in typical method study process.	08
8	a	The tasks to be studied in time study are broken down into tiny details called elements. Why it is done? What rules govern such a break-up?	08
	b	The end results of a work sampling study are given below: Observations of the machine working is 3500, observation of machine idle is 1500. The total number of observation is 5000. Calculate the limit of accuracy. Also evaluate whether the number of observations made are sufficient to achieve desired accuracy of plus or minus 5%.	08
		OR	
9	a	A company has 50 general purpose machines in their workshop. Work sample study is carried out in the shop to establish ineffective time in its various forms. First two days analysis revealed that the machine idle time was 40%. How many observations will be necessary to ensure that the observed time has the accuracy of plus or minus 2% with confidence limit of 95%. How many founds will be necessary to complete the required number of observations?	08
	b	Critically evaluate the relevance of the work measurements data in the fields of Manning employees and Measurement of employee's progress.	08
10	a	Critically analyze different systems of rating used in the establishment of time standards.	08
	b	Grinding of jib bushes at a bore grinding machine involves following nine elements and their average observed times and the related details are as below:	

Element Code	*Element Description*	*Obs.Time(min)*	*Rating (%)*
A	*Secure job in a three jaw chuck*	0.15	80
B	*Set chuck into the motion and advance work to wheel*	0.05	90
C	*Grind bore to rough size*	1.20	100
D	*Clear work from wheel*	0.03	100
E	*Inspect bore with plug gauge*	0.15	75
F	*Dress wheel*	0.20	80
G	*Finish grind bore to size*	0.25	100
H	*Check bore with bore dial gauge*	0.30	75
I	*Stop chuck movement, remove job and aside.*	0.15	80

Calculate the standard time of the job considering relaxation allowance to be 13% and contingency allowance to be 3%. 08

OR

11	a	Various allowances are considered in building up standard time of a job. Justify the need for considering these different allowances in the determination of standard time.	08
	b	A stop watch time study has been made of an operation which consists of 4 elements. The table below gives the continuous time reading in centiminutes and rating.	

Element	1	2	3	4
Observations	*Continuous watch*			
1	08	14	28	32
2	40	46	60	63
3	72	81	95	98
4	107	114	128	131
5	139	144	157	160
6	168	173	185	189
Rating (%)	110	120	105	90

Calculate the normal time. 08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

1	1.1	Productivity is the ratio of _____.	01
	1.2	The prime objective of lean manufacturing practice is _____.	01
	1.3	The father of time study is _____.	01
	1.4	Work study man should not give order to worker. Why?	01
	1.5	Irreducible minimum time theoretically required to produce one unit of output is known as _____.	01
	1.6	In a process chart, a circle within a square symbol denotes _____.	01
	1.7	A chart used to review the overall sequence of an operation by focusing either the movement of operators or materials is called _____.	01
	1.8	The symbol Δ is used to represent _____ in VSM.	01
	1.9	$SIMO$ charts are used in _____.	01
	1.10	_____ is the most appropriate recording technique used to understand and improve the movements of an operative at his workplace.	01
	1.11	The area that can be covered by the movement of forearm, hand and fingers with elbow as pivot is known as _____.	01
	1.12	_____ refers to motions of the human body at the workplace and to the mental activities associated with them.	01
	1.13	The technique of establishing the time for a qualified worker to carry out a specified job at a defined level of performance is known as _____.	01
	1.14	Define a qualified worker.	02
	1.15	Define rating.	02
	1.16	What is cumulative timing?	01
	1.17	What are the needs for allowances?	01
	1.18	Expand $PMTS$.	01

PART-B

2	a	Define industrial engineering as given by American Institute of Industrial Engineering. What are the functions of industrial engineering?	10
	b	Discuss the factors affecting productivity.	06
		OR	
3	a	Discuss the reaction of workers, supervisors and management towards the application of work study in their organization.	08
	b	Justify with reasons how work study is valuable to industry.	08
4	a	Explain the steps "Select" and "Record" in conducting method study.	08
	b	In a continuous production process, the following is the sequence of operations of a man-machine interface for one cycle.	

Sl. No.	*Elements*	*Time (min)*
1	*Castings picked up and brought to the machine*	0.09
2	*Job positioned and fastened to machine*	0.12
3	*Machine started*	0.04
4	*Feed changed and machining begun*	0.07
5	*Casting is machined automatically*	0.35
6	*Machine stopped*	0.04
7	*Job unfastened and removed*	0.10
8	*Casting taken back and deposited*	0.09

i) Draw a man-machine chart and find the cycle time and utilization of man and machine.

ii) Apply method study principles and arrive at an improved method for the operation. 08

OR

5	a	Explain the step "Critical Examination" of method study.	08
	b	Draw flow process chart and flow diagram for the following operation: A gardener, who is waiting at the porch of a house travels to the garage 85 $feet$ away, opens the garage door, walks 10 $feet$ to the locker, opens the locker, removes hose form the locker, walks to the rare garage door 15$feet$ away, opens door, walks to the water point 15$feet$ away, attaches the hose, waters the garden and retraces back to his place after replacing hose in the locker. (Refer fig 5b)	

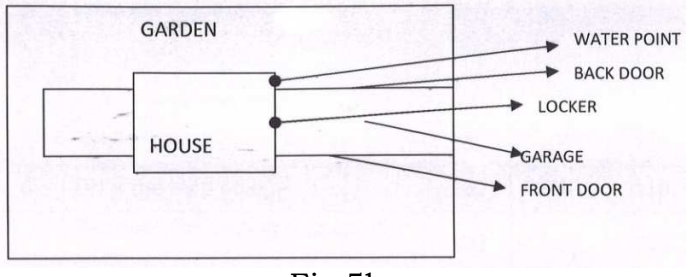


Fig 5b 08

6	a	List the principle of motion economy with respect to use of human body.	08
	b	Draw a two handed process chart for the activity of replacing 2 number of "dry battery cells" of a calculator. The calculator is placed on your right hand side and the cell towards your left hand side. The battery compartment is on the backside of calculator and has sliding cover.	08
		OR	
7	a	List the principle of motion economy with respect to arrangement to workplace.	08
	b	Write a note on Micromotion study.	04
	c	Distinguish between Cyclegraph and Chronocycle graph.	04
8	a	What is work sampling? Describe the procedure followed in conducting a work sampling study.	08
	b	Check whether the number of observations is sufficient for $\pm 5\%$ accuracy and 95% confidence limit, if Number of observations of machine working = 3000 Number of observations of machine idle = 1000 Total observations = 4000	08
		OR	
9	a	Explain briefly the steps to be followed in carrying out stopwatch time study.	08
	b	What are the reasons for breaking down the operations into elements? State the rules.	08
10	a	Explain the Westing house system of rating.	06
	b	The following readings in minutes were observed for 2 cycles using continuous timing method.	

Sl. No.	*Description of element*	*Cycle – I*		*Cycle – II*	
		Rating	*Watch reading*	*Rating*	*Watch reading*
1	*Picks up casting and positions it*	100	0.36	90	3.05
2	*Drills the first hole*	110	0.86	100	3.60
3	*Moves the casting*	110	1.06	100	3.82
4	*Drills the second hole*	100	1.61	110	4.32
5	*Moves the casting*	100	1.83	110	4.52
6	*Drills the third hole*	100	2.38	110	5.02
7	*Removes the casting and keeps it aside*	100	2.65	120	5.22

		Assuming an allowance of 20% calculate the standard time.	10
		OR	
11	a	Explain the different types of allowances added while computing standard time.	08
	b	Write a note on Standard data and *MTM*.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

1	1.1	What is meant by the term 'Productivity'?	02
	1.2	Why should method study precede work measurement?	02
	1.3	Name the chart that you will use to study unoccupied time of the operator in a given work cycle. Why?	02
	1.4	Identify the symbol to be used to record the activity of "Tea leaves being graded into different varieties".	02
	1.5	List the techniques of work measurement.	02
	1.6	Which is the most suitable technique for measuring manual operations confined to one work centre? Why?	02
	1.7	What are therbligs? How many symbols are defined as past of therbligs.	02
	1.8	Which timing method do you suggest to study an operation which involves fixed sequence of elements? Why?	02
	1.9	Mention the classification of motions under Method time measurement.	02
	1.10	What are Relaxation allowances? Why they are provided?	02

PART-B

2	a	Discuss the three ways of achieving productivity. Supplement your answer with necessary examples.	08
	b	What is meant by workstudy? Explain with a suitable illustration how workstudy technique leads to improved productivity.	08
		OR	
3	a	Provide an overview on evolution of Industrial Engineering approach.	08
	b	What are the constituents of work content? List the techniques used to reduce work content.	08
4	a	Explain the steps involved in conducting method study.	06

	b	The operator engaged on the machine performs the following operations: i) Pick up the job, place it between jaws of hydraulic vice($0.2\ min$). ii) Make the switch ON to tightly hold the part($0.08min$). iii) Make the switch on start automatic cycle of operation ($0.08min$). iv) Machining of the part on autocycle.($1.5\ min$). v) Wait till the vice opens automatically ($0.08min$). vi) Pick up the machined job from the vice ($0.05min$). vii) Keep it in the tray ($0.05\ min$). Construct the multiple activity chart for the machining operation. Also identify the cycle time and utilization of both the resources.	10
		OR	
5	a	According to the nature of job being studied and the purpose for which the record is required the recording techniques fall into two broad categories: Explain them in brief.	08
	b	One of the component manufacturing process involves the following activities. Component was brought from the stores which is $10m$ away. Then the component was loaded on the machine. The component was machined. It was then moved to inspection bench which is 12 meters away. It has to wait for 15 minutes for the inspector to be free from previous job. The component was checked for accuracy. It was then moved back to the store which is $12\ m$ away from the inspection bench and stored in rack. Represent the above process using a material type flow process chart.	08
6	a	Distinguish cycle graph and chronocycle graph.	06
	b	Draw a two handled process chart for the task of "Assembly of nut and bolt". Begin the chart when both hands are free before assembly. End charting when both hands are free after assembly. Also illustrate the work bench layout.	10
		OR	
7	a	Explain the principle of motion economy as applied to arrangement of workplace and design of tools and equipment.	10
	b	Discuss briefly the implementation and maintenance aspects of new methods in method study.	06
8	a	Elaborate the steps to be followed in conducting stop watch time study.	08
	b	Explain various types of elements in Time study. Provide one example to each type of element.	08
		OR	
9	a	What is work sampling? Discuss the procedure for conducting a work sampling study.	08

b The following data refers to a sampling study of production of one component. Duration of data collection on 5 days at eight hours per day. Number of operators = 10; Allowances given for the process is 15%. The production quantity in 5 days is 6000 units. The sampling data collected is as follows:

Days	1	2	3	4	5
Number of observations	230	240	200	180	225
Occurrence of activity	200	190	170	150	210

Calculate the standard time of production of the component if average performance rating of the operator is 120% and the entire operation is manual. 08

10 a What is meant by performance rating? List the factors affecting performance rating. 06

b The following data refers to the study conducted for an operation. The actual time for elements in minutes is shown below:

Cycle	1	2	3	4	5
Elements					
1	2.5	2.1	2.2	5.4	2.5
2	6.2	6.0	6.1	5.9	5.9
3	2.3	2.0	2.1	2.1	2.2
4	2.4	2.1	2.8	3.0	2.3

i) Element 2 is machine element.
ii) Consider the observations as abnormal and delete the same if they are more than 2% of average time of that element.
iii) Take performance rating as 120.
iv) Take following allowances personal allowances as 30 minutes in a shift of 8 hours. Fatigue allowance is 15%, contingency allowance is 2%. Estimate the standard time of operation and production per 8 hours shift. 10

OR

11 a What are different systems of rating? Explain any one system in detail. 06

b Turning gear blanks on centre lathe involves the following elements. The stop watch data is given. Assuming the rest and personal allowance as 13% and contingency allowance as 2%, Calculate the standard time.

Note:->Element No: 5, observation No: 2, foreign element 2.54 to 5.02

Element	*Description*	*Observation*				
		1	2	3	4	5
1	*Pick and place*	0.2	1.46	5.22	6.49	14.25
2	*'on' m/c and tool approach*	0.3	1.55	5.30	13.10	14.35
3	*Turn diameter*	1.05	2.31	6.05	13.84	15.10
4	*Withdraw tool and stop m/c*	1.13	2.38	6.14	13.92	15.17
5	*Release part and palce it side*	1.28	2.54 *	6.29	14.06	15.32

The elemental ratings are as follows:

Element	1	2	3	4	5
Rating%	90	110	110	110	95

10

Manufacturing Processes

III Semester SEE Question Papers

Course codes: 21IM34, IM234AI, 18IM36, 16IM36, 12IM34

- Jun/Jul 2025 Supplementary (21IM34)
- Feb/Mar 2025 (IM234AI)
- Apr/May 2024 (IM234AI)
- Apr/May 2023 (21IM34)
- Fast Track Jan/Feb 2023 (18IM36)
- Mar 2021 (18IM36)
- Fast Track Oct 2020 (16IM36)
- Fast Track Oct 2020 (18IM36)
- Nov/Dec 2019 (16IM36)
- Nov/Dec 2019 (18IM36)
- Fast Track Jul 2019 (16IM36)
- Nov/Dec 2018 (16IM36)
- Fast Track Jul 2018 (16IM36)
- Nov/Dec 2017 (16IM36)
- Fast Track Jul 2017 (12IM34)
- Nov/Dec 2016 (12IM34)
- Fast Track Jul 2016 (12IM34)

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Supplementary Examinations June/July – 2025

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Differentiate between job and batch production.	02	1	1
	1.2	Why is casting considered suitable for complex shapes?	02	1	1
	1.3	List the two common applications of rolling.	02	2	2
	1.4	What is the principle behind centrifugal casting?	02	2	2
	1.5	When would you choose adhesive bonding over welding?	02	3	3
	1.6	Suggest ways to reduce welding defects in MIG welding.	02	3	3
	1.7	What type of machining would you use for a flat surface with key slots?	02	2	2
	1.8	Why is excessive cutting temperature harmful?	02	2	2
	1.9	Why are advanced machining processes needed in modern manufacturing?	02	2	2
	1.10	Suggest a product where LBM is suitable.	02	2	2

PART-B

2	a	Discuss the recent developments in manufacturing and their significance.	08	1	1
	b	List the advantages and limitations of Expendable-Mould Casting Processes.	08	2	2
3	a	Compare hot working and cold working process.	08	2	2
	b	What is press forging? How does it differ from drop forging?	08	2	2
		OR			
4	a	Describe the principle of Investment Casting and list its application.	08	3	3
	b	Explain the steps of centrifugal casting process with a neat sketch.	08	3	3
5	a	Which fusion-welding process would you select for welding thick steel plates, and why?	08	3	3
	b	Evaluate how poor weldability can affect the quality and performance of a welded structure.	08	3	3
		OR			
6	a	Explain the process of soldering and identify its common areas of application.	08	4	4
	b	List and discuss welding defects and with their causes.	08	4	4
7	a	Explain the role of built-up edge in the cutting process and how it affects machining performance.	–	–	–
	b	State assumptions of merchant circle diagram with a neat sketch.	–	–	–
		OR			

8	a	List and describe the various tool angles used in single-point cutting tools.	–	–	–
	b	Explain the significance of tool life in metal cutting operations and its effect on production cost and quality.	–	–	–
9	a	Which advanced machining process would you recommend for machining intricate holes in hard metals without tool wear? Justify your choice.	–	–	–
	b	Explain the working principle of Electrochemical Machining (ECM). How is it different from conventional?	–	–	–
OR					
10	a	Discuss the limitations of Abrasive-Jet Machining (AJM) for precision machining applications.	–	–	–
	b	Explain the concept of hybrid machining. Why is it used instead of conventional or single advanced processes?	–	–	–

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Regular/Supplementary Examinations Feb/Mar – 2025

Industrial Engineering and Management

MANUFACTURING PROCESSES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Describe the importance of pattern color coding and provide examples.	02	1	1
	1.2	What are chaplets and describe their purpose in casting.	02	1	2
	1.3	Name the differences between hot working and cold working.	02	2	2
	1.4	What is precision investment casting and list its advantages.	02	1	2
	1.5	Define electron beam welding (EBW) and state where it is used.	02	1	1
	1.6	List the factors influencing the weldability.	02	1	3
	1.7	Write the different types of chips formed in metal cutting.	02	1	2
	1.8	What is machinability and list the factors that affect it.	02	1	3
	1.9	Find the machining time required for drilling a 25 mm hole through a 50 mm thick plate at 600 rpm with a feed rate of 0.2 mm/rev.	02	1	3
	1.10	Write about the purpose of indexing in milling.	02	1	4

PART-B

2	a	Discuss recent developments in manufacturing.	08	2	1
	b	Illustrate the steps involved in sand moulding process with neat sketch.	08	3	2
3	a	Explain the process of forging and compare smith forging, drop forging and press forging process.	08	4	2
	b	Explain the principle of extrusion, discuss its types according to direction of flow of materials and illustrate with a neat sketch.	08	4	2
		OR			
4	a	Discuss the principle of shell moulding process with neat sketch.	08	2	2
	b	Explain the hot chamber die casting process with neat sketch.	08	4	2
5	a	Explain the principles of arc welding with basic welding circuit and its applications.	08	4	1
	b	Identify the welding process that uses heat generated from burning a combination of different gases to melt and fuse metals and how does it work? Explain with neat sketch.	08	4	1
		OR			
6	a	Distinguish between fusion welding and solid state welding with examples.	–	–	–
	b	Discuss the factors affecting weld quality and explain the causes and remedies of common welding defects.	–	–	–
7	a	Explain single point cutting tool nomenclature and its geometry with neat sketch.	–	–	–

	b	In an orthogonal cutting test with a tool of rake angle 10°, the following observations were made: chip thickness ratio = 0.3, cutting force = 1290 N, feed force = 1650 N. From Merchant's theory, calculate shear plane angle, coefficient of friction, velocity of chip relative to tool at the chip tool interface.	–	–	–
		OR			
8	a	Explain the desired properties and types of cutting tool materials.	–	–	–
	b	Discuss the different types of cutting fluids and their application.	–	–	–
9	a	With a neat sketch, explain the nomenclature of twist drill bit.	–	–	–
	b	Explain the different types of drilling related operations with neat sketch.	–	–	–
		OR			
10	a	With neat sketch, differentiate between up milling and down milling operation.	–	–	–
	b	A gear blank on the milling machine in order to cut 73 teeth, differential indexing is used to rotate the part. The machine has a 40:1 ratio index head. Calculate the gear ratio necessary between the spindle and index plate.	–	–	–

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Examinations April/May – 2024

Industrial Engineering and Management

MANUFACTURING PROCESSES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	The value of chip reduction coefficient depends upon cutting tool material and _____.	2	1	1
	1.2	In the Taylor's tool life equation, $VT^n = C$, the value of n = 0.5. The tool life has 180 minutes at a cutting speed of 18 m/min. If the tool life is reduced to 45 minutes, then the cutting speed will be _____.	2	3	3
	1.3	The working of tin and lead even at room temperature is termed as hot working. Give reasons.	2	3	2
	1.4	List any two defects of welding.	2	2	2
	1.5	State any two differences between orthogonal cutting and oblique cutting.	2	2	1
	1.6	Define porosity.	2	2	3
	1.7	List any two limitations of extrusion process.	2	1	2
	1.8	List any two applications of hydro forming process.	2	3	1
	1.9	Circumference of a work piece is to be divided into 60 equal divisions using simple indexing method having 40 worm wheel teeth. Calculate indexing movement.	2	3	2
	1.10	List the cutting parameters that greatly affects the tool life.	2	2	2

PART-B

2	a	Define pattern. List the types of patterns and explain any one briefly.	06	2	2
	b	Explain the process of casting with a neat sketch.	10	1	1
3	a	With a neat sketch, explain the centrifugal casting process which is best suited for casting of hollow pipes and tubes.	10	3	3
	b	List the advantages and disadvantages of extrusion process.	06	2	2
		OR			
4	a	How are cup shaped vessels being manufactured? Explain the metal spinning process with a neat sketch.	10	3	3
	b	List the advantages and disadvantages of hydro forming process.	06	2	2
5	a	With a neat sketch, explain resistance seam welding process which is best used to rectangular or square or even in circular shapes.	–	–	–
	b	State the differences between fusion and solid state welding processes.	–	–	–
		OR			
6	a	Explain with the help of sketch, working principle of Thermit welding.	–	–	–
	b	What is weld quality? Explain the factors that affect weldability.	–	–	–

7	a	List cutting tool materials. Describe in brief, stating its principal characteristics and applications of any two cutting tool materials.	–	–	–
	b	Derive an expression for the shear angle in orthogonal cutting in terms of rake angle and chip thickness ratio.	–	–	–
		OR			
8	a	What are the different types of chips that are formed in metal machining?	–	–	–
	b	An orthogonal cutting operation is performed using a rake angle of 15°, chip thickness before the cut = 0.012 inch and width of cut = 0.100 inch. The chip thickness ratio is measured after the cut to be 0.55. Determine (i) the chip thickness after the cut (ii) Shear angle (iii) Friction angle (iv) Coefficient of friction (v) Shear strain.	–	–	–
9	a	Define indexing. Explain the following indexing methods. (i) Direct indexing (ii) Compound indexing.	–	–	–
	b	A drilling operation is performed on a workpiece made of mild steel using a twist drill with a diameter of 10 mm. The cutting speed is set to 60 m/min and the feed rate is 0.1 mm/rev. The drill needs to penetrate the workpiece to a depth of 30 mm. Assume that the drill needs to be changed after every 50 holes and each tool change takes 2 minutes. Calculate the total machining time required to drill 200 holes in the workpiece.	–	–	–
		OR			
10	a	State different types of drilling operations and explain any two with a neat sketch.	–	–	–
	b	Differentiate between up milling and down milling.	–	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations April/May-2023

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	What are the applications of casting?	02
	1.2	State any four types of patterns.	02
	1.3	List the examples of metal forming.	02
	1.4	Name the type of materials typically used for permanent-mold casting processes.	02
	1.5	Which is the most common type of welding and is usually referred to as stick welding?	02
	1.6	What is the purpose of flux?	02
	1.7	Write Taylor's tool life equation.	02
	1.8	Name the various cutting tool materials.	02
	1.9	What is the need for unconventional machining processes?	02
	1.10	Identify the mechanism of material removal, transfer media and energy source for EDM.	02

PART-B

2	a	Explain the steps involved in casting processes.	08
	b	What is moulding sand? What properties are required for perfect moulding sand?	08
3	a	Discuss the properties of hot working process of metal forming.	08
	b	Classify forging and explain any one process of it.	08
		OR	
4	a	Describe the principle of shell moulding and list its application.	08
	b	Differentiate between hot chamber and cold chamber die casting process.	08
5	a	With a basic welding circuit, explain the principle of arc welding.	08
	b	Explain the diffusion and pressure phenomena with respect to solid state bonding.	08
		OR	

6	a	Discuss the importance of weldability with the factors affecting the weldability of a material.	10
	b	List atleast two causes and remedies of defects in welding.	06
7	a	Discuss the importance of built up edge (BUE) and its tendency of reducing the BUE formation.	08
	b	State assumptions of Merchant's circle diagram with a neat sketch.	08
		OR	
8	a	List the different types of drilling related operations.	08
	b	Explain crater wear and flank wear of a tool life with a neat sketch.	08
9	a	Explain the different methods of applying Maskant.	12
	b	List the parameters affecting the efficiency of chemical machining.	04
		OR	
10	a	Discuss the design consideration for laser beam machining.	08
	b	What is virtual prototyping and how does it differ from additive methods?	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast track Examinations Jan/Feb-2023

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ type of chips form while machining of brittle materials.	01
	1.2	The angle between side cutting edge and end cutting edge is called as _____.	01
	1.3	_____ type of fusion welding process is used for welding sheet metals of all engineering metals (except Cu,Ag) in automobile and air craft.	01
	1.4	Jolt machine gives better packing of sand near_____.	01
	1.5	_____ is the type of welding defect caused due to poor deposition of weld rod.	01
	1.6	Hardness of green sand mould increases with decrease in _____.	01
	1.7	Draft allowance is also known as _____.	01
	1.8	_____ wood is most widely used for making pattern.	01
	1.9	Using the Taylor equation for tool life and letting n=0.5 and C=120, calculate the percentage increase in tool life when the cutting speed is reduced by 50%.	02
	1.10	The forces acting on the moulding flasks is _____.	01
	1.11	The tool made of cemented carbide wear out faster at _____ speeds.	01
	1.12	The trade name of nonferrous cast alloy composed of cobalt, chromium and tungsten is called _____.	01
	1.13	If t1 denotes the uncut chip thickness and t2 denotes cut chip thickness ratio then, write an equation for chip thickness ratio 'r'?	01
	1.14	In metal cutting operation, maximum heat is generated in _____ zone.	01
	1.15	For drilling operation, the drill rotates with _____ pressure if job is held on earth.	01
	1.16	Point angle of drill is also known as ____	01
	1.17	Circumference of a work piece is to be divided into 35 equal divisions using simple indexing method having 40 worm wheel teeth. Indexing movement will be?	02
	1.18	_____ milling operation can be used for machining of flat surface which is at some angle to the axis of cutter.	01

PART-B

2	a	"Jolt and Squeeze moulding machine is used for preparing the moulds at faster rate and with high quality". Explain with a neat sketch.	10
	b	"Moulding processes to produce internal cavities and reentrant angles". What is the process called and list the different types.	06

3	a	What are the reasons for an undersized weld in resistance welding?	10
	b	With a neat sketch, explain which resistance welding process is best used to fan covers and hollow metal doors.	06
		OR	
4	a	With a neat sketch, explain which resistance welding process is best used for rectangular or square or even in circular shapes.	10
	b	List the advantages and disadvantages of resistance welding.	06
5	a	What is the effect of rake angle and explain the types ?	08
	b	What is chip and explain the different types of chips?	08
		OR	
6	a	Differentiate between orthogonal and oblique cutting.	06
	b	An orthogonal cutting operation is performed using a rake angle of 15^0, chip thickness before the cut =0.012 in and width of cut = 0.100 in. The chip thickness ratio is measured after the cut to be 0.55. Determine. i. The chip thickness after the cut. ii. Shear angle iii. Friction angle iv. Coefficient of friction v. Shear strain.	10
7	a	Draw a neat diagram of a radial drilling machine. Name all the parts and explain the principle of operation.	10
	b	State the difference between capstan and turret Lathe from the Engine lathe.	06
8	a	List and explain the different indexing methods in popularity.	10
	b	Explain briefly the different types of abrasives used for grinding.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations March-2021

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Yellow strips on black background are provided on core prints for _____ surface.	01
	1.2	The distortion in casting cannot be prevented even by providing proper _____ allowance.	01
	1.3	In resistance welding, two electrodes are made of _____.	01
	1.4	The cutting tool removes the metal from work piece in the form of _____.	01
	1.5	In the Taylor's tool life equation, $VT^n = C$, the value of $n = 0.5$. The tool life has a life of 180 $minutes$ at a cutting speed of 18m/min. If the tool life is reduced to 45 $minutes$, then the cutting speed will be _____.	01
	1.6	Ceramic tools are fixed to the tool body by _____.	01
	1.7	Turret head in turret lathe is generally mounted on _____.	01
	1.8	On bar type barrel lathes, work to be machined is gripped in _____.	01
	1.9	Form cutting can be performed more effectively by _____ milling machine.	01
	1.10	Cutting T-slots can be performed more effectively by _____ milling machine.	01
	1.11	Enumerate the factors to be considered for selection of manufacturing process.	02
	1.12	State the example for distortion allowances.	02
	1.13	List any four defects in casting.	02
	1.14	Find the rpm of the drill spindle when the cutting speed is 26 $meters\ per\ min$, having a hole of 20mm diameter.	02
	1.15	List the common work holding devices used on milling machines.	02

PART-B

2	a	Discuss the causes for the following pattern allowances: i. Shrinkage allowances ii. Machining allowances.	08
	b	List and explain the factors that are to be considered for selecting the pattern material.	08

3	a	Describe with a neat sketch, the working principle of resistance seam welding.	08
	b	Enumerate the pro's and con's of seam and spot welding.	08
		OR	
4	a	Explain the working principle of projection welding with a neat sketch.	08
	b	Analyze the causes and remedies for the following welding defects. i. Weld crack ii. Porosity	08
5	a	Differentiate between orthogonal and oblique cutting with respect to the following aspects: i. Inclination of the cutting edge of the tool ii. Components of the cutting force iii. Maximum chip thickness occurrence iv. Tool life.	08
	b	During an orthogonal machining (turning) operation of $C-40$ steel, the following data were obtained: — Chip thickness = $0.45mm$ — Width of cut = $2.5mm$ — Feed = $0.25mm/rev$ — Tangential cut force = $1130N$ — Feed thrust force = $295N$ — Cutting speed = $2.5m/s$ — Rake angle = 10^0 Calculate Chip thickness ratio, shear angle, shear force and friction angle.	08
		OR	
6	a	Identify the different variables that affect the tool life.	06
	b	When machining at $50m/min$, the tool life noted as $100min$. Calculate the tool life if a similar tool cuts at 20% higher speed. Also find out the cutting speed if the tool is required to machine for $170min$ without failing. Take $n = 0.27$ in Taylor's tool life equation.	10
7	a	Explain with a neat sketch, the turret head indexing mechanism of capstan lathe.	10
	b	A hole with $40mm$ diameter and $50mm$ depth is to be drilled in mild steel component. The cutting speed can be taken as $65m/min$ and the feed rate as $0.25mm/rev$. Calculate the machining time and the metal removal rate.	06
8	a	State clearly, the Plate Number and the indexing movements to be made using a Universal dividing head, to divide the periphery of a work into 69 equal divisions. Following indexing plates are available: Plate No. : 15, 16, 17, 18, 19, 20 Plate No. : 21, 23, 27, 29, 31, 33 Plate No. : 37, 38, 41, 43, 47, 49	12
	b	Specify the grinding wheel for the following as per the Indian Standards Systems of making Grinding Wheel. $250 \times 25 \times 32\ W\ A\ 46\ L\ 4\ V\ 17$	04

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering & Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The taper provided on pattern for its easy and clean withdrawal from the mould is called _____.	01
	1.2	_____ machine gives better packing of sand near pattern	01
	1.3	A pattern is shaken by striking it with wooden piece. A negative allowance is provided for this. Which allowance is it?	01
	1.4	In a two piece pattern molding arrangement, _____ keeps the aligned between the parts of the pattern.	01
	1.5	The forces acting on the moulding flasks is _____.	01
	1.6	The property of sand to allow the gases to pass through is known as _____.	01
	1.7	The working temperature of cold working is below a certain temperature. That temperature is known as _____.	01
	1.8	In _____ forging operation, repeated hammering and closed die is used.	01
	1.9	Heat is created by chemical reaction in _____ welding.	01
	1.10	_____ type of chips are produced during machining of brittle materials.	01
	1.11	The angle between side cutting edge and end cutting edge is called as _____ angle.	01
	1.12	Crater wear occurs at _____ of the cutting tool.	01
	1.13	In the Tylor's tool life equation, $VT^n = C$, the value of $n = 0.5$. The tool life has a life of 180 $minutes$ at a cutting speed of $18m/min$. If the tool life is reduced to 45 $minutes$, then the cutting speed will be _____.	02
	1.14	For drilling operation, the drill rotates with _____ pressure if job is held on earth.	01
	1.15	_____ is an operation of finishing previously drilled hole in order to bring to accurate size and have good surface finish.	01
	1.16	The metal removal takes place in drilling process due to _____ & _____ of metal.	02
	1.17	The value of chip reduction coefficient depends upon _____ cutting tool material and _____.	02

PART-B

2	a	With a neat sketch, explain the working principle of Jolt squeeze moulding machine.	10

	b	What properties are desirable of molding sand from the stand point of sound casting? Explain any two.	06
3	a	How are cup shaped vessels being manufactured. Explain the process with a sketch.	10
	b	List defects in Welding Process.	06
		OR	
4	a	Explain the stages required for obtaining the final shape of the crank by using drop forging process.	08
	b	Illustrate the working principle of Projection welding.	08
5	a	What are the different types of chips that are formed in metal machining and give the conditions that are favorable to their formation?	10
	b	Identify the various cutting tool materials? Describe in brief, stating its principal characteristics & application of any two cutting tool materials.	06
		OR	
6	a	Define machinability? What are the factors affecting machinability of materials.	06
	b	In an orthogonal cutting process, the following data were obtained: chip length obtained = $96mm$, uncut chip length = $240mm$, rake angle used = 20^0. Depth of cut = $0.6mm$, horizontal component of cutting force = $2400N$ and vertical component of cutting force = $240N$. Calculate for the given data: i) Shear plane angle ii) Chip thickness iii) Friction angle iv) Resultant cutting force.	10
7	a	Differentiate between an engine lathe and capstan / turret lathe.	10
	b	State different types of drilling operations and explain any two with a neat sketch.	06
8	a	Differentiate between Upmilling and down milling.	10
	b	Explain briefly the different types of abrasives used for grinding.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast track Examinations October-2020

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Irregular depressions on the surface of the forging is example for _____ defect in forging.	01
	1.2	In metal forming process, working on steel at a temperature of 800^0C is _____ forming process.	01
	1.3	Positive allowance: machine allowance:: Negative allowance: _____.	01
	1.4	Why chaplets are used in molding process?	02
	1.5	The _____ is machine tool which is also known as mother of all machines.	01
	1.6	In drilling, depth of cut is _____ the drill diameter.	01
	1.7	In capstan and turret lathe, carriage is also called as _____.	01
	1.8	Balanced core is also called as _____ core.	01
	1.9	Positive rake angle: Right hand flute :: Zero rake angle: _____.	01
	1.10	What should be the percentage change in cutting speed to give 50% reduction in tool life? Take $n = 0.2$	02
	1.11	What is meant by choke area in Gating system?	02
	1.12	The working of tin and lead even at room temperature is termed as hot working. Give reasons.	02
	1.13	State any two specifications of lathe.	02
	1.14	The two common shapes produced by form milling are _____ and _____.	02

PART-B

2	a	Outline the production steps of typical sand casting operation with an aid of flowchart.	08
	b	List and explain the various types of Pattern Allowances.	08
3	a	Outline the advantages and disadvantages of Projection Welding.	06
	b	Explain the working principle of resistance seam welding.	10
		OR	
4	a	What are limitations of Flash welding?	08
	b	What are different types of Welding Defects?	08

5	a	A seamless tube $32mm$ outside diameter is turned on lathe. Cutting velocity of the tool relative to the workpiece is $10m/min$, rake angle= 350, depth of cut= $0.125mm$, length of chip = $60mm$, horizontal cutting force of the tool on the workpiece= $200N$, vertical cutting force required to hold the tool against the work= $80N$. Calculate: i) Coefficient of friction ii) Chip thickness ratio iii) Shear plane angle iv) Velocity of chip relative to the tool and v) Velocity of chip relative to the workpiece.	10
	b	Identify the desired properties of cutting tool materials and explain any two in brief.	06
		OR	
6	a	Derive an expression for the shear angle in orthogonal cutting in terms of rake angle and chip thickness ratio.	10
	b	Explain the factors affecting tool life in machining processes.	06
7	a	With neat sketch, explain the constructional features of Capstan lathe.	10
	b	Explain the methods adopted to make holes in component.	06
8	a	Define Indexing. Explain the following Indexing methods. i) Direct indexing ii) Compound indexing.	10
	b	What is the difference between Dressing and Truing in grinding?	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is gated pattern?	02
	1.2	Explain why the sprue should be tapered?	02
	1.3	List any two advantages of cold working process.	02
	1.4	Give a brief classification of various fabrication processes.	02
	1.5	Differentiate between flank wear and crater wear.	02
	1.6	State any two important functions of cutting fluids.	02
	1.7	Differentiate between capstan and turret lathes.	02
	1.8	What is the importance of point angle in twist drill?	02
	1.9	Differentiate between up milling and down milling.	02
	1.10	What do you mean by grinding wheel structure and wheel grade?	02

PART-B

2	a	Explain the following pattern allowances: i) Shrinkage allowance ii) Distortion allowance	08
	b	Explain Caine's method for riser design.	08
3	a	Explain the principle of rolling process with neat sketch. Explain any four different rolling stand arrangements with sketches.	08
	b	Explain any four welding defects.	08
		OR	
4	a	Explain the following forging methods: i) Smith forging, ii) Drop forging.	08
	b	Explain the working principle of resistance spot welding with a neat sketch.	08

5	a	Derive an expression for shear plane angle in terms of chip thickness ratio and tool rake angle during orthogonal cutting.	08
	b	Explain the following cutting tool materials with their importance & applications in machining field: i) Cemented coated carbide tools, ii) Ceramics.	08
		OR	
6	a	In an orthogonal cutting, the following observations were made: $Feed = 0.2mm/rev; chip\ thickness = 0.67mm;\ cutting\ force = 1200N;$ $Thrust\ force = 650N; Back\ rake\ angle = 10°; width\ of\ cut = 2mm;$ $Cutting\ speed = 200mpm.$ Determine the following: i) Coefficient of friction between tool and work, ii) Shear force, iii) Shear velocity, iv) Chip flow velocity and v) Specific cutting energy.	10
	b	What is machinability? Explain any four factors affecting machinability.	06
7	a	Sketch and explain the nomenclature of a twist drill.	08
	b	Explain the following with neat sketch: i) Centerless grinding, ii) Internal cylindrical grinding.	08
8	a	Explain the constructional features of a turret lathe with a neat sketch.	08
	b	Explain the following indexing methods with examples: i) Compound indexing, ii) Differential indexing.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	_____ process is the only technique in which metal is in its liquid form.	01
	1.2	Casting process requires preparation of cavity using _____ type of material.	01
	1.3	The dividing line between the cope and the drag is called _____.	01
	1.4	The moulding sand that contains moisture is termed as _____ sand.	01
	1.5	The expression to determine the permeability of sand specimen is P = _____.	01
	1.6	_____ is used for making cavities that normally cannot be produced by pattern alone.	01
	1.7	_____ technique is the most economical method and relies on one of the primary manufacturing process for raw materials.	01
	1.8	Fabrication using _____ are temporary in nature.	01
	1.9	In resistance welding _____ voltage and _____ current is employed.	02
	1.10	Expression for heat generated in resistance welding is given by H = _____.	01
	1.11	Disc type of electrodes are employed in _____ welding.	01
	1.12	In _____ welding one of the sheets to be joined is provided with number of projections.	01
	1.13	_____ is a type of welding defect.	01
	1.14	Flux upon solidification is called _____.	01
	1.15	Chip thickness ratio is defined as the ratio between the _____ and the chip thickness after _____	02
	1.16	Example for a tool and work holding device are _____ and _____ respectively.	02
	1.17	If 'd' is the diameter of the drill bit in 'mm' and 'n' the rpm, then the cutting speed is given by V = _____ m/min	01

PART-B

2	a	Sketch the cross section of a sand mould and label various parts.	08
	b	List major advantages and limitations of sand casting process. How do you overcome the limitations?	08

3	a	Suggest the type of fabrication technology for the following applications: i) Aircraft sheet metal assembly process ii) A temporary metal structure at civil construction site iii) Joining of irregular shaped plastic components in automobile assembly iv) Permanent join of two heavy metal structures.	08
	b	Explain how a tube can be manufactured from a sheet metal by a suitable welding technique.	08
		OR	
4	a	Explain with a neat sketch the principle of resistance welding.	08
	b	What are the defects that are generally found in welding? Discuss their course and remedy.	08
5	a	Mention the basic requirements for chip formation	03
	b	Explain various types of chips produced during a metal cutting process.	05
	c	With the help of Merchants circle diagram deduce an expression for shear force.	08
		OR	
6	a	List different type of cutting tool materials and explain any two, its composition and usage.	08
	b	If in turning of a steel rod by a given cutting tool at a given machining condition (feed and depth of cut) under a given environment (cutting fluid application) the tool life decreases from $80\,min$ to $20\,min$ due to increase in cutting velocity V from $60\,m/min$ to $120\,m/min$. Then at what cutting velocity the life of that tool under the same condition and environment will be $40\,min$?	08
7	a	Describe various parts of Capstan and Turret lathes.	08
	b	A drilling operation is to be performed with a $12.7\,mm$ diameter twist drill in a block of magnesium alloy. The hole is a blind hole at a depth of $60\,mm$. The cutting speed V is $25\,m/min$ and feed $0.3\,mm/rev$. Determine the: i) Drill time ii) Material removal rate iii) Power, if the average unit power for magnesium is $0.5\,\mathrm{W.S}/mm^3$.	08
8	a	A $20 \times 5\,cm$ cast Iron is to be faced on a milling machine with a cutter having a diameter of $10\,cm$ and 16 tooth for the cutting speed and feed are $50\,m/min$ and $5\,cm/min$ respectively. Determine the milling time, rpm and feed / tooth.	08
	b	Write short note on the following: i) Grinding Abrasives ii) Dressing of grinding wheel.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Fast Track ExaminationsJuly-19

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3.

PART-A

1	1.1	_____ pattern is used for generating axi-symmetrical and prismatic shapes.	01
	1.2	Balanced core is also called a _____ core.	01
	1.3	_____ casting uses Wax or similar sacrificial pattern with a ceramic mold.	01
	1.4	The helical grooves in the body of the drill is called as _____.	01
	1.5	Spot welding : cylindrical electrode::seam welding:_____.	01
	1.6	The junction between the base metal and the weld face is known as _____.	01
	1.7	The _____ acts as a reservoir for molten metal.	01
	1.8	Drilling is an example of _____ cutting.	01
	1.9	When the cutting edge is dull, _____ type of chip is formed during machining.	01
	1.10	The milling cutter is rotated in the same direction as the work piece is called _____ milling.	01
	1.11	A mild steel rod has a length of 100mm and tapered portion of length of 60mm, large and small diameter of taper is 70mm and 50mm respectively. Find the angle to which compound rest should be set?	02
	1.12	Why chaplets are used in molding process?	02
	1.13	Distinguish between upset and flash welding.	02
	1.14	A drill diameter 30mm makes a hole on a steel part at a cutting speed of 35m/min. Find out the spindle speed.	02
	1.15	Define indexing.	02

PART-B

2	a	List and explain the main constituents of molding sands.	08
	b	What do understand by gating system? What are the basic features of good gating system?	08

3	a	Sate the difference between hot working and cold working processes.	08
	b	Indicate by means of flow diagram the different stages in the manufacturing of 50mm diameter rod from a steel ingot. (supplement your answer with appropriate sketches).	08
		OR	
4	a	Explain the stages required for obtaining the final shape of the crank shaft by using drop forging process.	10
	b	What are the common defects that occur in welding?	06

5	a	Derive an expression for the shear angle in orthogonal cutting, in terms of rake angle and chip thickness ratio.	10
	b	Define cutting fluid. State the functions of a cutting fluid.	06
		OR	
6	a	In an orthogonal turning operation of a mild steel rod of $55mm$diameter, cutting speed was $25m$/min, rake angle of tool $30degrees,$feed rate $0.12mm/rev$, tangential force $2900N$, feed force $1200N,$length of continuous chip in one revolution$95mm$, determine coefficient of friction, shear plane angle, velocity of chip along tool face and chip thickness.	10
	b	Differentiate between crater wear and flank wear.	06

7	a	State different types of drilling operations and explain any two with neat sketches.	06
	b	A hole with 50mm diameter and 60mm depth is to be drilled in mild steel component. The cutting edge speed can be taken as 70m/min and the feed rate as 0.3 mm/revoln Calculate the machining time and the metal removal rate.	10

8	a	List and explain the different indexing methods applied in practice.	10
	b	Explain briefly the different types of abrasives used for grinding.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is gated pattern? State its advantage.	02
	1.2	Define Refractoriness and permeability as related to moulding materials.	02
	1.3	State any two specific merits of cold working over hot working.	02
	1.4	Give a brief classification of various fabrication processes.	02
	1.5	Differentiate between orthogonal cutting and oblique cutting (Any two).	02
	1.6	What is machinability? State any two factors affecting machinability.	02
	1.7	Differentiate between Collet chuck and four jaw chuck used in lathe.	02
	1.8	A mild steel rod has a length of $100mm$ and tapered portion of length $-60mm$, large and small diameter of taper is $70mm$ and $50mm$ respectively. Find the angle to which compound rest should be set.	02
	1.9	Sketch up milling and down milling.	02
	1.10	What do you mean by shellac bond? What is the advantage of using this bonding material for grinding wheel?	02

PART-B

2	a	Sketch and explain the following pattern allowances: i) Draft ii) Distortion.	08
	b	Explain Caine's method of riser design.	08
3	a	List the different stages in drop forging process in production of a lever component. Explain each stage in brief with a neat sketch.	08
	b	Sketch and explain any four rolling stand arrangements.	08
		OR	

4	a	Sketch and explain the projection welding process. State any two advantages and disadvantages in using projection welding.	08
	b	Explain the following welding defects: i) Incomplete fusion ii) Slag inclusion iii) Cold cracking iv) Lamellar fearing.	08
5	a	Establish the relationship between shear plane angle , chip thickness ratio and tool rake angle in orthogonal cutting.	06
	b	The following data were recorded during orthogonal cutting using a carbide tool. Rake angle $= 25°$, cutting speed $= \frac{40m}{min}$, uncut chip thickness $= 0.2mm$, cuttingforce $= 420N$, thrust force $= 109N$, length of uncut chip$= 250mm$, chip length$= 125mm$, width of cut$= 1mm$. Determine shear strength, coefficient of friction and shear strain.	10
		OR	
6	a	Explain any two cutting tool materials.	04
	b	Explain the following cutting fluids: i) Straight cutting oils ii) Soluble oils	04
	c	Explain indirect tool life criteria.	08
7	a	Explain the constructional features of a turret lathe with neat sketch.	08
	b	Sketch the nomenclature of a twist drill. Explain any four parts.	08
8	a	What is indexing? Explain any two indexing methods with examples.	06
	b	Sketch and explain the following grinding processes. i) External centreless grinding ii) Internal centreless grinding.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Fast Track July - 18

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A replica of the object to be made by the casting process can be termed as _____.	01
	1.2	Opening through which the molten metal enters the mould cavity are called_____.	01
	1.3	State one advantage and one limitation of casting process.	02
	1.4	The provision made to support the core inside the mould cavity is known as _____.	01
	1.5	During solidification of molten metal the liquid shrinkages is taken care of by _____.	01
	1.6	Hot working is done above _____ temperature.	01
	1.7	In forging the process if increasing the cross-sectional area of the stock at the expense of its length is known as _____ operation.	01
	1.8	Define resistance welding.	02
	1.9	$VT^n = C$is known as _____ tool life equation.	01
	1.10	Differentiate between orthogonal and oblique cutting.	02
	1.11	List any two properties of cutting tool material.	02
	1.12	List any four work holding devices.	02
	1.13	Drilling is an example of _____ cutting.	01
	1.14	Identify any two nomenclature of plain milling cutter.	02

PART-B

2	a	Define manufacturing and briefly explain the classification of manufacturing process.	08
	b	Explain the various steps involved in making casting.	08
3	a	Differentiate between hot working and cold working.	04
	b	Sketch and explain i) Three high rolling mills. ii) Tandem rolling mills.	06
	c	Discuss the causes of any three forging defects.	06

		OR	
4	a	Explain the working principle of resistance welding, along with a schematic representation of the welding setup.	08
	b	With a neat sketch explain flash welding.	08
5	a	List the types of chips. Explain with the help of a neat sketch.	08
	b	Briefly describe different forms of tool wear.	08
		OR	
6	a	Explain any two types of tool materials.	04
	b	What are the desirable properties of cutting fluid?	04
	c	Define machinability. Explain the factors affecting machinability.	08
7	a	Sketch and explain various parts of capstan lathe.	08
	b	Classify the drills and with neat sketches explain any four operations of drilling.	08
8	a	Differentiate between upmilling and downmilling.	04
	b	Explain: i) Straddle milling. ii) Gang milling with neat sketches.	06
	c	Briefly explain types of abrasives used in grinding wheels.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

MANUFACTURING PROCESS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Enumerate the factors to be considered for selection of manufacturing process.	02
	1.2	State the example for distortion allowances.	02
	1.3	Give examples for mechanical working of metal.	02
	1.4	List any four defects in casting.	02
	1.5	State the effect of back rake angle on cutting tool and mention the types.	02
	1.6	A tool having 7,8,6,7,5,6,0.1 as designation (Signature) in *ASA* system, specify the angles and nodes radius.	02
	1.7	List two types of semiautomatic lathe.	02
	1.8	Find the rpm of the drill spindle when the cutting speed is 26 meters per minute, having a hole of 20 mm diameter.	02
	1.9	List the common work holding devices used on milling machines.	02
	1.10	What are the types of wheels that are mounted on way grinders?	02

PART-B

2	a	List and explain at least six desirable characteristics of cores.	06
	b	Calculate the optimum pouring time for a casting whose mass is 20 kg and having average section thickness of 15 mm. the materials of casting are grey cast iron and steel. Consider the fluidity of iron as 28 inches.	10
3	a	Explain with neat sketch, basic working principle of rolling process. State its applications in industry.	10
	b	Discuss the different types of forging operations with examples.	06
		OR	

4	a	State the principle of resistance welding. List the applications of resistance welding in an automobile industry.	10
	b	Analyze the causes and remedies for the following welding defects. i) Incomplete fusion ii) Splatter.	06
5	a	Explain the basic elements of machining with a neat sketch.	04
	b	State the assumptions made while constructing a Merchant diagram.	04
	c	Explain the three possible modes by which a cutting tool can fail in machining.	08
		OR	
6	a	Identify the different variables that affect the tool life.	06
	b	An experiment was conducted on a MS tube $200\,mm$ diameter and $3\,mm$ thick. An orthogonal cut was taken with a cutting speed of $80\,mm$ and $0.15\,mm$ per revolution, feed with a cutting tool having back rake angle of -10°. It was determined that cutting force $= 150\,kg$, feed force $= 40\,kg$, net horse power for cutting was $3\,hp$ and chip thickness was $0.25\,mm$. Calculate the shear strain and strain energy per unit volume.	10
7	a	Explain at least four parts of a geometry of twist drill with a neat sketch.	08
	b	A hole with $40\,mm$ diameter and $50\,mm$ depth is to be drilled in mild steel component. The cutting speed can be taken as $65\,m/min$ and the feed rate as $0.25\,mm/rev$. Calculate the machining time and the metal removal rate.	08
8	a	State clearly, the Plate No. and the indexing movements to be made using a Universal dividing head, to divide the periphery of a work into 69 equal divisions. Following indexing plates are available.	12
	b	Write a note on Dressing and Truing of a Grinding wheel.	04

Plate No.	15	16	17	18	19	20
Plate No.	21	23	27	29	31	33
Plate No.	37	38	41	43	47	49

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R. V. COLLEGE OF ENGINEERING

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III Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

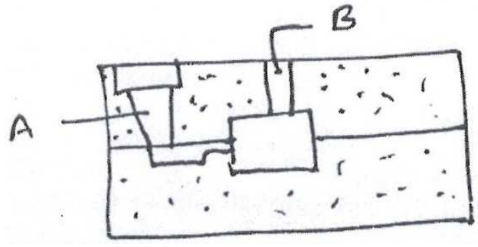
MANUFACTURING PROCESS – I

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

1	1.1	A _____ is a model of the desired casting, that can be used for forming an impression called _____.	02
	1.2	For each of the following applications, suggest most suitable primary manufacturing process (like welding, casting, forming, etc) a) Connecting Rod b) Automobile body c) Lathe bed d) A ceramic tipped cutting tool.	02
	1.3	A sample of Londha (Maharastra) sand of $50\,g$ is weighed after washing and drying and found to weigh $41.6\,g$. What would be the clay percentage in this sand sample?	01
	1.4	A sandcasting mould assembly is shown in figure 1.4. The elements marked A and B are _____ and _____ respectively.  Fig 1.4	01
	1.5	Gating ratio of a typical pressurized gating system is sprue: runner: ingate: : _____ : _____ : _____.	01
	1.6	The solidification time (t) of casting can be estimated in terms of its volume (v) and area (A) by using the _____ equation.	01
	1.7	The working of lead and tin even at room temperature is termed as hot working. Give reasons.	02
	1.8	What is meant by 'grain flow' in the case of forged or rolled components?	01
	1.9	What is extrusion ratio? The typical values of the extrusion ratio are _____.	01

	1.10	In case of resistance welding _____ electrodes are used for welding nonferrous materials such as aluminum and magnesium alloys.	01
	1.11	What is forge welding?	02
	1.12	State the applications of pressure die casting.	01
	1.13	Give two advantages of the investment casting.	02
	1.14	Differentiate between ingot, slab, bloom and billet.	02

PART-B

2	a	What do you understand by the term 'manufacturing'? Classify manufacturing processes.	04
	b	Explain the various properties of moulding sand.	08
	c	List the various pattern allowances and explain any one.	04
		OR	
3	a	Distinguish between green sand moulds and dry sand moulds from the point of view of: i) Process ii) Application.	04
	b	Identify the various types of patterns used in foundry and describe any three in detail with neat sketches.	12
4	a	Highlight the relevance of the facing sand towards the casting quality? Describe the facing sands normally used.	06
	b	Draw a neat sketch of a gating systems and explain all its elements.	10
		OR	
5	a	What types of binders are used in core making? Explain carbon dioxide moulding.	08
	b	Calculate the size of a cylindrical riser (height and diameter equal) necessary to feed steel slab casting $25 \times 25 \times 5cm$ with a side riser, casting poured horizontally into the mould.	08
6		Describe the construction and operation of a cupola furnace for melting cast iron with the help of a neat sketch.	16
		OR	
7	a	Highlight the categories under which the product design process of sand casting can be studied. Explain any two of them in detail.	10
	b	Enumerate the nature of heating and the other important features of the following furnaces. Substantiate your answer with the proper sketches. i) Reverberatory furnace ii) Crucible furnace iii) Induction furnace.	06

8	a	Give examples of a typical products manufactured by the following forging processes i) Machine forging ii) Drop forging.	04
	b	With neat sketches enumerate the working principle of forward and backward extrusion.	12
		OR	
9	a	What is the significance of recrystallization temperature in metal working?	06
	b	With a neat sketch explain the principle, construction and working of a seam welding process. Mention its advantages and limitations.	10
10		Explain the working principle of different types of centrifugal casting with necessary sketches. Mention the critical design issues in each of the type of centrifugal casting.	16
		OR	
11	a	Draw neat sketch and explain the working principle of Electron Beam Welding.	10
	b	Differentiate between thermochemical welding and radiant energy welding.	06

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R. V. COLLEGE OF ENGINEERING

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III Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

MANUFACTURING PROCESS – I

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Identify the methods producing components with Tubular shapes.	02
	1.2	Mention the two basic principles of fluid flow relevant to gating design.	02
	1.3	Identify the conditions under which Microporosity develops in castings.	02
	1.4	What is the function of core prints in core making?	02
	1.5	Chocolate forms are available in hollow shapes. What process is used to make these candies?	02
	1.6	Identify the metal forming process involved in making bolts and threads.	02
	1.7	Identify the factors which enable you to tell whether a part is forged or cast.	02
	1.8	What is meant by the choke area in Gating system?	02
	1.9	What are the two common ways in which slag inclusions can be eliminated/avoided in castings?	02
	1.10	Distinguish between Upset welding and Flash welding.	02

PART-B

2	a	Based in the property of the metal, provide an overview on the classification of production processes.	08
	b	Identify any four parts in a typical automobile manufactured by casting method. Also elaborate on the functions of those parts.	08
		OR	
3	a	Outline the production steps in a typical sand-casting operation with the aid of a flow chart.	08
	b	Identify major components and their functionalities of a sand mold with an appropriate schematic illustration.	08

4	a	Examine critically the influence of molten metal characteristics and casting parameters on the fluidity of the molten metal.	08
	b	Explain the three basic types of sand molds used for casting purposes.	08
		OR	
5	a	What is meant by Carbon dioxide moulding? Analyze the reasons for generally using carbon dioxide in core making.	08
	b	Calculate the size of a cylindrical riser (height and diameter equal) necessary to steel slab casting of $25 \times 25 \times 5cm$ with a side raiser, casting poured horizontally into the mould (note: Use Caine's equation)	08
6	a	With an appropriate illustration, examine how the melting process takes place in various zones of a cupola.	08
	b	Explain the working principle of Induction Furnace.	08
		OR	
7	a	Examine the varieties of defects occurring in the castings due to improper ramming of mould.	08
	b	Elaborate on the standardized nomenclature of casting defects designed by international committee of Foundry Technical Association.	08
8	a	With the aid of a schematic diagram, explain the principle involved in Rolling process.	08
	b	Distinguish between the principle involved in forward hot extrusion and backward hot extrusion. Bring clarity to your answer with appropriate line diagrams.	08
		OR	
9	a	Explain the principle involved in Resistance welding, along with a schematic representation of the welding setup.	08
	b	What is meant by Metal spinning? Provide an overview of three basic types of spinning techniques.	08
10	a	With suitable illustrations, discuss the steps involved in shell molding casting process.	08
	b	Analyze how the heat necessary for the joining process is obtained in Thermit welding process.	08
		OR	
11	a	Analyze how the atmosphere around the work piece affects the weld obtained in electron beam welding process.	08
	b	Critically review the three types of centrifugal casting processes.	08

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R. V. COLLEGE OF ENGINEERING

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III Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

MANUFACTURING PROCESS - I

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Casting is the only process where _____ is used.	01
	1.2	Why are pattern allowances provided?	02
	1.3	What are the casting defects?	02
	1.4	List out the applications of Rolling.	02
	1.5	What is hot working and cold working?	02
	1.6	Cold working is a process, which effects changes in _____.	01
	1.7	Define welding.	02
	1.8	What is die-casting?	02
	1.9	What are chaplets?	02
	1.10	List the types of moulding sand.	02
	1.11	What are additives?	02

PART-B

2	a	Explain why the understanding of manufacturing processes is essential for better products.	08
	b	List and explain the different steps involved in casting process.	08
		OR	
3	a	Define manufacturing process and briefly explain the classification of manufacturing process in brief.	08
	b	With a neat sketch explain split pattern and gated pattern.	08
4	a	List and explain the desirable properties of moulding sand.	08
	b	With a neat sketch, explain jolt type moulding machine.	08
		OR	
5	a	Explain the desirable characteristics of core in sand casting.	08
	b	Explain any two elements involved in gating system with neat sketch.	08

6	a	With neat sketch, explain the operation of cupola furnace for melting operation.	08
	b	Why are the shrinkage cavities caused in castings? Discuss the methods used for eliminating them.	08
		OR	
7	a	Explain any four types of defects under the category of moulding material defects.	08
	b	What are the problems caused by bosses in sand castings? Explain the methods available to solve them.	08
8	a	With a neat sketch, explain the following: i) Forward extrusion ii) Backward extrusion.	08
	b	Explain the four types of forging methods which are generally used.	08
		OR	
9	a	With neat schematic, explain the working principle of Resistance Welding.	08
	b	Explain any four types of welding defects.	08
10	a	With neat sketch, explain briefly the steps involved in a shell moulding process.	08
	b	List out the advantages and limitations of permanent mould casting.	08
		OR	
11	a	Write a short note on the following: i) Thermit welding ii) Forge welding.	10
	b	What are the advantages claimed of the friction welding along with its applications.	06

Digital Metrology / Metrology and Measurements

III Semester SEE Question Papers

Course codes: IM235AI, 21IM52, 18IM34, 16IM35

- Feb/Mar 2026 (IM235AI)
- Feb/Mar 2025 (IM235AI)
- Apr 2024 (IM235AI)
- Model Question Paper (2021 Scheme) (21IM52)
- Model Question Paper (2022 Scheme) (IM235AI)
- Fast Track Jan 2023 (18IM34)
- Mar 2021 (18IM34)
- Fast Track Oct 2020 (16IM35)
- Fast Track Oct 2020 (18IM34)
- Nov/Dec 2019 (16IM35)
- Nov/Dec 2019 (18IM34)
- Fast Track Jul 2019 (16IM35)
- Nov/Dec 2018 (16IM35)
- Fast Track Jul 2018 (16IM35)
- Nov/Dec 2017 (16IM35)

Question Paper ID: 853772

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM235AI

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Digital Metrology

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	state any two functions of a variable manipulation element.	02	1	1
1.2	What is meant by data presentation element?	02	1	1
1.3	State the principle of eddy current proximity sensor.	02	2	2
1.4	What is Hall voltage?	02	2	1
1.5	Why is it impossible to produce a part to exact size?	02	3	1
1.6	What is tolerance accumulation?	02	4	2
1.7	State the principle of autocollimator.	02	3	1
1.8	In fit and tolerance systems, what do the symbols H and h indicate?	02	4	1
1.9	Why is diffraction pattern technique suitable only for small diameter measurements?	02	4	2
1.10	Why are lasers preferred for alignment testing of machine tools?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Describe briefly the three stages of a generalized measurement system with a neat block diagram. Illustrate how the concept of a generalized measurement system is applied in pressure monitoring using a Bourdon-tube-based gauge on an automated production line.	08	1	4
2b	Explain resistive transducers with working principle and applications	08	1	4
3a	Explain the role of sensors in modern manufacturing systems	08	2	2
3b	Justify strain gauge load cell for 5 MN force measurement.	08	2	3
	OR			

4a	Explain construction and working of Flash ADC with example	08	2	2
4b	Explain principle and working of strain gauge using Wheatstone bridge.	08	2	3
5a	What are basic dimensions and toleranced dimensions?	08	4	1
5b	Determine the actual dimensions to be provided for a shaft and hole of 50 mm size for H7g6 type of fit. Given: i)Size 50 mm falls in the diameter step 30 mm – 50 mm ii) Tolerance values: IT7 = 16i, IT6 = 10i iii) Fundamental deviation: For shaft 'g' = $-2.5D^{0.34}$ Calculate the maximum and minimum clearance, type of fit, and represent graphically.	08	4	3
	OR			
6a	With the help of neat sketches, describe how different types of fits are obtained in each system.	08	4	3
6b	Determine the limits and fit for a 90 mm nominal size using F7h6 fit. Given: • Diameter step: 80–120 mm • IT7 = 16i, IT6 = 10i • Fundamental deviation: F hole = $5.5D^{0.41}$ Also determine its clearances	08	4	3
7a	With suitable sketches, explain how flatness error is determined using fringe count.	08	3	2
7b	Explain different elements of gear tooth geometry with neat sketches.	08	3	2
	OR			
8a	Explain how pitch of a screw thread is measured using Tool Maker's Microscope.	08	3	3
8b	Explain the principle, construction and working of an autocollimator.	08	3	2
9a	Explain the principle and working of a Coordinate Measuring Machine with a neat sketch.	08	4	3
9b	Analyze how CMMs help in reducing operator influence and improving productivity in inspection.	08	4	3
	OR			
10a	Analyze the advantages of laser interferometer over conventional optical interferometers.	08	4	3
10b	Describe laser viewers and explain their application in inspection of complex profiles.	08	4	3

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Regular/Supplementary Examinations Feb/Mar – 2025

Industrial Engineering and Management

DIGITAL METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Differentiate between Static and Dynamic Characteristics of a measurement system.	02	2	1
	1.2	State the importance of calibration in a measurement system.	02	1	1
	1.3	List any four key specifications of a sensor.	02	1	2
	1.4	Mention the main components of a data acquisition system.	02	1	2
	1.5	Define tolerance.	02	1	3
	1.6	Highlight any two functions of optical flats.	02	2	3
	1.7	What is meant by Fit?	02	1	3
	1.8	What is the Root Mean Square (RMS) value in surface finish measurement?	02	1	3
	1.9	Mention the type of CMM which is best suited for inspecting large and heavy workpieces.	02	2	4
	1.10	State the primary use of a laser in laser metrology.	02	2	4

PART-B

2	a	Explain the elements of a generalized measurement system with a neat sketch.	08	2	1
	b	Explain the importance of calibration in measurement systems.	08	2	1
3	a	Explain the working of a potentiometer as a displacement sensor.	08	3	2
	b	State and discuss the working principle of a tacho generator.	08	3	2
		OR			
4	a	Discuss the working of piezoelectric sensors for pressure measurement.	08	3	2
	b	Explain the elements of a data acquisition system and the role of A/D and D/A converters.	08	2	2
5	a	With suitable examples, differentiate between interchangeability and selective assembly.	08	3	3
	b	Differentiate between the hole basis system and the shaft basis system. Identify which is more commonly preferred, and provide the fits designation as per IS 919 – 1963.	08	3	3
		OR			
6	a	Explain the significance of tolerance in manufacturing. Discuss the various types of tolerances.	–	–	–
	b	What is Geometric Dimensioning and Tolerancing (GD&T)? And elaborate on the concept of position tolerance.	–	–	–

7	a	Explain the gear tooth Vernier method used to measure the thickness of gear teeth.	–	–	–
	b	Discuss the application of optical flats in interferometry for flatness measurement.	–	–	–
OR					
8	a	With the help of a neat sketch, explain the working and applications of a Tool Maker's Microscope.	–	–	–
	b	Elaborate on the use of Tomlinson Surface Meter in measuring surface finish.	–	–	–
9	a	Describe the constructional features and working principle of a Coordinate Measuring Machine.	–	–	–
	b	Explain the role of digital devices in computer-aided inspection with a Coordinate Measuring Machine.	–	–	–
OR					
10	a	Identify the different types of Coordinate Measuring Machine (CMMs) and their specific applications.	–	–	–
	b	Mention the principle and applications of the Michelson Interferometer in laser metrology.	–	–	–

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Examinations Apr – 2024

Industrial Engineering and Management

DIGITAL METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Discuss the differences between static and dynamic characteristics of a measurement system.	02	1	1
	1.2	List the Errors in measurements.	02	2	2
	1.3	Explain the functioning of a Piezoelectric Sensor and provide example for its applications.	02	2	2
	1.4	List the applications of a Capacitive Transducer in Proximity Sensing.	02	2	2
	1.5	What is meant by “Basic Hole” and “Basic Shaft”?	02	2	3
	1.6	How will you write the fit: Shaft ‘m’ of grade 5 and hole ‘H’ of grade 11 and the basic size of 40 mm?	02	1	3
	1.7	An optical flat can be employed to measure height differences in which range?	02	2	2
	1.8	Describe the working principle of an Autocollimator.	02	1	2
	1.9	Describe the working principle of a Michelson Interferometer.	02	2	4
	1.10	List four applications of Laser Metrology in Industry.	02	2	4

PART-B

2	a	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams.	08	3	1
	b	Describe briefly the working principle of a Capacitive Transducer and also explain the measurement of linear displacement in a surface using the change in the area of plates.	08	2	1
3	a	Discuss the specifications of sensors and their importance in metrology. Elucidate various types of sensors employed in measurement procedures.	08	3	2
	b	Explain the Velocity and motion sensors and their application in metrology for the detection of infrared radiation and temperature variations.	08	2	2
		OR			
4	a	Describe the factors influencing the selection of sensors for a particular application. How can one ensure the compatibility between sensors and data acquisition systems?	08	3	2
	b	Compare and contrast position sensors and proximity sensors, highlighting their working principles and applications.	08	3	2

5	a	In a limit system the following limits are specified to give a clearance fit between a shaft and a hole: Shaft 30 $^{-0.005}_{-0.018}$ mmϕ; Hole 30 $^{+0.020}_{-0.000}$ mmϕ. Determine: (i) Basic size (ii) Shaft and hole tolerances (iii) The shaft and hole limits (iv) The maximum and minimum clearance.	08	–	–
	b	Define Fit. Explain briefly the various types of fits with examples.	08	–	–
		OR			
6	a	Enumerate on the concept: (i) Interchangeability (ii) Hole basis systems.	–	–	–
	b	Design the 'General' type 'GO' and 'NOGO' gauges for the components having 50H8c8 fit. Given the following with usual notations: (i) Upper deviation for Shaft = (95 + 0.8D) microns (ii) 50 mm falls in the diameter step of 50 – 80 mm (iii) IT8 = 25i. Take wear allowance as 10% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	–	–	–
7	a	Enumerate on the working of a floating carriage micrometer and explain how the minor and effective diameter of a screw thread can be measured.	–	–	–
	b	With the help of a neat sketch briefly explain the method of determining the chordal thickness of a gear tooth using gear tooth Vernier Calliper.	–	–	–
		OR			
8	a	Explain the constant chord method for measuring the gear tooth thickness.	–	–	–
	b	Describe briefly the working principle of a Tool Makers Microscope and explain how the minor and effective diameter of a screw thread can be measured.	–	–	–
9	a	Enumerate on the different types of main structure of a Coordinate Measuring Machine with neat sketches, and list out the industrial applications of a CMM.	–	–	–
	b	Describe briefly the working of a Michelson Interferometer and list out its applications.	–	–	–
		OR			
10	a	Describe the role of CMM in Computer Aided Manufacturing and explain briefly the advantages of CMM.	–	–	–
	b	Explain the basic principle of LASER. List the various uses of LASER in metrology and explain any two in detail.	–	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
V Semester B.E. April – 2024 Examinations
Model Question Paper (2021 Scheme)
Industrial Engineering and Management
DIGITAL METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Differentiate between static and dynamic errors in the context of measurements.	02	1	1
	1.2	Give two examples each for resistive and capacitive transducers.	02	2	2
	1.3	Differentiate between A/D and D/A converters in the context of a data acquisition system.	02	2	2
	1.4	Give two applications of velocity and motion sensors.	02	2	2
	1.5	Differentiate hole and shaft basis system.	02	2	3
	1.6	List the important types of fits.	02	1	3
	1.7	Give the working principle of optical flats and Tool Makers microscope.	02	2	2
	1.8	Define best size wire.	02	1	2
	1.9	List any four applications of CMM.	02	2	4
	1.10	Differentiate between bridge and gantry type CMM.	02	2	4
		PART-B			
2	a	Discuss the components and functioning of the generalized measurement system, providing justification through an illustrative example, clear sketches, and block diagrams.	08	3	1
	b	Describe briefly the working principle of a Capacitive Transducer and also explain the measurement of linear displacement in a surface using the change in the area of plates.	08	2	1
3	a	Discuss the fundamental concept of sensors and their importance in metrology. Include examples to elucidate various types of sensors employed in measurement procedures.	08	3	2
	b	Explain the operational principles of pyroelectric sensors and their application in metrology for the detection of infrared radiation and temperature variations.	08	2	2
		OR			
4	a	Classify sensors based on their functionality into displacement, position, and proximity sensors. Discuss any one type, providing examples and applications in metrology.	08	3	2
	b	Explain how piezoelectric sensors are employed for fluid pressure measurements in metrology. Provide examples of their usage and advantages in pressure sensing applications.	08	3	2

5	a	Calculate the limits of tolerance and allowance for a 25 mm shaft and hole pair designated by H8d9 and sketch the fit. Given: 25 mm lies between 18 and 30, fundamental deviation of 'H' hole = 0, fundamental deviation of 'd' shaft = $-16D^{0.44}$, IT8 = 25i, IT9 = 40i.	08	4	4
	b	Discuss the concepts of (i) Interchangeability and (ii) Selective Assembly, highlighting their significance in engineering and manufacturing processes.	08	2	3
OR					
6	a	Define the concept of fit and provide a concise overview of the different types of fits, supplemented with illustrative examples.	08	2	3
	b	Calculate the relevant dimensions of 35H7f8 fit, dimension 35 falls in the steps of 30–50 mm. Fundamental deviation for f shaft is $-5.5D^{0.41}$, IT7 = 16i, IT8 = 25i.	08	4	4
7	a	Illustrate the terminologies utilized to denote surface roughness through a clear diagram, elucidating each term's significance.	08	3	2
	b	Illustrate the operational mechanism of a floating carriage micrometer using a clear diagram and provide a concise explanation of its functionality.	08	2	2
OR					
8	a	Enumerate and describe five frequently employed techniques for measuring surface roughness in engineering contexts.	08	3	2
	b	Illustrate the operational principles of a tool maker's microscope through a clear diagram, providing a brief overview of its functionality.	08	2	2
9	a	Illustrate and describe two variations of Coordinate Measuring Machines (CMMs) using sketches to elucidate their distinguishing features and operational methodologies.	08	2	4
	b	Describe briefly the working of a Michelson Interferometer and list out its applications.	08	2	3
OR					
10	a	Discuss the significance and applications of Coordinate Measuring Machines (CMMs) in various industries.	08	3	4
	b	Explain the fundamental principle underlying LASER technology. Enumerate the diverse applications of LASER in metrology and elaborate on two specific applications in depth.	08	2	3

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RV COLLEGE OF ENGINEERING®

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III Semester B.E. Examinations

MODEL QUESTION PAPER (2022 SCHEME)

Industrial Engineering and Management

DIGITAL METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	List any two desirable static and dynamic characteristics of measuring instruments.	02	2	1
	1.2	Mention the working principle of an Inductive sensor.	02	1	2
	1.3	Categorize the two applications of a Tactile sensor and analyze the principles underlying the use of tactile sensors in each application.	02	3	2
	1.4	Identify the two techniques used for converting Digital Signals into Analog signals.	02	2	2
	1.5	Compare and contrast the Hole basis system and shaft basis system in terms of their application in geometric dimensioning and tolerancing (GD&T).	02	3	1
	1.6	Define Maximum clearance and Fundamental deviation.	02	1	1
	1.7	Evaluate the importance of optical flats in measuring surface finish, considering their role in precision measurement techniques.	02	3	3
	1.8	Evaluate the significance of determining a Ten-point Height Average Value in surface metrology.	02	3	3
	1.9	Classify the primary components of a Coordinate Measuring Machine (CMM) according to their functions.	02	2	4
	1.10	Classify the different geometrical checks performed on machine tools using LASER technology and analyze the principles behind each check.	02	3	4

PART-B

2	a	Describe briefly the three stages of a generalized measurement system with block diagram. Provide two specific examples of measurement systems and identify how these examples demonstrate the three stages of measurement according to the categorized stages mentioned.	08	3	1
	b	Illustrate the operational principles of two distinct resistive transducers, namely the strain gauge and the thermistor, and compare their applications in real-world scenarios.	08	3	2
3	a	Explain the functioning of tactile sensors detailing its basic operation and how it detects contact with surfaces. Evaluate the advantages and limitations of this sensor in different scenarios, such as robotics, medical devices, touch-sensitive electronics, and industrial monitoring.	08	4	2

	b	Describe the operational principles of Potentiometer Sensors based on resistance change. Explain how these sensors function to measure position, displacement, or angle by varying resistance. Provide examples of their applications in volume control knobs, servo motors, joysticks, and industrial position feedback systems.	08	4	2
		OR			
4	a	Describe the operational principles of a tachogenerator and its role in measuring rotational speed. Evaluate the role of tachogenerators in measuring and monitoring rotational speed in automotive systems and automation/robotics.	08	4	2
	b	Enumerate briefly on the classification of sensors based on different criteria and explain the working principle of Hall effect sensors. Discuss how Hall effect sensors generate a voltage difference in response to a magnetic field and how this principle is applied in various applications such as proximity sensing, position sensing, and speed measurement.	08	4	2
5	a	Compare and contrast the characteristics, advantages, and applications of different types of fits in engineering assemblies. Also evaluate their suitability for specific assembly requirements.	08	4	1
	b	Determine the actual dimension to be provided for a shaft and hole of 90 mm size for h8E9 type of fit. (i) Size 90 mm falls in the diameter step of 80 mm–100 mm. (ii) Value of tolerances for IT8 and IT9 are 25i and 40i. (iii) Value of fundamental deviation for 'e' type shaft is $11D^{0.41}$. Also mention the type of fit. Represent the dimensions graphically.	08	4	1
		OR			
6	a	Explain briefly the Hole basis system and shaft basis system in geometric dimensioning and tolerancing (GD&T). Analyze the principles behind each system, evaluate their advantages and limitations, and assess their effectiveness in ensuring proper fit and functionality in mechanical components, such as engine parts or aerospace components.	08	4	1
	b	Enumerate on the concepts of interchangeability and selective assembly in manufacturing processes. How do these principles contribute to efficient production and product quality? Provide examples to illustrate your explanation.	08	3	1
7	a	Apply the principle of interference to demonstrate the functioning of an autocollimator? Furthermore, discuss how autocollimators are practically utilized in manufacturing settings to assess the straightness of components.	08	4	3
	b	Define Crest, Flank and Pitch and schematically represent on a screw thread. Explain how the major diameter of a screw thread can be measured using Bench micrometer.	08	3	3
		OR			
8	a	Discuss the Tomlinson Surface Meter, its construction, principles of operation, and applications in surface finish measurement. Explain how this specialized instrument contributes to accurate surface finish analysis in engineering and manufacturing processes, highlighting its advantages and limitations.	08	4	3
	b	Define Pitch Circle diameter, addendum and dedendum and represent on a gear profile. Assess the effectiveness of using vernier calipers for gear tooth thickness measurement compared to other measurement methods, and discuss potential sources of error in gear measurements.	08	3	3

9	a	Explain briefly the 4 main elements of a Coordinate Measuring Machine (CMM). Evaluate how each component contributes to the accurate measurement and inspection capabilities of the CMM in various industrial applications.	08	3	4
	b	How does Coordinate Measuring Machine (CMM) integration with Computer-Aided Manufacturing (CAM) enhance quality control processes in manufacturing? Discuss with examples.	08	3	4
OR					
10	a	Describe the key components of a laser interferometry system in detail, focusing on its different elements. Explain the function of each component and how they collectively enable precise distance measurements and displacement sensing through interference phenomena.	08	3	4
	b	Analyze the components and functions of a control system integrated with a data conversion system. Provide a diagram illustrating signal flow and conversion between analog and digital formats.	08	3	4

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Fast Track Examinations January-2023

METROLOGY AND MEASUREMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	_____ denotes the smallest change in the measured variable to which the instrument responds.	01
	1.2	A shaft has a dimension $35mm$, upper limit -0.009 and lower limit -0.025 respectively. The respective values of fundamental deviation and tolerances are _____ and _____.	01
	1.3	What is the principle of operation of LVDT?	01
	1.4	Expressing a dimension as $25.3\ ^{\wedge}\ (\pm 0.05)mm$ is the case of _____ tolerance.	01
	1.5	Johansson mickrokator is a type of _____ comparator.	01
	1.6	_____ fit joints are provided for assembling bush bearing in housing.	01
	1.7	Piezo-electric transducers are used to measure _____.	01
	1.8	Horizontal boring mill type CMM is most suited for _____ workpiecese.	01
	1.9	Dynamic measurements are done with the help of _____ transducer.	01
	1.10	Identify the signal conditioner in a bourdon tube pressure gage.	01
	1.11	List the use of beam benders in LASER interferometer.	01
	1.12	Thermocouples work on the principle of _____ effect.	01
	1.13	List any two devices used for measurement of roundness of a specimen.	01
	1.14	List any two features that the CMM software should include.	01
	1.15	The distance between the crest and root of the thread is called _____.	01
	1.16	List any two applications of hall effect sensors.	01
	1.17	List the different types of gauges according to the form of testing surface.	01
	1.18	_____ is a new standard in communication used in CAM.	01
	1.19	_____type of construction of CMM is suited for very large components.	01
	1.20	Mention the principle involved in bimetallic strip.	01

PART-B

2	a	Describe briefly the different static and dynamic characteristics of measurement systems.	08
	b	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams.	08

3	a	Design the 'general' type 'GO' and 'NOGO' gauges for the components having $50H_8C_8$ fit. Given the following with usual notations. i. Upper deviation for shaft = $(95 + 0.8D)microns$ ii. $50mm$ falls in the diameter step of $50 - 80mm$ iii. $IT8 = 25i$. Take wear allowance as 10% of the gauge tolerance. Also mention the type of fit. Represent the dimensions graphically.	10
	b	Differentiate between hole basis system and shaft basis system. (Any three.)	06
		OR	
4	a	Briefly explain the Taylor's principle of gauge design with neat sketches.	08
	b	With examples, list and explain the various types of fits.	08
5	a	Explain briefly the working principle of a sine bar and list out the uses of it.	10
	b	Describe briefly the basic elements of a comparator. What are its characteristics?	06
		OR	
6	a	'LVDT works on the principle of mutual induction'. Describe in detail with neat sketches.	10
	b	Describe briefly the different errors of a screw thread and the reasons for such errors.	06
7	a	List the various uses of LASER in metrology and briefly explain any three.	08
	b	With the help of neat sketches, enumerate the main elements of a Coordinate Measuring Machine.	08
8	a	State and explain the two laws of thermocouple.	08
	b	With the help of a neat sketch, briefly explain the working of bimetallic strip thermometer.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations March-2021

Industrial Engineering and Management

METROLOGY AND MEASUREMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

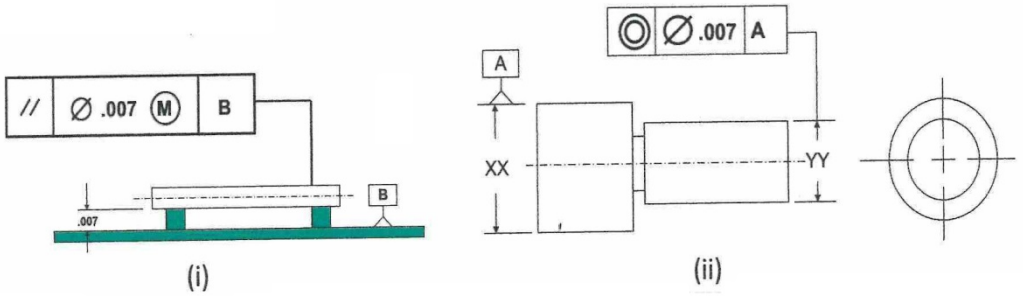
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The cross section of international prototype meter is _____.	01
	1.2	Parallax effect is a disadvantage of _____ standards.	01
	1.3	An $M\,112$ set of slip gauges will have _____ number of gage blocks in steps of $0.5mm$.	01
	1.4	The time lag between the application of input signal and output measurement is known as _____.	01
	1.5	The intentional difference between hole dimension and shaft dimension is known as _____.	01
	1.6	In the calculation of standard tolerance unit, the value $'D'$ is calculated as _____ of the step size.	01
	1.7	For a hole designated by $'H'$, the fundamental deviation is _____.	01
	1.8	The limit gauge used to check both cylindrical and non-cylindrical work is known as _____.	01
	1.9	The type of mechanical comparator which uses principle of 'Button spinning in an in loop' is _____.	01
	1.10	The displacement transducer which uses three symmetrically placed AC coils is known as _____.	01
	1.11	The type of optical flat which has only one flat surface is called _____.	01
	1.12	An optical device which is a combined 'telescope at infinity' and a 'collimating lens' is known as _____.	01
	1.13	The ratio of 'information' delivered by a pickup to the 'information' received by the pickup is known as _____.	01
	1.14	In a floating carriage micrometer, V-shaped prisms are used for measurement of _____ of screw thread.	01
	1.15	The ability of an instrument to give consistent or repetitive readings is known as _____.	01
	1.16	The type of transducer which does not require an external power source is known as _____.	01
	1.17	The acronym 'LASER' stands for _____.	01
	1.18	For measurement of torque, the device generally used is known as _____.	01

	1.19	The working principle of a thermocouple is based on _____.	01
	1.20	The % composition of Constantan used in thermocouple is _____.	01

PART-B

2	a	Define static and dynamic characteristics of measuring instruments. List the desirable and undesirable characteristics under each type.	06
	b	What is error in measurements? List the types of error and sources of error.	06
	c	Explain 'Hall effect' and mention the applications of hall effect transducers.	04
3	a	What are limit gauges? Mention the types of limit gauges. Sketch and explain the Taylor's principle of design of limit gauges.	08
	b	Determine the actual dimensions to be provided for a shaft and hole $90mm$ size for H_8e_9 type clearance fit. Size $90mm$ falls in the diameter step of $80-100mm$. Value of standard tolerance unit $i = 0.45\sqrt[3]{D} + 0.001D$. The values of tolerances for $IT8$ & $IT9$ grades are $25i$ & $40i$ respectively. Value of fundamental deviation for $'e'$ type shaft is $-11D^{0.41}$	08
		OR	
4	a	Write a note on i. Interchangeability ii. Selective assembly iii. Hole basis system iv. Shaft basis system v. Geometrical tolerances.	10
	b	Interpret the GD and T feature control frames shown in Fig 4b (i) and Gig 4b (ii) below.  Fig 4b	06
5	a	Discuss the principle of interferometry as applied to optical flats for flatness testing.	08
	b	With a neat sketch, explain the working of a Johansson's Mikrokator.	08
		OR	
6	a	Illustrate with a sketch, the radius measurement of a semicircular bearing cap.	06
	b	Sketch and explain tooth thickness measurement using gear tooth Vernier.	10

7	a	List the important elements of CMM. Also mention the advantages of CMM.	08
	b	What are MEMS? List the types of MEMS sensors and mention their applications.	08
8	a	With a schematic diagram, explain the principle of working of pressure thermometer.	08
	b	What are thermistors? How do they differ from RTD's? Name the types of thermistors and their applications.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

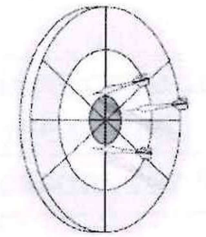
MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Identify the signal conditioner in a bourdon tube pressure gage.	01
	1.2	What does the illustration represent with respect to accuracy and precision in Fig 1.2?  Fig 1.2	01
	1.3	_____ is the minimum value of the input quantity to which an instrument responds.	01
	1.4	A voltmeter might show a reading of $1\,volt$ even when it is disconnected from any electromagnetic influence. Which type of error does it signify?	01
	1.5	With $M87$ slip gauges, build the dimensions of $45.985mm$.	01
	1.6	List different types of gauges according to the form of testing surface.	01
	1.7	$LVDT$ works on the principle of _____.	01
	1.8	The optical flat works on the principle of _____.	01
	1.9	The distance between the crest and root of the thread is called _____	01
	1.10	The error that repeats itself at regular intervals along the thread is called _____.	01
	1.11	The radial distance of a gear tooth from the pitch circle to the bottom of the tooth is called _____.	01
	1.12	List any two devices used for measurement of roundness of a specimen.	01
	1.13	_____ is provided to activate the drive for manual measurement in CMM.	01
	1.14	List any two features that the CMM software should include.	01

	1.15	The replacement of mirrors by cube-corner reflector is to _____.	01
	1.16	List the use of Beam Benders in *LASER* Interferometer.	01
	1.17	Sketch an Unequal arm balance.	02
	1.18	Sketch a basic thermocouple circuit and state its working principle.	02

PART-B

2	a	Explain briefly the generalized measurement system with a block diagram and give examples for each stage. Consider a bourdon tube as an example and identify the different stages of measurement.	08
	b	Describe briefly the different static and dynamic characteristics of Measurement systems.	08
3	a	Describe with neat sketches, various line standards. Also enumerate on the characteristics of the line standards.	10
	b	Describe briefly the wringing phenomenon in slip gauges with neat sketches. Build the dimensions of $35.4875mm$ using $M112$ set of slip gauges.	06
		OR	
4	a	'*LVDT* works on the principle of Mutual Induction'. Describe the same with neat sketch.	08
	b	Explain briefly the working principle of a sine bar and with neat sketches, describe the uses of it.	08
5	a	Describe briefly the working principle of a Tool Makers Microscope and explain how the minor and effective diameter of a screw thread can be measured.	10
	b	Describe briefly different errors of a screw thread and the reasons for such errors.	06
		OR	
6	a	Explain the constant cord method for measuring the gear tooth thickness.	10
	b	Describe briefly the different errors of a Gear and the reasons for such errors.	06
7	a	Describe the role of *CMM* in Computer Aided Manufacturing and explain briefly the advantages of *CMM*.	08
	b	Explain the basic principle of *LASER*. List various uses of *LASER* in metrology and explain any two in detail.	08
8	a	Describe with neat sketch the working of Hydraulic dynamometer.	08
	b	With neat sketches, explain briefly the working of Bimetallic strip thermometer. List the applications of it.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

METROLOGY AND MEASUREMENTS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	The first stage in a generalized measurement system is _____.	01
	1.2	A difference in readings when the value of the measured quantity is approached from higher value or from a lower value is due to _____.	01
	1.3	The time lag between the application of input signal and output measurement is known as _____.	01
	1.4	The type of gauge used to check the limits of size of a shaft is _____.	01
	1.5	Development of a transverse electric field in a solid material when it carries an electric current and is placed in a magnetic field that is perpendicular to the current is known as _____.	01
	1.6	Specifications of form, profile, orientation, location and run out are known as _____.	01
	1.7	An optical device which is a combined 'telescope focused at infinity; and a 'collimator' is known as _____.	01
	1.8	The standard tolerance unit $'i'$ is calculated using the formula _____.	01
	1.9	When one part in an assembly can be substituted for a similar part which has been made to the same drawing, it is known as _____.	01
	1.10	The type of mechanical comparator which uses a perforated and twisted metal strip is _____.	01
	1.11	A device that measures the geometry of physical objects by sensing discrete points on the surface of the object with a probe is known as _____.	01
	1.12	Base tangent method is used for measurement of _____.	01
	1.13	The minimum size of a hole and max size of a shaft correspond to _____ material condition.	01
	1.14	The phenomenon of development of an emf between the junctions of two dissimilar metals is known as _____.	01
	1.15	The fringe pattern obtained when an optical flat is placed on a convex surface is _____.	01

	1.16	The comparator which is used to check roundness and taper of cylinder bores is _____.	01
	1.17	The electrical comparator which can be used as a displacement transducer is _____.	01
	1.18	The intentional difference between hole dimension and shaft dimension is known as _____.	01
	1.19	The acronym $MEMS$ stands for _____.	01
	1.20	A temperature-sensitive electrical contact consisting of two bands of different metals joined lengthwise is known as _____.	01

PART-B

2	a	How are the performance characteristics of a measuring system classified? Define each type and mention atleast four examples for each type of characteristics.	08
	b	Briefly explain: i) DAQ and its components ii) Smart sensors.	08
3	a	Determine the term 'fit'. Explain with help of neat sketches, the classification of fits and their applications.	08
	b	Determine the tolerance on the hole and the shaft for a precision running fit designated by $50\,H_7g_6$, given; $50mm$ lies between $30-50mm$, $i\ (in\ microns) = 0.45\ (D)^{1/3} + 0.001D$, Fundamental deviation for $'H'\ hole = 0$, Fundamental deviation for g shaft $= -2.5D^{0.34}, IT7 = 16i$ and $IT6 = 10i$. State the actual maximum and minimum sizes of the hole and shaft and maximum and minimum clearances.	08
		OR	
4	a	Write a note on: i) Interchangeability ii) Selective assembly iii) Hole basis system iv) Shaft basis system.	08
	b	What are limit gauges? Give their classification. With a suitable example, explain Taylor's principle of design of limit gauges.	08
5	a	Explain with neat sketch, the construction of $LVDT$ and its characteristic curves.	08
	b	Define a comparator. How does it differ from a measuring instrument? With a neat sketch, explain Solex pneumatic gauge.	08
		OR	
6	a	Discuss the measurement of surface roughness using Tomlinson surface meter.	08
	b	List the various methods used for gear tooth thickness measurement. Explain the method used in David Brown's tangent comparator.	08

7	a	Discuss the different types of arrangement of main structure of CMM with neat sketches.	08
	b	Explain the different types of $MEMS$ sensors and their applications.	08
8	a	Mention the devices used for measurement of force. Sketch and explain a device used for force measurement which is in the form of a circular ring.	08
	b	What is thermoelectric effect? Explain the law of intermediate metal and law of intermediate temperatures as applied to temperature measurement by thermocouple.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Distinguish between Accuracy and Precision with a suitable example.	02
	1.2	Identify the cause for loading effect in measurement system.	01
	1.3	_____ is the degree of closeness with which a given value may be repeatedly measured.	01
	1.4	Define Threshold.	01
	1.5	The supports for the bars are determined by _____.	01
	1.6	With $M87$ slip gauges, build the diemsnions of $45.985mm$.	01
	1.7	Identify the gear mechanism by which a dial indicator works.	01
	1.8	How do two slip gauges wrung adhere to each other perfectly?	01
	1.9	The optical flat works on the principle of _____.	01
	1.10	Differentiate between Periodic error and progressive error in screw threads.	01
	1.11	Define module in gear.	01
	1.12	Define Run Out in a gear error.	01
	1.13	_____ is a new standard in communication used in CAM.	01
	1.14	_____ type of construction of CMM is suited for very large components.	01
	1.15	The replacement of mirrors by cube-corner reflector is to _____.	01
	1.16	List the use of beam benders in $LASER$ Interferometer.	01
	1.17	Mention the principle involved in bimetallic strip.	01
	1.18	Sketch a basic thermocouple circuit and state its working principle.	02

PART-B

2	a	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams.	08
	b	Describe briefly the different static and dynamic characteristics of measurement systems.	08
3	a	Describe with neat sketches the various line standards. Also enumerate on the characteristics of the line standards.	06

	b	Design the 'General' type $'GO'$ and $'NOGO'$ gauges for the components having $50H_8c_8$ fit. Given the following with usual notations. i) Upper deviation for shaft= $(95 + 0.8D)microns$ ii) $50mm$ falls in the diameter step of $50 - 80mm$ iii) $IT8 = 25i$ Take wear allowance as 10% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	10
		OR	
4	a	Describe briefly the basic elements of a comparator? What are its characteristics? With a neat sketch, explain the working principle of Johnson's mikrokator comparator.	08
	b	Explain briefly the working principle of a sine bar and list out the uses of it.	08
5	a	Describe briefly the working of a floating carriage micrometer and explain how the minor and effective diameter of a screw thread can be measured.	10
	b	Describe briefly the different errors of a screw thread and the reasons for such errors.	06
		OR	
6	a	Explain the constant cord method for measuring the gear tooth thickness.	10
	b	Describe briefly the different errors of a Gear and the reasons for such errors.	06
7	a	Enumerate on the main elements of a Coordinate Measuring Machine with neat sketches and list out the industrial applications of a CMM.	10
	b	Describe the role of CMM in Computer aided Manufacturing.	06
8	a	Describe with then neat sketch, the working of Hydraulic dynamometer.	08
	b	With neat sketches, explain briefly the working of Pressure thermometer. List out the applications of it.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

METROLOGY AND MEASUREMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Apply the knowledge of accuracy and precision to comment on the statement given below. "Clocks in a watch shop are set at 8:30 but not working".	01
	1.2	A voltmeter might show a reading of 1 volt even when it is disconnected from any electromagnetic influence – Which type of error does it signifies.	01
	1.3	Piezoelectric Transducer works with the principle of _____.	01
	1.4	List any two applications of Hall effect sensors	01
	1.5	Steel rings in wooden bullock carts is an example of _____ Fit	01
	1.6	List the different type of gauges according to the form of testing surface.	01
	1.7	If the gauge is exact copy of mating part of the component to be checked, it is called _____ gauges.	01
	1.8	If the components are measured and sorted into groups by dimension it is called _____.	01
	1.9	The _____ comparator utilizes the principle of button spinning on a loop of a string.	01
	1.10	The optical flat works on the principle of _____.	01
	1.11	Differentiate between periodic error and progressive error in screw threads.	02
	1.12	_____ is a new standard in communication used in CAM.	01
	1.13	_____ type of construction of CMM is suited for very large components.	01
	1.14	The replacement of mirrors by cube corner reflector is to_____.	01
	1.15	List the use of Beam Benders in $LASER$ Interferometer.	01
	1.16	Mention the principle involved in bimetallic strip.	02
	1.17	Sketch the basic thermocouple circuit and state its working principle.	02

PART-B

2	a	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams.	08
	b	Describe briefly the working principle of a Capacitive Transducer and also explain the measurement of linear displacement using the change in the Area of Plates.	08
3	a	Define Fit. Explain briefly the various types of fits with examples.	10
	b	Enumerate on the concept: i) Interchangeability ii) Hole basis systems	06
		OR	
4	a	Explain briefly the Taylor's principle of gauge design with neat sketches.	08
	b	Design the 'General' type 'GO' and '$NOGO$' gauges for the components having $50H_8C_8$ fit. Given the following with usual notations. i) $Upper\ deviation\ for\ shaft\ =\ (95 + 0.8D) microns$ ii) $50mm\ falls\ in\ the\ diameter\ step\ of\ 50 - 80mm$ iii) $IT8 = 25i$. Take wear allowance as 10% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	08
5	a	Describe briefly the basic elements of a comparator? List any four of its characteristics? With a neat sketch explain the working principle of Johnson's mikrokator comparator	08
	b	Explain briefly the working principle of a sine bar and list out the uses of it.	08
		OR	
6	a	Enumerate on the working of a floating carriage micrometer and explain how the minor and effective diameter of a screw thread can be measured.	10
	b	Explain the constant cord method for measuring the gear tooth thickness	06
7	a	Enumerate on the different types of Main structure of a Coordinate Measuring Machine with neat sketches, and list out the industrial Applications of a CMM.	10
	b	Describe briefly the working of a Michelson Interferometer and list out its applications.	06
8	a	Describe with neat sketch the working of Hydraulic dynamometer.	08
	b	With neat sketches explain briefly the working of pressure thermometer. List the applications of it.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw neat sketches wherever necessary.

PART-A

1	1.1	Apply the knowledge of accuracy and precision to comment on the statement given below. "clocks in a watch shop are set at 8:30 but not working".	01
	1.2	A voltmeter might show a reading of 1 volt even when it is disconnected from any electromagnetic influence. Which type of error does it signify?	01
	1.3	_____ denotes the smallest change in the measures variable to which the instrument responds.	01
	1.4	Identify the signal conditioner in a bourdon tube pressure gage.	01
	1.5	With $M87$ slip gauges, build the dimensions of $45.985mm$.	01
	1.6	The supports for the bars are determined by _____.	01
	1.7	The _____ comparator utilizes the principle of button spinning on a loop of a string.	01
	1.8	The optical flat works on the principle of _____.	01
	1.9	Differentiate between periodic error and progressiveerror in screw threads.	02
	1.10	Define diametral pitch in a gear.	01
	1.11	Define Run Out in gear error.	01
	1.12	_____ is a new standard in communication used in CAM.	01
	1.13	_____ type of construction of CMM is suited for very large components.	01
	1.14	The replacement of mirrors by cube-corner reflector is to _____.	01
	1.15	List the use of beam benders in $LASER$ interferometer.	01
	1.16	Mention the principle involved in bimetallic strip.	02
	1.17	Sketch a basic thermocouple circuit and state its working principle .	02

PART-B

2	a	Enumerate on the generalized measurement system. Justify the answer with an example, neat sketches and block diagrams	08
	b	Describe briefly the different static and dynamic characteristics of measurement systems	08
3	a	Describe with neatsketches the various line standards. Also enumerate on the characteristic of the line standards.	06
	b	Design the 'General' type 'GO and '$NOGO$' gauges for the components having $50H_8C_8$ fit. Given the following with usual notations i) Upper deviation for shaft = $(95 + 0.8D)$ microns ii) $50mm$ falls in the diameter step of $50 - 80mm$ iii) $IT8 = 25i$. Take wear allowance as 10% of the gauge tolerance. Also mention the type of fit. Represent the dimensions graphically	10
		OR	
4	a	Describe briefly the basic elements of a comparator? What are the characteristics? With a neat sketch, explain the working principle of Johnson's mikrokator comparator.	08
	b	Explain briefly the working principle of a sine bar and list out the uses of it.	08
5	a	Describe briefly the working of a floating carriage micrometer and explain how the minor and effective diameter of a screw thread can be measured.	10
	b	Describe briefly the different errors of a screw thread and the reasons for such errors.	06
		OR	
6	a	Explain the constant chord method for measuring the gear tooth thickness	10
	b	Describe briefly the different errors of a gear and the reasons for such errors	06
7	a	Enumerate on the main elements of a coordinate measuring machine with neat sketches, and list out the industrial applications of a CMM.	10
	b	Describe the role of CMM in computer aided manufacturing.	06
8	a	Describe with the neat sketch, the working of hydraulic dynamometer.	08
	b	Withneat sketches, explain briefly the working of pressure thermometer. List out the applications of it.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Reading on a car speedometer is not observed till the speed reaches $15km/hr$. relate the value 15 to a static characteristics of speedometer.	01
	1.2	Distinguish between Accuracy and precision.	02
	1.3	Define error.	01
	1.4	When airy points are used to support a length standard at two points they will be apart at a distance of _____.	01
	1.5	Why is interferometer considered the extension of the application of the optical flat.	01
	1.6	Johnson's Mikrokator comparator works on the principle of _____.	01
	1.7	State the working principle of Sine bar.	01
	1.8	Write the expression for best wire for screw thread measurement with usual notations.	02
	1.9	Define: a) Module b) Effective diameter.	02
	1.10	_____ type of construction of CMM is suited for very large components.	01
	1.11	The replacement of mirrors by cube- corner reflector is to _____.	01
	1.12	_____ probes are suitable for automatic tool changing.	01
	1.13	_____ is a new standard in communication used in CAM.	01
	1.14	Sketch a basic thermocouple circuit.	02
	1.15	List the different devices used to measure force and torque.	02

PART-B

2	a	With an example, enumerate on the generalized measurement system. Justify the answers with neat sketches and block diagrams.	08
	b	Describe between different static and dynamic characteristics of measurement system.	08

3	a	Describe with a neat sketch the working principle of a vernier micrometer. Represent the reading of $12.535mm$ on it, also discuss about its accuracy compare to another micrometers.	08
	b	Design the general type "GO" and "$NOGO$" gauges for the hole designed as $25Hz$. Given the following with usual notation. i) Fundamental deviation for Hole= 0 ii) $25mm$ falls in the diameter steps of $18 - 30mm$. iii) $IT7 = 16i$. Take wear allowance as 10% of the gauge allowance. Represent the dimensions graphically.	08
		OR	
4	a	With a neat sketch explain briefly the working of a sorex pnuematic comparator.	08
	b	Describe briefly the working principle of a sine bar with sketches. Also explain briefly the uses of sine bar to measure known and unknown angles.	08
5	a	Enumerate on the measurement of screw thread parameters using floating carriage micrometer.	08
	b	What is best size wire? Derive an expression for the same.	08
		OR	
6	a	Describe a gear tooth-vernier caliper and illustrate the use of it to measure tooth thickness.	10
	b	Define the following and represent these terminologies on a Gear profile: i) Pitch circle diameter ii) Addendum iii) Tooth thickness iv) Diametral pitch.	06
7	a	Enumerate on the main elements of a CMM with neat sketches and list the applications of it.	08
	b	Explain the basic principle of $LASER$. List the various uses of $LASER$ in metrology and explain any two in detail.	08
8	a	With a neat sketch describe briefly the working of a proving ring.	06
	b	Hydraulic dynamometers are used for the measurement of torque-explain in detail with a sketch.	06
	c	State and explain the different laws of thermocouple.	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. ExaminationsFast Track July-18

Industrial Engineering and Management

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Define SI unit of length.	01
	1.2	List out the examples of line standard.	01
	1.3	Differentiate between reproducibility and calibration.	02
	1.4	With $M87$ slip gauges build the dimensions of $29.758\,mm$.	01
	1.5	Identify the present definition of metre. What is the repeatability of this standard?	01
	1.6	Explain the term 'magnification of dial indicator.	01
	1.7	Why the slip gauges are termed as End Standard?	01
	1.8	Differentiate between environmental errors and dynamic errors.	02
	1.9	Define Tooth thickness.	01
	1.10	Define run out in a gear error.	01
	1.11	A comparative type length measuring machine is called as _____.	01
	1.12	Measuring machines are used for measuring _____.	01
	1.13	Optical flats are made of _____ or _____.	01
	1.14	The phenomenon of interaction of light waves is called _____.	01
	1.15	Mention the desirable properties of thermo-couple materials.	02
	1.16	Enumerate standardized calibrated device which causes a variety of effects.	02

PART-B

2	a	Elaborate the significance of the following terms used with reference of measurement. i. Accuracy ii. Precision iii. Readability	06
	b	Discuss with block diagrams generalized measurement system with examples for each stage elements.	10
3	a	Discuss the procedure for the calibration of endbars.	08
	b	Write a brief note on the manufacture of slip gauges.	04
	c	What is the role of standards? Discuss length standard.	04
		OR	
4	a	What are the advantages and disadvantages of LVDT?	08
	b	Describe with a neat sketch the construction and working of mechanical – optical comparator.	08
5	a	Explain with a schematic diagram, the method of measuring the major diameter of a screw thread using bench micrometer.	10
	b	How do you measure major diameters of internal screw threads?	06
		OR	
6	a	Explain the gear tooth measurement using gear tooth vernier caliper.	10
	b	Illustrate the use of gear tooth vernier caliper to measure tooth thickness.	06
7	a	Describe briefly co-ordinate measuring machine.	10
	b	State the advantage and possible sources of errors in CMM.	06
8	a	What are the methods of force measurement?	08
	b	What are thermistors? With sketches explain their construction.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw near sketches wherever necessary.

PART-A

1	1.1	Define Line Standard and End Standard	02
	1.2	List the advantages of Wavelength Standard.	02
	1.3	Define tolerance and its types.	02
	1.4	Define calibration and its importance.	02
	1.5	Define accuracy and precision.	02
	1.6	Using $M112$ slip gauges build the dimension $35.4875\ mm$.	02
	1.7	Define Measurement. Write the requirements to be satisfied for a measurement to be meaningful.	02
	1.8	Define True Value and Error.	02
	1.9	Define repeatability and Readability.	02
	1.10	Define Thermister and its use.	02

PART-B

2	a	Using block diagram, describe the stages of a measuring system. Using a measuring instrument identify the various elements of the instrument into different stages.	10
	b	Write the general classification of error and define each error.	06
3	a	Define a Slip Gauge. Write the classification of slip gauges. Describe the wringing phenomenon of slip gauges using neat sketch.	08
	b	Using neat sketches describe the limit gauges used to check the external and internal dimensions.	08
		OR	

4	a	Describe the working of Solex Pneumatic Comparator using neat sketch.	08
	b	Illustrate the process of setting an angle using Sine bar and measuring slope of a component using Sine bar with neat sketches.	08
5	a	Explain the process of measuring thread diameter using floating carriage micrometer using neat sketch.	08
	b	List and define the various types of errors in a screw thread. How these errors can be eliminated?	08
		OR	
6	a	Using neat sketch describe the process of measuring tooth thickness using gear tooth vernier caliper.	08
	b	List and define the various types of errors in a gear and reasons for the same. Describe how these errors can be eliminated.	08
7	a	Using neat sketches describe the main elements of Co-ordinate Measuring Machine. Write the important applications of CMM.	08
	b	What is the use of Co-ordinate Measuring Machine in Computer Aided Manufacturing? Describe briefly.	08
8	a	Using neat sketch describe the working of Proney Brake Dynamometer, what are its applications.	08
	b	Using neat sketch, describe the working of a Thermocouple to measure unknown temperature. State the laws of thermocouple on which this instrument work.	08

Engineering Materials and Mechanics

III Semester SEE Question Papers

Course codes: 21IM32

- May 2024 Supplementary (21IM32)
- Apr/May 2023 (21IM32)

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Supplementary Examinations May-2024

Industrial Engineering and Management

ENGINEERING MATERIALS & MECHANICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Those metals which do not contain iron as base metal are known as _____.	01	1	1
	1.2	The ability of a material to resist elastic deformation is known as _____.	01	2	1
	1.3	Medium carbon steel contains carbon between _____.	01	1	2
	1.4	The property of metal by which it can be drawn into thin wires is called _____.	01	1	1
	1.5	Materials which have same elastic properties along three mutually perpendicular directions are known as _____.	01	2	2
	1.6	When a ferromagnetic material is heated above the curie temperature it becomes_____.	01	1	1
	1.7	The acronym CFRP stands for _____**.**	01	2	1
	1.8	Lever rule is used to determine _____ of each phase in a binary equilibrium diagram.	01	1	1
	1.9	The number of phases present in equilibrium at eutectic point is _____.	01	2	1
	1.10	In nodular iron, graphite is present in the form of _____.	01	1	1
	1.11	Polar modulus of a shaft is the ratio of _____ of a shaft to its _____.	02	2	3
	1.12	For the same material, length and given torque, a _____shaft weighs less than a _____ shaft.	02	2	3
	1.13	The shear stress in a shaft is zero at _____ and maximum at _____.	02	3	3
	1.14	In a thin cylinder, the _____ stress is twice the _____stress.	02	2	4
	1.15	Lame's equations for thick cylinder are used to obtain the variation of _____ and _____ along the radius of the cylinder.	02	3	4

PART-B

2	a	Define a composite material and explain the classification of composite materials.	08	2	1
	b	Discuss the composition, properties and applications of i) Stainless steel ii) Brass iii) Bronze.	08	3	1

3	a	Define i) Engineering Stress and strain ii) True stress and strain. Establish the relationship between true stress and strain.	08	1	2
	b	With neat sketches, explain the stages of moderately ductile fracture under the action of load.	08	3	2
		OR			
4	a	What is fatigue? Explain the various factors affecting fatigue life.	08	3	2
	b	Explain fatigue life testing with neat sketch	08	2	2
5	a	What is a solid solution? Discuss Hume-Rothary rules for formation of solid solution.	08	4	2
	b	Explain with a neat sketch, nucleation & growth of dendrites.	08	3	2
		OR			
6	a	Draw the iron-iron carbide phase diagram and show the region where the following reactions take place. i) Eutectic ii) Peritectic iii) Eutectoid.	08	4	2
	b	Draw the $T-T-T$ diagram for 08% C Eutectoid steel by superimposing cooling curves on it.	08	3	2
7	a	With usual notations, derive the torsion equation in its general form. Also, state clearly the assumptions made.	10	2	3
	b	Determine the diameter of a sold steel shaft to transmit $100KW$ at $150\,rpm$, if the maximum torque exceeds the mean by 20%. The shear stress is not to exceed $60MPa$. Also find the angle of twist in a length of $4m$ of the shaft. Take $G = 80GPa$	06	3	3
		OR			
8	a	Prove that a hollow shaft is stronger & stiffer than a solid shaft of the same material, length and weight.	10	2	3
	b	Determine the diameter of a solid circular shaft to transmit $192\,KW$ at $450rpm$. Allowable shear stress may be taken as $70MPa$. If the solid shaft is replaced by a hollow shaft of same material & length, whose diameter ratio is 0.8, find the percentage reduction in weight achieved.	06	3	3
9	a	With usual notations and neat sketches, obtain expressions for the circumferential and longitudinal stresses induced in a thin cylinder subjected to internal pressure p.	08	6	4
	b	A cylindrical pressure vessel of 1m inner diameter and $1.5m$ long is subjected to internal pressure p. Thickness of the cylinder wall is $15mm$. Taking the allowable stress for the cylinder material as $90MPa$, determine i) Magnitude of pressure p ii) Change in dimensions & volume. Take $E = 200GPa$ & μ=0.3	08	5	4
		OR			

10	a	With usual notations derive Lame's equation for thick cylinders.	08	6	4
	b	A thick cylindrical vessel of $500mm$ inner diameter is subjected to an internal pressure of $9MPa$. Taking allowable stress for the material of the cylinder as $40MPa$, determine: i) Wall thickness of the cylinder ii) Circumferential stress at the outer radius and iii) % error involved if thickness is calculated based on thin vessel theory.	08	5	4

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations April/May-2023

Industrial Engineering and Management

ENGINEERING MATERIALS & MECHANICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	The tendency of a deformed solid to regain its actual proportions instantly upon unloading known as _____.	01
	1.2	The ability of a material to resist plastic deformation known as _____.	01
	1.3	Deformation that occurs due to stress over a period of time is known as _____.	01
	1.4	Carbon grey iron castings usually is present in the form of _____.	01
	1.5	Materials which have same elastic properties in all directions are known as _____.	01
	1.6	The non-metal used in vulcanization of rubber is _____.	01
	1.7	The acronym $GFRP$ stands for _____.	01
	1.8	Lever rule is used to determine _____ of each phase in a binary equilibrium diagram.	01
	1.9	The number of phases present in equilibrium at eutectic point is _____.	01
	1.10	Maximum percentage of carbon in austenite is _____.	01
	1.11	The two criteria used in design of shafts subjected to torsion are _____ and _____.	02
	1.12	If the diameter of shaft is doubled, its torque carrying capacity increases by _____ times.	02
	1.13	Calculate the torque which a shaft of $300mm$ diameter can safely transmit, if the shear stress is $48\ N/mm^2$.	02
	1.14	A penstock of $10m$ diameter carries water under a pressure head of $100m$. If the wall thickness is $9mm$, what is the maximum stress in the pipe wall?	02
	1.15	In Lame's equations for thick cylinders, the unit of constant a is _____ and that of constant b is _____.	02

PART-B

2	a	List the basic steps in selection of materials and briefly explain them.	08
	b	How density is related to Young's modulus? Sketch and label the material selection chart showing density vs. young's modulus.	08
3	a	Sketch the stress-strain curve for a ductile material. Define the salient points and also mark them on the curve.	06
	b	Explain the principle of fatigue testing with the sketch.	10

		OR	
4	a	What is fracture? Discuss the types of fracture with appropriate sketches.	08
	b	Define endurance limit of a material. With the help of S-N curve, explain high cycle and low cycle fatigue.	08
5	a	What is fracture? Discuss Hume-Rothary rules for formation of solid solution.	08
	b	With the help of TTT and CCT diagram, explain martempering.	08
		OR	
6	a	Explain the following: i) Gibb's phase rule ii) Different types of stainless steels.	08
	b	Explain the steps to construct TTT diagram. Draw a labeled sketch of TTT diagram for a eutectoid steel.	08
7	a	Define the terms: i) Pure torsion ii) Polar Moment of inertia iii) Polar modulus iv) Torsional rigidity	08
	b	Determine the diameter of a solid circular shaft to transmit 192 KW at 450 rpm. Allowable shear stress may be taken as 70 MPa. If the solid shaft is replaced by a hollow shaft of same material and length, whose diameter ratio is 0.8. Find the percentage reduction in weight achieved and the ratio of torsional stiffness of hollow and solid shafts.	08
		OR	
8	a	Starting from the fundamentals, derive the torsion equation in the general form.	08
	b	A solid shaft rotating at 500rpm transmits 30KW. Maximum torque is 20% more than the mean torque. Material of the shaft has an allowable stress of 65 MPa and the modulus of rigidity is 81MPa. Angle of twist in the shaft should not exceed 1° per meter length. Determine the size of the shaft.	08
9	a	With usual notations, obtain an expression for change in volume of a thin cylindrical shell subjected to internal pressure P.	08
	b	A thin cylinder 2m long and 200mm diameter with 10 mm wall thickness is filled completely with a fluid at atmospheric pressure. If an additional 25000 mm^3 is pumped in, find the longitudinal and hoop stress developed. Also determine the change in diameter and length if $E = 2 \times 10^5 N/mm^2$ and Poisson's ratio = 0.3.	08
		OR	
10	a	With usual notations, derive Lame's equation for thick cylinders.	08
	b	A thick-walled cylindrical pressure vessel has inner radius of 150mm and outer radius of 185 mm. If the internal pressure applied is 10MPa and the external pressure is zero, sketch the radial and hoop stress distribution across the section of the cylinder wall.	08

Fluid Mechanics and Thermodynamics

III Semester SEE Question Papers

Course codes: 18IM33, 16IM34, 16IM34 / 12IM36, 12IM36

- Fast Track Jan 2023 (18IM33)
- Mar 2021 (18IM33)
- Fast Track Oct 2020 (16IM34)
- Fast Track Oct 2020 (18IM33)
- Nov/Dec 2019 (18IM33)
- Fast Track Jul 2019 (16IM34)
- Nov/Dec 2018 (16IM34)
- Fast Track Jul 2018 (16IM34 / 12IM36)
- Nov/Dec 2017 (16IM34)
- Fast Track Jul 2017 (12IM36)
- Nov/Dec 2016 (12IM36)
- Fast Track Jul 2016 (12IM36)

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Fast Track Examinations January - 2023

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

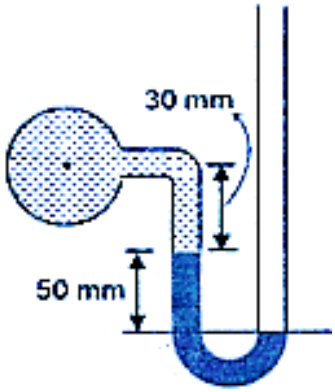
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

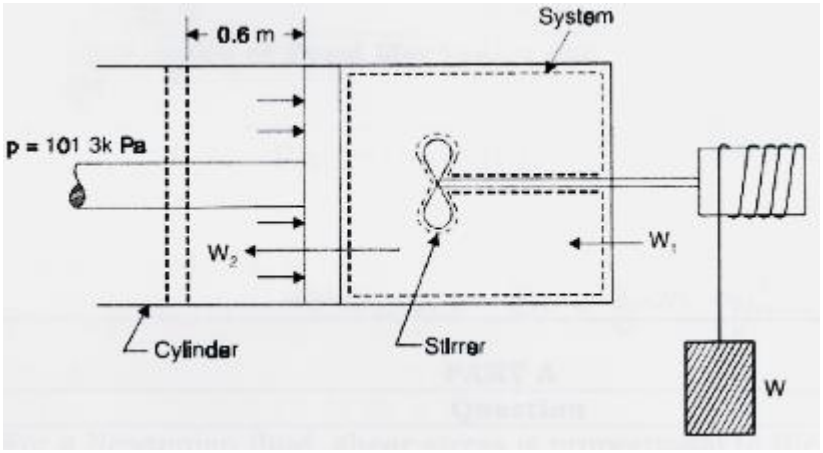
PART-A

1	1.1	For a Newtonian fluid, shear stress is proportional to the _____.	01
	1.2	The SI unit of kinematic viscosity (v) is _____.	01
	1.3	A pipe, through which water is flowing is having diameter of 20cm and 10cm at the cross sections 1 and 2 respectively. The velocity of water at section 1 is 4m/s. The velocity head at section 2 in m of water is _____.	02
	1.4	Co-efficent of discharge for a orifice plate lies in the range of _____.	01
	1.5	For a fully developed laminar flow in a pipe of diameter 10cm, the fluid is flowing at a velocity of 0.2m/s and the kinematic viscosity of water is 10^{-5} m^2/s. The value of Darcy's friction factor is _____.	02
	1.6	Centrifugal pump is an example of _____ thermodynamic system.	01
	1.7	For a thermocouple, _____ is the thermometric property.	01
	1.8	A cylinder contains $5m^3$ of an ideal gas at a pressure of 1 bar. This gas is compressed in a reversible isothermal process till its pressure increases to 5 bar. The work in kJ required for this process is _____.	02
	1.9	PMM-I is a violation of _____.	01
	1.10	The COP of a Carnot heat pump operating between 6℃ and 37℃ is _____.	02
	1.11	Air at a temperature of 620℃ and a velocity of 55 m/s enters an adiabatic nozzle and leaves at a temperature of 510℃ and velocity of _____m/s. (CP=1.005kJ/kgK).	02
	1.12	The power transmitted through the shaft of a car when the torque applied is 200Nm and the shaft rotates 4000 rpm is _____kW.	02
	1.13	Change in internal energy for a thermodynamic cycle is _____.	01
	1.14	A $5m^3$ rigid tank contains nitrogen gas at 700 kPa and 300K. Now heat is transferred to the nitrogen in the tank and the pressure of nitrogen rises to 900kPa. Displacement work done during the process is _____.	01

PART-B

2	a	The velocity distribution for flow over a plate is given by $u = 3y - y^2$, where u is the velocity in m/s at a distance y metre above the plate. Find the velocity gradient and shear stress at the boundary and 0.2m from it when $\mu = 0.86 Ns/m^2$.	06
	b	Assumed bulk modulus of elasticity of water is $2.07 \times 10^6 kN/m^2$ at standard atmospheric conditions; determine the increase of pressure necessary to produce 1% reduction in volume at the same temperature.	04
	c	Negative pressure in water pipe was measured using U-tube manometer shown in Fig. 2b containing mercury (Specific gravity 13.6). The left limb was connected to the water pipe and right limb was exposed to the atmosphere. Height of water upto centre of pipe in the left limb was 30mm. Difference between the mercury levels in the two limbs was 50 mm. Determine vacuum presume in the pipe.  Fig. 2b	06
3	a	Derive for Euler's' equation of motion for a fluid element flowing along a streamline. Reduce the equation further to Bernoulli's equation. State all the assumptions made from the above derivation.	08
	b	5000 litres/min of petrol density 0.8kg/litre flows through a 15 cm dia by 7.5cm dia vertical Venturimeter. The flow is in the upward direction and the Cd of Venturimeter is 0.991. The length of the converging section is 30cm. Draw the Venturimeter arrangement and compute the pressure difference between the inlet and throat sections and the deflection that would be recorded by a mercury differential U gauge connected between the sections.	08
		OR	
4	a	A hostel consisting of 1000 students is supplied water by a pumping station situated at 2500m from the hostel. The total amount of water required is to be supplied within 8 hours and each student requires 200litres of water per day. If the friction loss through pipe is 40m of water, then find the diameter of the pipe required. Use Dacry-Weisbach formula and take f=0.001.	08
	b	Explain minor energy losses in pipe flows.	08

5	a	Differentiate between i. Intensive and extensive property ii. Adiabatic and diathermic wall. iii. Path and point function	06
	b	A temperature scale of certain thermometer is given by the relation $t = a \ln p + b$ where a and b are constants a and p is thermometric property of the fluid in the thermometer. If at the ice point and steam point the thermometric properties are found to be 1.5 and 7.5 respectively what will be the temperature corresponding to the thermometric property of 3.5 on Celsius scale.	05
	c	Explain "quasi-static process" using an example of expansion of gas in a piston cylinder arrangement.	05
		OR	
6	a	In a gas turbine unit, the gases flow through the turbine at 15kg/s and the power developed by the turbine is 12000kW. The enthalpies of gases at the inlet and the outlet are 1260kJ/kg and 400kJ/kg respectively, and the velocity of gases at the inlet and outlet are 50m/s and 110m/s respectively. Calculate: i. The rate at which heat is rejected to the turbine, and ii. The area of the inlet pipe given that the specific volume of the gases at the inlet of 0.45 m^3/kg.	08
	b	0.15 m^3 of an ideal gas at a pressure of 15 bar and 500K is expanded isothermally to 4 times the initial volume. It is then cooled to 290K at constant volume and then compressed back polytropically to its initial state. Calculate the net work done and heat transferred during the cycle.	08
7	a	State Kelvin and Planck statement of second law of thermodynamics, Prove that they are equivalent.	08
	b	A reversible heat engine operates between two reservoirs at temperatures at 600℃ and 40℃. The engine drives a reversible refrigerator which operates between reservoirs at temperatures of 40℃ and -20℃. The heat transfer to the heat engine is 2000 kJ and the net-work output of the combined engine refrigerator plant is 360kJ. i. Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir 40℃. ii. Reconsider (i) given that the efficiency of the heat engine and the COP of the refrigerator are each 40% of their maximum possible values.	08
8	a	On the same P-v plot, starting from the same initial pressure and specific volume, show the following expansion processes. i. Isochoric process ii. Isobaric process iii. Isothermal process iv. Polytropic process with n=2.	04

b	A piston and cylinder machine containing a fluid system has a stirring device as shown in Fig. 8b. The piston is frictionless, and it is held down against the fluid due to atmospheric pressure of 101.3 kPa. The stirring device is turned 9500 revolutions with an average torque against the fluid of 1.25Nm. Meanwhile the piston of 0.65m diameter moves out 0.6m.Find the net-work transfer for the system  Fig. 8b	05
c	It is required to melt 5 tonnes/h of iron from a charge at 15℃ to molten metal at 1650℃. The melting point is 1535℃, and the latent heat is 270kJ/kg. The specific heat in solid state is 0.502 and in liquid state (29.93/ atomic weight) kJ/kg K. If an electric furnace has 70% efficiency, find the kW rating needed. If the density in molten state is 6900kg/m^3 and the bath volume is three times the hourly melting rate, find the dimensions of the cylindrical furnace if the length to diameter ratio is 2. The atomic weight of iron is 56.	07

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations March-2021

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Zeroth law provides the concept for _____.	01
	1.2	A thermocouple works on _____.	01
	1.3	Gauge pressure is written as _____ (in terms of absolute pressure and atmospheric pressure).	01
	1.4	As thermodynamic properties are point function their change is given by _____ differentials.	01
	1.5	Heat flow takes place when there is a difference in _____.	01
	1.6	Heat transferred to a closed stationary system at constant volume is equal to _____.	01
	1.7	If a process can be stopped at any stage and reversed so that the system and surroundings are exactly restored to their initial states, it is known as _____.	01
	1.8	A gas which obeys kinetic theory perfectly is known as _____.	01
	1.9	In a process, if every state passes by the system is an equilibrium, then it is called _____.	01
	1.10	Free expansion of gas within a system is _____ process.	01
	1.11	The concept of continuum in fluid flow assumes that the characteristic length of the flow is _____ than the mean free path of the molecules.	01
	1.12	A column of 30 cm of petrol of specific gravity 0.680 is equivalent to _____cm of mercury.	01
	1.13	Newton's law of viscosity for a fluid states that the shear stress is _____ to rate of angular deformation.	01
	1.14	The dimensions of the coefficient of dynamic viscosity in $[M, L, T]$ notation system are _____.	01
	1.15	If a Mohr circle is drawn for a fluid element inside a fluid body at rest, it would be _____on the normal stress axis.	01
	1.16	A barometer at a given location always shows _____ which may change with time.	01
	1.17	An oil of kinematic viscosity 0.25 stokes flows through a pipe of diameter 10 cm. The flow is critical at velocity of _____ m/s.	02

	1.18	Oil of viscosity 1.5 Pa.s and relative density 0.9 flows through a circular pipe of diameter 5cm with a mean velocity of 1.2 m/s. The shear stress at the wall in Pa is _____	02

PART-B

2	a	What do you mean by Viscosity? State and explain Newton's law of viscosity.	04
	b	A flat plate having an area of 0.64 m^2 slides down the inclined plane at an angle of 30° to the horizontal with a speed of 0.35 m/s. A lubricant layer of 1.6 mm thickness is placed between the plane and the plate. Determine the viscosity of the lubricant used if the weight of the plate is 250N.	06
	c	A disc of diameter 80 mm is rotated on a spindle and is enclosed in a small chamber filled with oil of viscosity 5 poise. Determine the torque required to rotate disc at 60 rpm when the clearance at the top and the bottom of the disc is 1mm. Assume linear velocity profile in the oil film.	06
3	a	Derive Euler's equation of motion. Also obtain Bernoulli's equation from it.	08
	b	The smaller diameter 0.1 m of a 4 m long pipe at lower level is inclined at an angle of 30° with the horizontal and its large diameter is 0.3 m. If the velocity of water at the smaller diameter section is 2 m/s , then calculate the difference of pressure between the smaller and larger diameter sections of the pipe.	08
		OR	
4	a	What is an orificemeter and what are its merits and demerits? Also derive an expression for discharge through it.	08
	b	A venturimeter inclined at 60° to the vertical is fitted to a 300 mm diameter pipe and its 150 mm diameter throat is 1.3m from the entrance along its length. The gasoline (specific gravity = 0.78) flows upwards at a rate of 230 litres per second. Determine (i) the discharge coefficient of venturimeter if the pressure gauges fitted at the entrance and throat indicate pressures of 150 kPa and 80 kPa, respectively and (ii) if pressure gauges fitted at the entrance and throat of the meter are replaced by a U-tube mercury manometer, then determine the reading in differential mercury column. Take specific gravity of mercury as 13.6.	08
5	a	What is the different between thermal equilibrium and thermodynamic equilibrium? Explain the following terms: i) Thermodynamic Property ii) Intensive Property iii) Specific Property iv) Thermodynamic state	04

	b	In an unknown temperature scale freezing point of water is $0°X$ and boiling point of water is $1000°X$. Obtain a conversion relation between degrees X and degree Celsius. Also determine the absolute zero in degree X.	06
	c	A temperature scale is being developed using the following relation. $t = a * ln\|p\| + \frac{b}{2}$ Where 'p' is thermometric property and 'a' and 'b' are constants. Determine Celsius temperature corresponding to thermometric property of 6.5, if ice point and steam point give thermometric property value of 3 and 8.	06
		OR	
6	a	One mol of air at 0.5 MPa and 400 K, initially undergoes following processes, sequentially (a) Heating at constant pressure till the volume gets doubled. (b) Expansion at constant temperature till the volume is six times of initial volume. Determine the work done by air.	08
	b	A compressed air bottle of volume $0.15m^3$ contains air at 40 bar and $27°C$. It is used to drive a turbine which exhausts to atmosphere at 1 bar. If the pressure in the bottle is allowed to fall to 2 bar, determine the amount of work that could be delivered by the turbine.	08
7	a	Explain Kelvin Planck & Claussius statements for the second law of thermodynamics.	06
	b	Write the difference between reversible and irreversible thermodynamic processes.	04
	c	An engine has a heat input of 500 kJ at $437°C$. It rejects 200 kJ at $82°C$. The engine develops 250 kJ of work. Check whether such as engine is possible.	06
8	a	Prove that the heat transferred to an ideal gas for a polytropic process $pv^n = c$ is expressed as $$Q_{1\to2} = \frac{\gamma - n}{\gamma - 1} W_{1\to2}$$ Where γ represents ratio of specific heats and n is polytropic index.	08
	b	A system consisting of 1.2 kg of an ideal gas at 4.8 bar pressure and $0.02\,m^3$ volume executes a cyclic process comprising the following three distinct operations: i) Reversible expansion to $0.08\,m^3$ volume, 1.6 bar pressure, presuming pressure to be a linear function of volume ($p = a + bV$), ii) Reversible cooling at constant pressure and iii) Reversible hyperbolic compression according to law pV = constant. This brings the gas back to initial conditions. iv) Sketch the cycle on p – V diagram. v) Calculate the work done in each process starting whether it is done on or by the system and evaluate the net cyclic work and heat transfer.	08

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of Thermodynamics tables is permitted.

PART-A

1	1.1	How does viscosity varies with temperature.	01
	1.2	Define Kinematic viscosity.	01
	1.3	Capillary rise or fall of a liquid is given by _____.	01
	1.4	The specific volume of a liquid is the reciprocal of _____.	01
	1.5	A beaker is filled with a liquid up to mark of one litre and weighed. The weight of the liquid is found to be $6.5N$. Find the specific gravity of the liquid.	01
	1.6	State Bernoulli's theorem.	01
	1.7	State any two assumptions made in the derivation of Bernoulli's equation.	01
	1.8	Find the loss of head when a pipe of diameter $200mm$ is suddenly enlarged to a diameter of $400mm$. The rate of flow of water through the pipe is $250\ litres/s$.	01
	1.9	The discharge through a triangular notch is given by _____.	01
	1.10	The mass flow of liquid is _____.	01
	1.11	The definite area or a space where some thermodynamic process takes place is known as _____.	01
	1.12	When neither mass nor energy is allowed to cross the boundary of a system, it is then known as _____.	01
	1.13	Temperature is the example for _____ property of a thermodynamic system.	01
	1.14	Carnot cycle consists of _____ and _____ processes.	01
	1.15	Otto cycle is also known as _____ cycle.	01
	1.16	For the same compression ratio, the efficiency of diesel cycle is _____ than Otto cycle.	01
	1.17	State Zeroth law of thermodynamics.	01
	1.18	For a reversible adiabatic process, the change in entropy is _____.	01
	1.19	_____ mean that the rates of flow of mass and energy across the control surface are constant.	01
	1.20	_____ point is that state at which all three phases of pure substance coexists in equilibrium.	01

PART-B

2	a	The large plane surface is $2.4cm$ apart. The space between the surfaces is filled with glycerin. What force is required to drag a very thin plate of surface area $0.5\ sq\ metre$ between the 2 large plane surfaces at a speed of $0.6m/s$ if the thin plate is at a distance of $0.8cm$ from one of the plane surfaces Take the dynamic viscosity of Glycerine = $8.10 \times 10^{-1} Ns/m^2$.	06
	b	Define capillarity and derive an expression for a capillarity rise of a liquid.	10
3	a	Derive Bernoulli's equation starting from fundamentals and state all the assumptions made for such derivation.	08
	b	A non-uniform part of pipe line $5m$ long is laid at a slope of $2\ in\ 5$. Two pressure gauges each fitted at upper and lower ends read $20\ N/cm^2$ and $12.5\ N/cm^2$. If the diameters at the upper and lower ends are $15cm$ and $10cm$ respectively, determine the quantity of water flowing per second.	08
		OR	
4	a	What is Venturimeter? Derive an expression for the discharge through a Venturimeter.	08
	b	An orifice meter with orifice diameter $15cm$ is inserted in a pipe of $30cm$ diameter. The pressure difference measured by a Mercury oil differential manometer on the 2 sides of the orifice meter gives a reading of $50cm$ of Mercury. Find the rate of flow of oil of specific gravity 0.9 when the coefficient of discharge of the meter=0.64	08
5	a	Describe the importance of Zeroth law in thermodynamics.	04
	b	Explain the difference between: i) Point function and path function ii) Heat and work iii) Closed system and open system.	06
	c	The end of thermocouple with test junction at t^0C on gas thermometer scale and reference junction at ice point gives the emf as $e = 0.20\ t - 5 \times 10^{-4} t^2\ mV$. The millivolt meter is calibrated at ice and steam points. What will be the reading on this thermometer where the gas thermometer reads 70^0C?	06
		OR	
6	a	Derive an expression for displacement work if the system follows the law $PV^n = C$.	08
	b	A mass of $1.5kg$ of air is compressed in a Quasi-static process from $0.1MPa$ to $0.7MPa$ for which $PV = constant$. The initial density of air is $1.16\ kg/m^2$. Find the work done by the piston to compress the air.	08
7	a	A mass of $8kg$ gas expands within a flexible container so that the $P - V$ relationship is of the form $PV^{1.2} = Constant$. The initial pressure is $100KPa$ and the Initial volume is $1m^3$. The final pressure is $5KPa$. If specific internal energy of the gas decreases by $40KJ/kg$; find the heat transfer in magnitude and direction.	06

	b	Write the steady flow energy equation and mention the terms involved in it.	06
	c	Prove that internal energy is a property of the system.	04
8	a	State and explain two statements of second law of thermodynamics.	06
	b	Prove that a relation between refrigerator and heat pump is given by $(COP)_{HP} = (COP)_R + 1$	04
	c	If a refrigerator is used for heating purposes in winter so that the atmosphere becomes the cold body and the room to be heated becomes the hot body, how much heat would be available for heating for each KW input to the driving motor? The COP of the refrigerator is 5 and the electromechanical efficiency of the motor is 90%. How does this compare with resistance heating.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Draw neat sketches wherever necessary.
4. Use of tables permitted.

PART-A

1	1.1	The specific volume of a fluid is the reciprocal of _____.	01
	1.2	The pressure head of a fluid is the ratio of intensity of pressure to _____.	01
	1.3	The loss of head in a sudden expansion in a pipe is given by _____.	01
	1.4	The ratio of actual discharge to theoretical discharge in a venturimeter is called _____.	01
	1.5	When two systems are each in thermal equilibrium with a third system, then they are 'also in thermal equilibrium' with each other. This statement is known as _____.	01
	1.6	In a _____ system, no transfer of mass takes place but only heat transfers across the boundaries.	01
	1.7	The heat flows from a cold body to a hot body with the aid of an external source. This statement is given by _____.	01
	1.8	Workdone during constant volume process is= _____.	01
	1.9	Newton's law of viscosity is a relation that gives the shear stress in terms of _____ and _____.	02
	1.10	The excess pressure inside a soap bubble of radius $'r'$ is _____.	02
	1.11	The head of water over the centre of an orifice of diameter $20mm$ is $1M$. The actual discharge through the orifice is $0.85litre/s$, then coefficient of discharge is _____.	02
	1.12	The barometer pressure at sea level is $760mm$ of mercury while that on a mountain top is $735mm$. If the density of air is assumed constant at $1.2kg/m^3$, then elevation of the mountain top is _____.	02
	1.13	When a gas is cooled at constant pressure, the temperature _____ and volume _____.	02
	1.14	In a reversible heat engine, the temperature of the source and sink are 700^0C and 50^0C respectively. The heat supplied is $5MJ/min$, then the power developed by the engine is _____.	02

PART-B

2	a	State and prove Hydrostatic law.	06
	b	What do you mean by single column nanometers? How are they used for the measurement of pressure?	04
	c	The dynamic viscosity of an oil used for lubrication between a shaft and sleeve is $6\,Poise$. The shaft is of diameter $0.4m$ and rotates at $190rpm$. Calculate the power lost in the bearing for sleeve length of $90mm$. The thickness of the oil film is $1.5mm$.	06
3	a	Derive Euler's equation of motion for the flow of an incompressible frictionless fluid from consideration of momentum.	06
	b	An orifice meter with orifice diameter $10cm$ is inserted in a pipe of $20cm$ diameter. The pressure gauges fitted upstream and downstream of the orifice meter gives readings of $19.62N/cm^2$ and $9.81N/cm^2$ respectively. Co-efficient of discharge for the orifice meter is given as 0.6 . Find the discharge of water through pipe.	06
	c	An oil of specific gravity 0.7 is flowing through a pipe of diameter $300mm$ at the rate of $500\ litres/s$ in the head lost due to friction and power required to maintain the flow for a length of $1000m$. Take kinematic viscosity=$0.29\ stokes$.	04
		OR	
4	a	What is the principle of a Pitot-tube? How will you determine the velocity at any point with the help of Pitot-tube?	04
	b	Show that the loss of head due to sudden expansion in pipeline is a function of velocity head.	08
	c	Water is flowing through a pipe having diameter $300mm$ and $200mm$ at the bottom and upper end respectively. The intensity of pressure at the bottom end is $24.525\ N/cm^2$ and the pressure at the upper end is $9.81\ N/cm^2$. Determine the difference in datum head if the rate of flow through pipe is $400\ litres/s$.	04
5	a	Distinguish between intensive and extensive properties. Give example to each.	04
	b	With the help of sketches, explain the Joule's experiment to state the first law of thermodynamics.	06
	c	The readings T_A and T_B of two Celsius thermometers A and B agree at the ice point (0^0C) and steam point (100^0C), but elsewhere they are related by the equation $T_A = L + mT_B + nT_B{}^2$ where $'L'$, m and n are constants. When both the thermometers are immersed in a well stirred bath, A registres 51^0C whereas B registers 50^0C. i) Determine the reading on B when A registers 25^0C and ii) Which thermometer is correct?	06
		OR	
6	a	Recall the definitions of the following: i) Thermal equilibrium ii) Isolated system iii) Thermodynamic process iv) Thermodynamic cycle.	04

	b	Sketch the block diagram and write the steady flow energy equation for system where two streams of fluid enters and one stream leaves. Also write the mass balance equation.	04
	c	Air at a temperature of 15^0C passes through a heat exchanger at a velocity of $30m/s$ where its temperature is raised to 800^0C. It then enters a turbine with the same velocity of $30m/s$ and expands until the temperature falls to 650^0C. On leaving the turbine, the air is taken at a velocity of $60m/s$ to a nozzle where it expands until the temperature has fallen to 500^0C. If the air flow rate is $2kg/s$, calculate: i) The rate of heat transfer to the air in the heat exchanger ii) The power output from the turbine assuming no heat loss and iii) The velocity at exit from the nozzle, assuming no heat loss. Take the enthalpy of air as $h = C_pT$ where C_p is the specific heat equal to $1.005\ kJ/kg\ K$ and T is temperature.	08
7	a	State the Kelvin-Plank's statement with reference to the second law of thermodynamics and show that the violation of Kelvin-Plank's statement leads to the violation of Clausius statement.	06
	b	Derive the expression for efficiency of a Carnot cycle considering the work done at all the process.	06
	c	A reversible engine operates between temperature limits of T_1 and T where T_1 is the higher temperature. The heat rejected by this engine is received by a second reversible engine at the same temperature T which in turn rejects heat to a sink at temperature T_2. i) If the engines have equal efficiencies, show that $T = \sqrt{T_1T_2}$ ii) If the two engines have equal work output, show that $T = (T_1+T_2)/2$	04
8	a	Considering battery as a system, justify that the thermodynamic definition of work transfer is more general.	06
	b	Write any two similarities between heat and work.	02
	c	A piston cylinder arrangement is containing fluid at $10 \times 10^5\ N/m^2$ and a volume of $0.05\ m^3$. Find the work done by the fluid when it expands reversibly for the following cases: i) At constant pressure to a final volume of $0.2\ m^3$ ii) According to a law $PV = C$ to a final volume of $0.2\ m^3$ iii) According to a law $PV^{1.3} = C$ to a final volume of $0.2\ m^3$ iv) When $'P'$ relates to V as $P = -300V + 630$ where $P\ in\ kPa$ and $V\ in\ m^3$, final volume is $0.2\ m^3$.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The specific volume is the ratio of volume of fluid to _____.	01
	1.2	Kinematic viscosity is defined as the ratio between the _____ and _____.	01
	1.3	State Hydrostatic law.	01
	1.4	In the inverted U-tube Differential manometer, specific gravity of manometric fluid used in _____ than relative to the fluid flowing in the pipes as the manometric fluid is at the top.	01
	1.5	The aircraft fly is based on _____ principle.	01
	1.6	Write the relationship between orificemeter diameter and pipe diameter.	01
	1.7	_____ property of the fluid accounts for the major losses in pipe.	01
	1.8	The discharge through a triangular notch is given by _____.	01
	1.9	A definite area or a space where some thermodynamic process takes place, is known as _____.	01
	1.10	When neither mass nor energy is allowed to cross the boundary of a system, it is then known as _____.	01
	1.11	Temperature is the example for _____ property of a thermodynamic system.	01
	1.12	Carnot cycle consists of _____ and _____ processes.	01
	1.13	Otto cycle is also known as _____ cycle.	01
	1.14	For the same compression ratio, the efficiency of Diesel cycle is _____ than Otto cycle.	01
	1.15	Heat and work are _____ functions.	01
	1.16	For a reversible adiabatic process the change in entropy is _____.	01
	1.17	A Quasi-static process is also called a _____ process.	01
	1.18	_____ means that the rate of flow of mass and energy across the control surface are constant.	01
	1.19	_____ law state that if two system thermal equilibrium with a third system then they are in thermal equilibrium with each other.	01
	1.20	_____ point is that state at which all the 3 phases of pure substance co exists in equilibrium.	01

PART-B

2	a	Define the following: i) Mass density ii) Newton's law of viscosity iii) Surface Tension iv) Specific Weight.	04
	b	Explain the phenomena of capillarity. Obtain expression for capillarity rise of liquid.	06
	c	The space between 2 square flat parallel plates is filled with oil. Each side of the plate is $60cm$. The thickness of the oil film is $12.5\,mm$. The upperplate, which moves at $2.5\,m/sec$, requires a force of $98.1N$ to maintain the speed. Determine the: i) Dynamic viscosity of the oil in poise and ii) Kinematic viscosity of the oil in stokes if the specific gravity of the oil is 0.95.	06
3	a	Derive Bernoulli's equation starting from fundamentals and state all the assumptions made for such derivation.	08
	b	The water flowing through a pipe having diameters $20cm$ and $10cm$ at section 1 and 2 respectively. The rate of flow through pipe is $35\,litre/sec$. The section 1 is $6m$ above datum and section 2 is $4m$ above datum. If thepressure at section 1 is $39.24\,N/cm^2$ find the intensity of pressure at section 2.	08
		OR	
4	a	What is Venturimeter? Derive an expression for the discharge through a Venturimeter.	08
	b	An oil of specific gravity 0.9 and viscosity 0.06 poise is flowing through a pipe of diameter $200\,mm$ at the rate of $60\,litre/s$. Find the head lost due to friction for a $500m$ length of pipe. Find the power required to maintain the flow.	08
5	a	Define: i) State ii) Path iii) Process iv) Cycle.	04
	b	Fahrenheit reading and Celsius thermometers are both immersed in a fluid. Fahrenheit reading is numerically twice that of the Celsius reading. Compute temperature of the fluid.	06
	c	The readings t_A and t_B of two thermometers A and B on the Celsius scale degree at the ice point and the steam point, and are assumed to be related by the expression $t_A = l + mt_B + nt_B{}^2$ Where l, m and n are constants. When both thermometers are immersed in a well stirred oil bath, A reads $51°C$ and B reads $50°C$. Determine the readings on thermometer A when thermometer B reads $25°C$. Which thermometer can be taken to be correct?	06
		OR	
6	a	With neat sketch, state and explain first law of thermodynamics with reference to Joules's experiment.	06

	b	Derive steady flow energy equation and explain each term.	04
	c	The relationship between pressure and volume in a non-flow process is prescribed by the expression: $P = \left(\frac{3}{V} + 2\right)$ where pressure P is in bar and volume V is in m^3. During the process $1600KJ$ of heat is added to the gas and the volume changes from $1.2m^3$ to $4m^3$. Determine the change in internal energy.	06
7	a	Write the Kelvin-Planck and Clasius statement of second law of thermodynamics.	06
	b	Compare a heat engine with that of heat pump.	04
	c	A heat engine working on Carnot cycle converts one-fifth of the heat input into work. When the temperature of the sink is reduced by $80°C$, the efficiency gets doubled. Make calculations for the temperature of source and sink.	06
8	a	State the similarities between work and heat.	04
	b	Derive an expression for workdone when the system undergoes polytrophic process when $n = 1.4$.	06
	c	A cylinder fitted with a movable piston contains $0.04m^3$ of air at 10 bar pressure and $400K$ temperature. The air expands according to the law $P = \left[\frac{A}{V^2} - \frac{B}{V}\right]$ to a final pressure of 1 bar and $0.2m^3$. Determine the work done.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Mass density of a fluid is defined as the ratio _____ of a fluid to its _____.	02
	1.2	_____ is defined as the ratio between the dynamic viscosity and density of fluid.	01
	1.3	Specific volume is the reciprocal of _____.	01
	1.4	_____ law states that pressure at a point in a static fluid is equal in all direction.	01
	1.5	Bernoulli's equation is obtained by integrating the _____ equation.	01
	1.6	Bernoulli's equation for real fluids _____.	01
	1.7	The properties whose value does not depend on the mass of the system are known as _____.	01
	1.8	The cyclic integral of any properties _____.	01
	1.9	The measurement of temperature based is on _____ law of $T.D$.	01
	1.10	The constant pressure is known as _____.	01
	1.11	The first law of thermodynamics was developed by _____.	01
	1.12	_____ system, both mass and energy crosses the boundary of the system.	01
	1.13	A quasistatic process is also called a _____ process.	01
	1.14	Steady flow means _____.	01
	1.15	A reservoir from which heat is transferred is called _____ and a reservoir to which heat is transferred is called _____.	02
	1.16	$PMM2$ voilates _____ law of thermodynamics.	01
	1.17	For a reversed heat engine acting as _____.	01
	1.18	State any two assumptions made in the derivation of Bernoulli's theorem.	01

PART-B

2	a	Distinguish between the following: i) Mass density and weight density. ii) Dynamic viscosity and kinematic viscosity. iii) Ideal fluid and real fluid.	06
	b	Explain phenomenon of capillarity. Obtain expressions for capillary rise of liquid.	04

	c	A cylinder of $0.6m^3$ in volume contains air at 50℃ and $0.3N/mm^2$ absolute pressure. The air is compressed to $0.3m^3$. Find: i) Pressure inside the cylinder assuming isothermal process. ii) Pressure and temperature assuming adiabatic process. Take $\gamma = 1.4$.	06
3	a	State and derive Bernoulli's equation for a incompressible steady flow and state the assumptions made.	08
	b	A horizontal venturimeter with inlet diameter $20cm$ and throat diameter $10cm$ is used to measure the flow of water. The pressure at inlet is $17.658N/cm^2$ and the vacuum pressure at the throat is $30cm$ of mercury. Find the discharge of water through venturimeter. Take $C_d = 0.98$.	08
		OR	
4	a	Explain the principle of venturimeter with a neat sketch. Derive the expression for the rate of flow of fluid through it.	08
	b	An orifice meter with orifice diameter $15cm$ is inserted in a pipe of $30cm$ diameter. The pressure difference measured by a mercury oil differential monometer on the two sides of the orifice meter gives the reading of $50cm$ of mercury. Find the rate of flow of oil of specific gravity 0.9 when $C_d = 0.64$.	08
5	a	State and explain zeroth law of thermodynamics.	04
	b	With suitable examples distinguish between: i) Closed and open system. ii) Point function and path function. iii) Intensive and extensive property.	06
	c	A temperature of thermometric scale is defined in terms of property P by relation $T = a \log_e P + b$, where a and b are constants. The temperature at ice point and stream points are 0℃ and 100℃ respectively. Instruments give value of P as 1.86 and 6.81 at ice point and steam point respectively. Evaluate temperature corresponding to a reading of $P = 2.5$.	06
		OR	
6	a	Define heat and work. Show that both heat and work are path function.	08
	b	Suggest the most economical process when V is desired to compress $1kg$ of air ($\gamma = 1.4$) from an initial state of $300K$ and 1bar to a final stage of $300K$ and 10bar from among the following processes: i) Isothermal compression. ii) Cooling at constant pressure followed by heating at constant volume. iii) Adiabatic compression followed by cooling at constant volume. Take $R = 287J/kg\ K$.	08
7	a	Clearly write the steady flow energy equation for an open system and explain the terms involved.	04
	b	Define internal energy and prove that it is a property.	04

	c	The mass flow rate of steam through a steam turbine unit is $5000kg/hr$ and heat loss to the surroundings is $31712\,kJ/hr$. Steam enters at a pressure of 21bar and superheated to 370℃ and leaves the unit at 1.1 bar as fully dry and saturated. The velocities of steam at entry and exit are $60m/s$ and $180m/s$. Entry point of steam is $2m$ above the exit point. If the enthalpy of steam at entry is $3181.3kJ/kg$ and exit is $2679\,kJ/kg$. What is the power developed?	08
8	a	Explain heat engine, heat pump and refrigerator with a block diagram.	06
	b	Show that friction makes the process irreversible.	04
	c	An inventor claims to have developed a refrigerator that maintains the refrigerated space at 2℃ while operating in a room where the temperature is 24℃ and that has a COP of 13.5. Is this claim reasonable?	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	$V = 2y^2, W = 2xyz$ represent the two velocity components. Determine the third component of velocity such that they satisfy the continuity equation.	02
	1.2	Find the head lost due to friction in a pipe of diameter $300\,mm$ and length $50\,m$, through which the water is flowing at a velocity of $3m/s$ using chezy's formula for which $C = 60$.	02
	1.3	Find the loss of head when a pipe of diameter $200\,mm$ is suddenly enlarged to a diameter of $400\,mm$. The rate of flow of water through the pipe is $250\,lts/s$.	02
	1.4	What is a Thermodynamic reason for feeling of discomfort by human being during winter and summer season?	02
	1.5	A candle is burning in a well-insulated room. Taking the room(the air plus the candle) as the system, determine: a) if there is any heat transfer during this process and b) if there is any change in the internal energy of the system.	02
	1.6	An engineer claims his engine can develop $3.75\,KW$. On testing, the engine consumes $0.44\,kg$ of fuel per hour having calorific value of $42000\,kJ/kg$. The maximum temperature recorded in the cycle is 1400^0C and minimum is $350^0\,C$. Find whether the engineer is justified in making the claim.	02
	1.7	A definite area or space where some thermodynamic process takes place is known as _____.	01
	1.8	When neither mass nor energy is allowed to cross the boundary of a system, it is then known as _____.	01
	1.9	Temperature is the example for _____ property of a thermodynamic system.	01
	1.10	Carnot cycle consist of _____ & _____ processes.	02
	1.11	Measurement of temperature using a thermometer is example of _____ approach of thermodynamics.	01
	1.12	Give two reasons why work is not to be considered as a thermodynamic property?	02

PART-B

2	a	Define and explain Newton's law of viscosity.	04
	b	State and prove the Pascal's law.	06
	c	A solid cylinder of diameter $4.0\,m$ has a height of 3 meters. Find the meta-centric height of the cylinder when it is flowing in water with its axis vertical. The specific gracity of the cylinder $= 0.6$.	06
3	a	The diameters of a small piston and a large piston of a hydraulic jack at $3\,cm$ and $10\,cm$ respectively. A force of $80\,N$ is applied on the small piston. Find the load lifted by the large piston when small piston is $40\,cm$ above the large piston.	04
	b	Define the equation of continuity. Obtain the expression for continuity equation for a three dimensional flow.	06
	c	The water is flowing through a taper pipe of length $100\,m$ having diameters $600\,mm$ at the upper end and $300\,mm$ at the lower end, at the rate of 50 litres/s. the pipe hav a slope of 1 in 30. Find the pressure at the lower end if the pressure at the higher level is 19.62 N/cm2.	06
		OR	
4	a	Explain the principle of venturimeter with a neat sketch. Derive the expression for the rate of flow of fluid through it.	10
	b	An orifice meter with orifice diameter $10\,cm$ is inserted in a pipe of $20\,cm$ diameter. The pressure gauges fitted upstream and downstream of the orifice meter give readings of $19.62\,N/cm^2$ and $9.81\,N/cm^2$ respectively. Coefficient of discharge for the meter is given as 0.6. Find the discharge of water through pipe.	06
5	a	Define Open, closed & isolated system. Identify the class of system given below: i) A wind mill. ii) Mixture of ice and water in a metal container. iii) Thermo flask filled with hot tea. iv) Pressure cooker. v) Storage cell producing electricity.	08
	b	State the first law of thermodynamics for cyclic process. Briefly explain Joule Experiments, show that internal energy is a property of system.	08
		OR	
6	a	There is an argument between two Industrial Engineering students regarding thermodynamic system and energy interaction. They have decided to meet their class topper and clarify their doubt. They wanted to know which type of thermodynamic system and energy interaction takes place in a freezing compartment of a refrigerator. As a class topper how do you justify the doubt of Industrial Engineering Students?	08
	b	Consider an alcohol and a mercury thermometer that read exactly 0^0C at the ice point and 100^0C at the steam point. The distance between the two points is divided into 100 equal parts in both thermometers. Do you think these thermometers will give exactly same reading at a temperature of say 60^0C? Explain.	08

7	a	In the month of May in Delhi a house wife gets an idea to beat the heat by closing the windows and doors of the dining hall and opening the door of refrigerator. In the beginning she feels cool and comfortable, but very soon this effect wears off. How would you explain the situation with the help of First law of thermodynamics? The room along with a refrigerator is treated as a system. The walls, doors & windows are treated as non-conducting.	10
	b	Baseboard heaters are basically electric resistance heater and are frequently used in space heating. A homeowner claims that her 5 year old baseboard heater has conversion efficiency of 100 percent, is this claim in violation of any thermodynamic laws. Explain.	06
8	a	Define the two statements of second law of thermodynamics. Show that violation of Kevin Planck statement leads to violation of Clausius statement.	10
	b	Show that COP of heat pump is greater that COP of a refrigerator by unity.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Fast Track Examinations July-18

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS & THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A fluid element cannot resist _____ force.	01
	1.2	The equivalent pressure head of mercury corresponding to $30cm$ column of kerosene of relative density 0.8 is _____.	02
	1.3	The space between two parallel plates kept $4mm$ apart is filled with an oil of dynamic viscosity 0.2 Pa-s. If the upper plate is moved with a velocity of $1.6m/s$, then the shear stress on the lower fixed plate is given by _____.	02
	1.4	Surface tension is due to _____.	01
	1.5	Bernoulli equation is law of conservation of _____.	02
	1.6	Bulk modulus for fluid elements is mathematically defined as _____.	01
	1.7	The continuity equation represents conservation of _____.	01
	1.8	First law of thermodynamics is the law of conservation of _____.	01
	1.9	A $PMM1$ is a $HYPOTHETICAL$ machine whose operation violates the _____ law of thermodynamics.	01
	1.10	A closed system undergoes a process $1-2$ for which the values of $Q1-2$ and $W1-2$ are $+20kJ$ and $+50kJ$ respectively. If the system is returned to state 1 and $Q2-1$ is $-10kJ$. The value of work $W2-1$ is _____.	02
	1.11	If a heat engine attains 100% thermal efficiency, it violates _____ law of thermodynamics.	01
	1.12	The expression for reversible work done by the system under steady flow condition between state 1 and state 2 is given by _____.	01
	1.13	A refrigerator and the heat pump operate between the same temperature limits. If the COP of the refrigerator is 4, the COP of the heat pump would be _____.	02
	1.14	An engine P operates between temperature limits of $900K$ and T_2 and another engine Q between T_2 and $400K$. If both engines are equally efficient, T_2 will be equal to _____.	02

PART-B

2	a	Define the following terms: i) Specific weight ii) Mass density iii) Specific volume iv) Specific gravity.	04
	b	Define Newtonian and non-Newtonian Fluids with examples. State Newton's law of viscosity.	04
	c	A $90N$ rectangular solid block slides down a 30° inclined plane. The plane is lubricated by a $3mm$ thick film of oil of relative density 0.90 and viscosity 8.0poise. If the contact area is $0.3m^2$, estimate the terminal velocity of the block.	08
3	a	What is vapour pressure/ What is its significance in flow problems? What is the significance of Cavitation?	08
	b	Define surface tension. Derive the expression for the pressure: i) Within the droplet of water ii) Inside a soap bubble.	08
		OR	
4	a	Explain the principle of Orifice meter with neat sketch.	06
	b	A non-viscous liquid of constant density $1000\,Kg/m^3$ flows in streamline motion along a tube of variable cross section. The tube is kept inclined in vertical plane such that smaller cross-section is at lower position. The smaller cross-section which is at height of $2m$ is $0.004m^2$ and bigger at $5m$ with cross-section $0.008m^2$. The velocity at the inlet at smaller area is $1m/s$. Find: i) The work done per unit volume by the pressure, and ii) The gravity forces as the fluid flows from inlet to outlet.	10
5	a	Differentiate between microscopic approach from macroscopic approach. Explain the difference between open system and closed system with suitable examples.	08
	b	A new temperature scale in degrees N is desired with freezing point at $100°N$ and boiling point at $400°N$. Establish a correlation between degrees Celsius and degrees N. What would be absolute temperature at $0°N$?	08
		OR	
6	a	Prove that the adiabatic process is represented as $pv^{\gamma} = C$, where, p =pressure, v =volume, γ=ratio of specific heat constants.	04
	b	One mole of a monoatomic ideal gas is taken through the cyclic process $ABCDA$ clockwise such that $A-B$: adiabatic expansion $B-C$: cooling at constant temperature $C-D$: adiabatic compression $D-A$: heating at constant volume	

		The pressure and temperature at A, B, etc., are denoted by p_A, T_A, p_B, T_B etc., respectively. Given that $T_A = 1000K$, $p_B = (2/3)p_A$ and $p_C = (1/3)p_A$, calculate: i) The work done by the gas in the process $A - B$ ii) The heat lost by the gas in the process $B - C$ iii) The temperature T_D.	12
7	a	What is steady flow process? State all assumptions made for such a flow process.	08
	b	An insulated and rigid tank contains $5cm^3$ of air at 10bar and $425K$. The air is then let off to atmosphere through a valve. Determine the work obtained by utilizing the kinetic energy of the discharged air. Take $c_p = 1kJ/kg.K$, $c_v = 0.714kJ/kg.K$, atmospheric air = $1bar$.	08
8	a	Write the statements of second law of thermodynamics. Prove that violating clausius statement also violated Kelvin-Plank statement.	06
	b	A reversible heat engine operated between two reservoirs at temperatures 600℃ and 40℃. The engine drives a reversible refrigerator which operates between reservoirs at temperatures 40℃ and −20℃. The heat transfer to the heat engine is $2000kJ$ and the net work output of the combined engine refrigerator plant is $360kJ$. Given that the efficiency of the heat engine and the COP of the refrigerator are each 40% of their maximum possible values. Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40℃.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

PRINCIPLES OF FLUID MECHANICS AND THERMODYNAMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The specific volume is the ratio of volume of fluid to _____.	01
	1.2	Kinematic viscosity is defined as the ratio between the _____ and _____.	02
	1.3	1bar is _____ Pascals.	01
	1.4	Reynolds number is the ratio of inertia force to _____.	01
	1.5	The mass flow of fluid is _____.	01
	1.6	The pressure at any point in a fluid at rest is obtained by the _____ law.	01
	1.7	Specific gravity is known as _____.	01
	1.8	Heat and work are _____ functions.	01
	1.9	The first law of thermodynamics was developed by _____.	01
	1.10	For a reversible adiabatic process the change in entropy _____.	01
	1.11	A quasi-static process is also called a _____ process.	01
	1.12	The unit of work is _____.	01
	1.13	_____ means that the rates of flow of mass and energy across the control surface are constant.	01
	1.14	A reservoir from which heat is transferred is called _____ and a reservoir to which heat is transferred is called a _____.	02
	1.15	_____ point is that state at which all the three phases of pure substance coexists in equilibrium.	01
	1.16	A carnot cycle is made of a pair of _____ and _____ process.	02
	1.17	_____ law state that if two system thermal equilibrium with a third system then they are in thermal equilibrium with each other.	01

PART-B

2	a	Define the following: i) Mass density ii) Surface tension iii) Newton's law viscosity iv) Specific weight	06
	b	Explain the phenomenon of capillarity. Obtain expression for capillarity rise of liquid.	04
	c	The space between two square flat parallel plates is filled with oil. Each side of the plate is 60 cm. the thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 m/sec, requires a force of 98.1 N to maintain the speed.	

		Determine the: i) dynamic viscosity of the oil in poise and ii) kinematic viscosity of the oil in stokes if the specific gravity of the oil is 0.95.	06
3	a	Derive Bernoull's equation starting from fundamentals and state all the assumptions.	08
	b	The water flowing through a pipe having diameters $20cm$ and $10cm$ at section 1 and 2 respectively. The rate of flow through pipe is $35litre/sec$. the section 1 is $6m$ above datum and section 2 is $4m$ above datum. If pressure at section 1 is $39.24N/cm^2$, find the intensity of pressure at section 1.	08
		OR	
4	a	What is venturimeter? Derive an expression for the discharge through a venturimeter.	08
	b	An orifice meter with orifice diameter $10cm$ is inserted in a pipe of $20cm$ diameter. The pressure gauges fitted upstream and down stream of the orifice meter gives reading of $19.62N/cm^2$ and $9.81N/cm^2$ respectively. Coefficient of discharge for the meter is given as 0.6. Find the discharge of water through pipe.	08
5	a	State Zeroth law of thermodynamics. Write its importance in thermodynamics.	04
	b	How are system classified? What is the importance of each type?	06
	c	The emf of thermocouple with test junction at t℃ on gas thermometer scale and reference junction at ice point and stream point is given by, $\epsilon = 0.2t - 5 \times 10^{-4}t^2$ mV. The milli-voltmeter is calibrated at ice point and stem point. What will this thermometer read in a place where gas thermometer reads 50℃?	06
		OR	
6	a	A system undergoes a process in which the pressure and volume are related by an equation of the form $PV^n = C$. Derive an expression for displacement work during this process.	06
	b	Clearly distinguish between heat and work in thermodynamics.	04
	c	200kJ of work is done on a system. If the initial volume is $0.5m^3$ and pressure changes according to $P = (6 - 6V)$ where P is bar and V is in m^3. Determine the final pressure and final volume of the system.	06
7	a	Write the steady flow energy equation and mention the terms involved in it.	06
	b	Prove that internal energy is a property of the system.	04
	c	A steam turbine receives a steam flow of $4500kg/hr$ and delivers $5000kW$. The heat transfer from the turbine is negligible. Find the change in enthalpy across the turbine if the velocity at the entrance is $400m/s$ and the exit is $60m/s$. the inlet pipe is $3m$ above that of the exit.	06
8	a	State and explain two statements of second law of thermodynamics.	06
	b	Prove that a relation between refrigerator and heat pump is given by $(COP)_{HP} = (COP)_R + 1$.	04
	c	A heat engine used to drive a heat pump. The heat transfer from the heat engine and from the heat pump is used to heat the water circulating through radiators of a building. The efficiency of the engine is 27% and COP of the heat pump is 4. Evaluate the ratio of the heat transfer to the circulating water to the heat transfer to the heat engine.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

BASICS OF THERMODYNAMICS AND FLUID MECHANICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6. Answer FIVE full questions from Part B.

PART-A

1	1.1	Thermos flask is an example of _____ system.	01
	1.2	Density is an example of _____ property.	01
	1.3	A system comprising single phase is called _____.	01
	1.4	If the system passes through a continuous series if equilibrium state it is called as _____.	01
	1.5	The measurement temperature is based on _____ law of thermodynamics.	01
	1.6	Mean effective pressure is the ratio of _____ and _____.	02
	1.7	The carnot cycle consists of _____ and _____ process.	02
	1.8	The efficiency of a otto cycle can be expressed in _____.	01
	1.9	Specific volume is the reciprocal of _____.	01
	1.10	The _____ states that intensity of pressure for a fluid at rest in all directions.	01
	1.11	_____ is a device used for measuring pressure at a point in a fluid.	01
	1.12	The point about which a body starts oscillating when the body is tilted is known as _____.	01
	1.13	If the Reynold's number in a pipe is more than 2000, the flow is said to be _____ and if Reynolds number is more than 4000, the flow is said to be _____.	02
	1.14	Continuity equation is written as _____.	01
	1.15	The discharge, Q through a Venturimeter or an orifice meter is given by _____.	01
	1.16	A _____ is a device used for measuring the rate of flow of a liquid through a small channel. A _____ is a concrete or masonry structure placed in the open channel over which the flow occurs.	02

PART-B

2	a	Distinguish between: i) Intensive and extensive properties ii) Point and path functions iii) Thermal and thermodynamic equilibrium iv) Macroscopic and microscopic approaches.	08

	b	A thermo couple with test junction at $t°C$ on a gas thermometer scale and reference junction at ice point gives the emf as $e = 0.3t - 4 \times 10^{-4}t^2\ mV$. The millivolt meter is calibrated at ice and steam points. What will he the reading on this thermometer when the gas thermometer reads $80°C$.	08
		OR	
3	a	State the zeroth law of thermodynamics, steam point and ice point.	06
	b	State the first law of thermodynamics for a cyclic and non-cyclic process.	04
	c	$200\ kJ$ of work is done on a system. If the initial volume is $0.5\ m^3$ and pressure changes according to $p = (6 - 6V)$ where P is bar and V is in m^3. Determine the final pressure and final volume of the system.	06
4	a	Derive an expression for the air standard efficiency of dual combustion cycle with usual notations and represent the cycle on P-V and T-S diagram.	08
	b	In an engine working on dual cycle, the temperature and pressure at the beginning of compression are $100°C$ and $1\ bar$ respectively. The compression ratio is 10. If the maximum pressure is limited to $70\ bar$, and $1700\ kj$ of the heat is added per kg of air. Determine the temperature at salient points of the cycle and air standard efficiency.	08
		OR	
5	a	For some state of air before compression and same maximum pressure and temperature in both the cycles. Using appropriate $P - V$ and $T - S$ diagrams. Compare the performance of air standard otto and diesel cycles.	08
	b	Air in a piston-cylinder device of bore = $200\ mm$, stroke = $300\ mm$ and clearance volume = 7% of stroke volume undergoes a diesel cycle. The pressure and temperature of air at the beginning of compression are $1bar$ and $27°C$. The maximum temperature in the cycle is $1900\ K$. calculate the following: i) Compression ratio ii) Cut off ratio iii) Heat transferred to air, kJ/kg iv) Cycle efficiency.	08
6	a	Define the following: i) Mass density ii) Specific weight iii) Specific gravity iv) Specific volume.	08
	b	The space between two square flat parallel plates is fitted with oil. Each side of the plate is $60\ cm$. The thickness of the oil film is $12.5\ mm$. the upper plate, which moves at $2.5\ m/s$ required a force of $98.1\ N$ to maintain the speed. Determine: i) The dynamic viscosity of the oil in poise and ii) The kinematic viscosity of the oil in stokes. If the specific gravity of the oil is 0.95.	08

		OR	
7	a	State and prove the Pascal's law.	08
	b	Find the volume of the water displaced and position of center of buoyancy for a wooden block of width $2.5\,m$ and depth $1.5\,m$. when it floats horizontally in water the density of the wood block is $650 kg/m^3$ and its length $6.0\,m$.	08
8	a	Sketch and derive the relation for actual discharge through an orifice.	08
	b	A pitot tube is inserted in a pipe of $100\,mm$ of Hg (vacuum). The stagnation pressure at the center of the pipe, recorded by the pitot tube is $0.981\,N/cm^2$. Calculate the rate of flow of water through pipe, if the mean velocity of flow is 0.85 times the central velocity. Take $C_v = 0.98$.	08
		OR	
9	a	What are the energy losses that occur in pipes?	08
	b	Calculate the discharge through a pipe of diameter $200\,mm$ when the differential of pressure head between two ends of a pipe $500\,m$ apart is $4\,m$ of water. Take the value $f = 0.009$ in the formula $h_f = \frac{4flv^2}{2gd}$	08
10	a	With a neat sketch, explain spool type DC valve.	08
	b	Explain the working of a simple pressure relief valve with a neat sketch.	08
		OR	
11	a	Classify the control components used in hydraulic system and name any six valve actuation symbols used in hydraulic system with simple sketch.	08
	b	Explain briefly with neat sketches the working of pressure compensated flow control valve used in hydraulic system.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

BASICS OF THERMODYNAMICS AND FLUID MECHANICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The properties whose values do not depend on the mass of the systems are known as _____.	01
	1.2	In a _____ system, only energy flow take place across the boundary but no mass flow take place.	01
	1.3	Density is _____ property of thermodynamics.	01
	1.4	_____ states that if two systems are in thermal equilibrium with a third system then they are in thermal equilibrium with each other.	01
	1.5	Cyclic integral of any property is _____.	01
	1.6	The efficiency of an engine using air as the working medium is known as _____.	01
	1.7	For the same compression ratio thermal efficiency of otto cycle is _____than that of diesel cycle.	01
	1.8	A carnot cycle is made of a pair of _____ and _____ process.	02
	1.9	The air standard efficiency of the otto cycle is given by η = _____.	01
	1.10	_____ is defined as the property of the fluid resistance to the movement of one layer over another layer of the fluid.	01
	1.11	Specific volume is the reciprocal of _____.	01
	1.12	_____ states that intensity of pressure for a fluid at rest is equal in all directions.	01
	1.13	The upward force exerted by a fluid on a body when the body is immersed in the liquid is known as _____.	01
	1.14	_____ is a device used for measuring the rate of a flow of a fluid flowing through a pipe.	01
	1.15	For a fluid flowing through an open channel the rate of flow is measured by _____.	01
	1.16	Energy loss due to friction is given by Darcy formula h_f = _____.	01
	1.17	_____ is defined as the line which gives sum of pressure head and datum head of a flowing fluid in a pipe with respect to some reference line.	01
	1.18	_____ is moving and operating equipment with oil pressure and _____ is moving and operating equipment with air under pressure.	02

PART-B

2	a	Explain the difference between i. Point function and path function ii. Heat and work iii. Closed system and open system	06
	b	State and explain Zeroth law of thermodynamics.	04
	c	Compute the work done by 1kg of a fluid system as it expands slowly behind a piston from an initial pressure of $6 \times 10^5 Pa$ and initial volume of $0.06m^3$ to final volume of $0.18m^3$ in the following process: i) Pressure remains constant ii) Volume remains constant iii) PV^{1-3} = constant.	06
		OR	
3	a	Write the thermodynamic definition of work with suitable example. Explain how it is more general than the definition of work in mechanics.	04
	b	Derive an expression for displacement work if the system follows the law $PV^n = c$.	06
	c	A spherical balloon of $1.0m$ diameter contains a gas at $200kPa$. The gas inside the balloon is heated until the pressure reaches $500kPa$. During the process of heating, the pressure of the gas inside the balloon is proportional to the cubic diameter of the balloon. Determine the work done by the gas inside the balloon.	06
4	a	With the help of $P-V$ and $T-S$ diagram, derive an expression for the air standard efficiency of a diesel engine.	06
	b	The compression curve in an indicator diagram taken on a diesel engine follows the law $PV^{1-4} = c$. At two points lying on the compression curve at $1/8^{th}$ and $7/8^{th}$ stroke, the pressures are 15 bar and 1.5 bar respectively. Find the compression ratio, if the cut off takes place at 5% of stroke. Calculate the air standard efficiency of the engine.	10
		OR	
5	a	Derive an expression for the air standard efficiency of otto cycle. Represent the cycle on $P-V$ and $T-S$ diagram.	06
	b	In a constant volume cycle the pressure at the end of compression is 1.5times that at the start, the temperature of air at the beginning of compression is 37^0C and maximum temperature attained in the cycle is 1950^0C. Find: i) The compression ratio ii) Thermal efficiency of the cycle iii) Heat supplied per kg of air iv) The work done per kg of air.	10
6	a	Define the following: i) Specific weight ii) Viscosity iii) Specific gravity.	06

	b	Define surface fusion and derive an expression for the pressure inside a soab bubble.	04
	c	The space between two square flat parallel plates is filled with oil. Each side of the plate is $60cm$. the thickness of the oil film is $12.5mm$. the upper plate which moves at $2.5m/s$ requires a force of $98.1N$ to maintain the speed. Determine: i) The dynamic viscosity of the oil in poise ii) The kinematic viscosity of the oil in stokes if the specific gravity of the oil is 0.95.	06
		OR	
7	a	Derive Bernoullis equation for a fluid flow and explain its significance.	06
	b	Define the following: i) Meta center ii) Buyoncy.	04
	c	A pipe of diameter $400mm$ carries water at a velocity of $2.5m/s$. the pressure at the points A and B are given as $29.43N/cm^2$ and $22.563N/cm^2$ respectively while datum head on A and B are $28m$ and $30m$. Find the loss of head between A and B.	06
8	a	Derive an expression for discharge through venturimeter. Why venturimeter is better than orifice meter?	06
	b	Define C_d, C_v and C_c for vertical orifice showing vena contracta.	04
	c	An oil of specific gravity 0.8 is flowing through a venturimeter having inlet diameter $20cm$ and throat diameter $10cm$. the oil-mercury differential manometer shows a reading of $25cm$. calculate the discharge of oil through the horizontal venturimeter. Take $C_d = 0.98$.	06
		OR	
9	a	How will you determine the loss of head due to friction in pipes by using Darcy formula?	06
	b	Give expressions for the minor loss.	04
	c	Crude oil of kinematic viscosity 0.4 stoke is flowing through a pipe of diameter $300mm$ at the rate of 300 liters.sec. Find the head lost due to friction for a length of $50m$ of the pipe.	06
10	a	Explain with sketch any one flow control valve. Write its symbolic representations.	08
	b	Explain the function of lubricator in a Pneumatic Systems.	04
	c	Explain the various ways of classifying the flow control valve.	04
		OR	
11	a	What is the purpose of a directional control valve? Differentiate between open center and closed center type.	08
	b	With a neat schematic diagram, discuss the method of controlling a double acting hydraulic cylinder with the help of a manually operated 4/2 flow control valve.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

BASICS OF THERMODYNAMICS AND FLUID MECHANICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6. Answer FIVE full questions from Part B.

PART-A

1	1.1	Which is the system boundary in the analysis of class room as a thermodynamic system?	01
	1.2	Measurement of temperature using a thermometer is an example of approach of thermodynamics.	01
	1.3	Give two reasons why work is not to be considered as a thermodynamic property.	02
	1.4	Otto cycle is also known as _____.	01
	1.5	For the same compression ratio, the efficiency of Diesel Cycle is _____ than Otto cycle.	01
	1.6	State hydrostatic law.	01
	1.7	Define rate of flow or discharge.	01
	1.8	How does the viscosity of fluid and gases varies with temperature.	02
	1.9	State Bernoulli's theorem.	01
	1.10	Venturimeter: Discharge :: Pitot tube: _____.	01
	1.11	Find the loss of head when a pipe of diameter $200mm$ is suddenly enlarged to a diameter of $400mm$. The rate of flow of water through the pipe is $250l/s$.	02
	1.12	The discharge through a triangular notch is given by _____.	01
	1.13	The directional control valves are used to control the _____ in a hydraulic circuit.	01
	1.14	Sketch a single solenoid actuated, 2 position, 4 way spring centered DCV.	02
	1.15	State the primary function of pressure relief valve.	01
	1.16	Draw the line diagram of a simple double acting hydraulic cylinder.	01

PART-B

2	a	Define Open, closed and isolated system. Identify and draw the system boundary for the systems as listed below: i) A wind mill ii) Mixture of ice and water in a metal container iii) Thermos flask filled with hot tea iv) Storage cell producing electricity	11
	b	Derive an expression for the work done during an isentropic process.	05
		OR	
3	a	Derive the two statements of second law of thermodynamics. Show that violation of Kelvin Planck statement leads to violation of Clausius statement.	10
	b	A quantity of gas has a volume of $0.14m^3$, pressure $1.5bar$ and temperature $100°C$. If the gas is compressed at constant pressure, until its volume becomes $0.112m^3$. Determine the temperature at the end of compression, work done during compression, decrease in internal energy and heat given out of the gas.	06
4	a	Prove that the efficiency of the Carnot engine working between the temperature limits T_1 & T_3 is equal to $\frac{T_1-T_2}{T_1}$. And show that no engine can be more efficient than this when working over the same temperature range.	08
	b	The compression ratio of an ideal air standard diesel cycle is 15. The heat transfer is $1465kJ/kg$ of air. Find the pressure and temperature at the end of each process.	08
		OR	
5	a	Explain briefly Otto cycle with the help of $p-v$ and $T-S$ diagram and derive an expression for the ideal efficiency of Otto cycle.	10
	b	A certain quantity of air at a pressure of $1bar$ and temperature $70°C$ is compressed reversibly and adiabatically until pressure is $7bar$ in an otto cycle engine. $460kJ$ of heat per kg of air is now added at constant volume. Determine: i) Compression ratio of the engine ii) Temperature at the end of compression and iii) Temperature at the end of heat addition.	06
6	a	What is Fluid Mechanics? Define and explain the practical application its classification in a day to day life with few examples.	07
	b	What is a surface tension? Show that surface tension acting on liquid droplet is doubled the surface tension acting on a hallow bubble.	09
		OR	

7	a	Define the equation of continuity. Obtain an expression for a continuity for a three-dimensional flow.	10
	b	The water is flowing through a taper pipe of length $100mm$ having diameters $600mm$ at the upper end and $300mm$ at the lower end, at the rate of $50l/s$. The pipe has a slope of 1 in 30. Find the pressure at the lower end if the pressure at the higher level is $19.62N/cm^2$.	06
8	a	What are the different devices used to measure flow rate in a pipe? Which is the appropriate device and why? Obtain expression for theoretical discharge in it.	10
	b	An orifice meter with orifice diameter $10cm$ is inserted in a pipe of $20cm$ diameter. The pressure gauges fitted upstream and downstream of the orifice meter give readings of $19.62N/cm^2$ and $9.81N/cm^2$ respectively. Coefficient of discharge for the meter is given as 0.6. Find the discharge of water through pipe.	06
		OR	
9	a	Derive Darcy-Weisbach equation from first principles and hence deduce Chezy's equation from it.	08
	b	A horizontal pipe line $40m$ long is connected to a water tank at one end and discharges freely into the atmosphere at the other end. For the first $25m$ of its length from the tank, the pipe is $150mm$ diameter and its diameter is suddenly enlarged to $300mm$. The height of water level in the tank is $8m$above the center of the pipe. Considering all losses of head which occur. Determine the rate of flow. Take $f = 0.01$ for both the sections of the pipe. Also draw Hydraulic gradient and total energy line.	08
10	a	Describe briefly the classification of Direction Control Valves.	04
	b	With a neat sketch explain briefly the operation of a regenerative circuit.	06
	c	A double acting cylinder is hooked up in the regenerative circuit. The relief valve setting is $70bars$. The piston area is $0.016m^2$ and the rod area is $0.0045m^2$. If the pump flow is $0.0013m^3/s$, find the cylinder speed and load carrying capacity for the extending stroke.	06
		OR	
11	a	An actuator forward speed is controlled by a meter in circuit. The pressure setting of relief valve of $50bar$ and the pump discharge = $30l/min$. The cylinder has to carry a load of $3600N$ during the forward motion. The area of piston is $15cm^2$ and rod area = $8cm^2$. The flow control valve is set to allow only $10l/min$. Calculate the power input the motor. Forward speed and return speed and efficiency of the circuit.	08
	b	Explain briefly the meter-in logic circuit.	08

Bridge Course Mathematics (lateral entry)

III Semester SEE Question Papers

Course codes: 18DMA37, 12DMA37, 16DMA37

- Mar 2021 (18DMA37)
- Nov/Dec 2019 (18DMA37)
- Fast Track Jul 2019 (12DMA37)
- Nov/Dec 2018 (16DMA37)
- Fast Track Jul 2018 (16DMA37)
- Nov/Dec 2017 (16DMA37)

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RV COLLEGE OF ENGINEERING

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations March-2021

Common to CV/BT/ET/CH/ME/EE/EC/IM/EI/AS

BRIDGE COURSE MATHEMATICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer FIVE full questions. Question numbers 1, 6, 7 are compulsory. Answer any one full question from 2 and 3 & one full question from 4 and 5.

PART-A

1	a	Find the partial derivatives $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}$ and $\frac{\partial^2 z}{\partial x \partial y}$ when $z = x^2 + \sin y + y^2 + \cos x + xy$.	05
	b	If $u = x^2 + y^2$, $v = 2xy$, then find $J\left(\frac{u,v}{x,y}\right)$.	05
2	a	If $\vec{r} = t^2\hat{\imath} + (t^3 + 5t)\hat{\jmath} + t\hat{k}$, compute i) Velocity, acceleration at $t = 2$. ii) Magnitude of velocity and acceleration at $t = 1$.	05
	b	If $\Phi = x^3 + y^3 + 3x^2y^2z^2$, then find $\nabla\Phi$ and $\lvert\nabla\Phi\rvert$ at $(1,0,-1)$.	05
		OR	
3	a	Compute $\nabla.\vec{F}$ at $(2,1,-1)$ when $\vec{F} = (x^2 + xy)\hat{\imath} + (y^2 + 2z)\hat{\jmath} + (z^2 + 2yz)\hat{k}$.	05
	b	Prove that a vector field given by $\vec{F} = (2xy + z^3)\hat{\imath} + x^2\hat{\jmath} + 3xz^2\hat{k}$ is irrotational and also find its scalar potential Φ such that $\vec{F} = \nabla\Phi$.	05
4	a	Solve $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$.	05
	b	Find the general solution of the differential equation $(D^2 + 2D + 1)y = e^x$	05
		OR	
5	a	Solve $\frac{d^2y}{dx^2} + 4y = \sin 2x$.	05
	b	Find the general solution of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = x$.	05

6	a	Find an approximate root of the equation $e^x = x^3 + \cos 25x$ which lies near $x_0 = 4.5$ by applying Newton-Raphson method. Carry out 6 iterations.	05
	b	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using Simpson's $\frac{3}{8}th$ rule.	05
7	a	Evaluate $\int_1^2 \int_{-1}^1 (x^3 + y^3) dx\, dy$.	05
	b	Evaluate $\int_2^3 \int_1^2 \int_0^1 x^2 y^3 z^4\, dx\, dy\, dz$	05

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations Nov/Dec-19

Common to AS / CV / BT / CH / ME / IM / EC / EE / EI / TE

BRIDGE COURSE MATHEMATICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all Questions.
2. Question number 1, 6 and 7 are compulsory. Answer any one full question from 2 and 3 and one full question from 4 and 5.

1	a	If $u = x^3 + y^3 + 3axy$, then find the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$ and $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^2 u}{\partial x \partial y}$, $\frac{\partial^2 u}{\partial y^2}$.	05
	b	If $x = u(1 - v)$ and $y = uv$, then find $\frac{\partial(x,y)}{\partial(u,v)}$.	05
2	a	Find the velocity and acceleration of a particle moving along the curve $\vec{r} = e^{-2t}\,\hat{\imath} + (2\cos 5t)\,\hat{\jmath} + (5\sin 2t)\,\hat{k}$. Also find the magnitude of velocity and acceleration at $t = 0$.	05
	b	If $\Phi(x, y, z) = xy^3z^2 - 4$, then find $\nabla\Phi$ and $\lvert\nabla\Phi\rvert$ at $(-1, -1, 2)$.	05
		OR	
3	a	If $\vec{F} = (x^2 + y^2)\hat{\imath} + (y^2 + 2x)\,\hat{\jmath} + (z^2 + xy)\hat{k}$, then find $\nabla.\vec{F}$ at $(1, -1, 1)$.	05
	b	A vector field is given by $\vec{F} = (x^2 - y^2 + x)\hat{\imath} - (2xy + y)\,\hat{\jmath}$. Show that the field is irroattional and also find its scalar potential Φ such that $\vec{F} = \nabla\Phi$.	05
4	a	Solve $(D^3 - 2D^2 - D + 2)y = 0$.	05
	b	Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = e^{2x}$.	05
		OR	
5	a	Solve $\frac{d^2y}{dx^2} + 9y = \cos 3x$.	05
	b	Solve $(D^2 + 2)y = x$.	05
6	a	Apply Newton-Raphson method to find an approximate root of the equation $x^3 - 3x - 5 = 0$ which lies near $x = 2$.	05
	b	Evaluate $\int_0^6 \frac{dx}{1+x}$ by using Simpson's $\frac{1}{3}rd$ rule.	05
7	a	Evaluate $\int_{y=1}^{2}\int_{x=3}^{4}(x + y)dx\,dy$.	05
	b	Evaluate $\int_2^3\int_0^1\int_1^2 xyz\,dx\,dy\,dz$.	05

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution Affiliated to VTU)

III Semester B. E. Fast Track Examinations July-19

Common to AS / CV / BT / CH / ME / IM

BRIDGE COURSE MATHEMATICS - I

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The nth derivative of $\log_e(ax+b)$ is_____.	01
	1.2	The Taylor series expansion of the function $y(x)$ about the point $x=b$ is _____.	01
	1.3	The nth derivative of $\sinh(3x)$ is _____.	02
	1.4	If $u=\log_e(2y+b)$ $then$ $\frac{\partial u}{\partial y}$ is_____.	01
	1.5	IF $u=f(x+ay)+g(x-ay)$ then $\frac{\partial^2 u}{\partial x^2}$ is _____.	01
	1.6	IF $u=x^2+y^2, v=xy$ then $J\left(\frac{u,v}{x,y}\right)$is _____.	02
	1.7	The general solution of first order linear differential equation $\frac{dy}{dx}+Py=Q$ is _____.	01
	1.8	The integrating factor of the differential equation $\frac{dy}{dx}+\frac{y}{x+1}=e^{3x}(x+1)$ is _____.	01
	1.9	The solution of $\frac{dy}{dx}=e^{2x+y}$ is _____.	02
	1.10	Particular integral of $\frac{d^2y}{dx^2}-3\frac{dy}{dx}-5y=e^{2x}$ is _____.	01
	1.11	The general solution of $(D^2-1)y=0$ is _____.	01
	1.12	The Wronskian of $x\ and\ e^x$ is _____.	02
	1.13	If $\vec{v}=\hat{\imath}+3\hat{k}\ then\ \lvert\vec{v}\rvert$ is _____.	01
	1.14	If $\vec{F}=t^3\hat{\imath}+\cos t\hat{\jmath}+2e^t\hat{k}$ then $\frac{d\vec{F}}{dt}$ =_____.	01
	1.15	If $\vec{a}=2\hat{\imath}+\hat{\jmath}-2\hat{k}, \vec{b}=-\hat{\jmath}+\hat{k}\ and\ \vec{c}=\hat{\imath}-2\hat{\jmath}+2\hat{k}$ then find $\vec{a}\cdot(\vec{b}\times\vec{c})$.	02

PART-B

2	a	Find the nth derivative of $y=e^{3x}\cos^3 x$.	08
	b	Obtain the Maclaurin's series expansion of $\log_e(1+x)$ upto terms containing x^4.	08
		OR	
3	a	Find the nth derivative of the function $\frac{x}{(x+1)(2x-1)}$.	08
	b	If $y=a\cos(\log x)+b\sin(\log x)$show that $x^2_{y_{n+2}}+(2n+1)xy_{n+1}+(n^2+1)y_n=0$.	08

4	a	If $z = e^{ax+by} f(ax - by)$, prove that $b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz$	08
	b	IF $u = f(y - z, z - x, x - y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	08
		OR	
5	a	IF $u = x(1 - y)$ *and* $v = xy$, then find $J = \frac{\partial(u,v)}{\partial(x,y)}$ and $J' = \frac{\partial(x,y)}{\partial(u,v)}$ and verify $JJ' = 1$.	08
	b	If $u = x^2 + y^2 + z^2$ where $x = e^t, y = \log t, z = t^3 + 3$, find $\frac{du}{dt}$ using partial derivatives.	08

6	a	Solve:$\frac{dy}{dx} = e^{3x-2y} + x^2e^{-2y}$.	08
	b	Solve:$\frac{dy}{dx} + y = x$.	08
		OR	
7	a	Solve:$x\frac{dy}{dx} + y = x^3y^6$.	08
	b	Solve:$(y - x^3)dx + (x + y^3)dy = 0$.	08

8	a	Solve:$(D^2 - 3D + 2)y = e^{3x} + \sin 2x$.	08
	b	Solve: by the method of variation of parameters $y'' + a^2y = \tan ax$.	08
		OR	
9	a	Solve:$(D^4 - 2D^3 + 5D^2 - 8D + 4)y = 0$.	08
	b	Solve:$(D^2 - 4D + 13)y = e^{2x}\cos 3x$.	08

10	a	Find the unit normal to both the vectors $4\hat{\imath} - \hat{\jmath} + 3\hat{k}$ and $-2\hat{\imath} + \hat{\jmath} - 2\hat{k}$. Find also the sine of the angle between them.	08
	b	Find the velocity and acceleration of particle moving along the curve $x = 3t^2, y = t^3 - 4t$ *and* $z = 3t + 4$. Also find their magnitude at $t = 2$.	08
		OR	
11	a	If $\vec{A} = 5t^2\,\hat{\imath} + t\hat{\jmath} - t^3\hat{k}$ *and* $\vec{B} = \sin t\hat{\imath} - \cos t\hat{\jmath}$then find i. $\frac{d}{dt} = (\vec{A} \cdot \vec{B})$ ii. $\frac{d}{dt}(\vec{A} \times \vec{B})$	08
	b	Find the angle between the tangents to the curve $x = t, y = t^2, z = t^3$ *at* $t = 1$ *and* $t = 2$.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution Affiliated to VTU

III Semester B. E. Examinations Nov/Dec-18

Common to AS / CV / BT / CH / ME / IM

BRIDGE COURSE MATHEMATICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The Maclaurins series expansion of e^x is _____.	01
	1.2	If $Z = e^x \cos y + x^3y^2 + x^2 + 3$ then $\frac{\partial z}{\partial y}$ = _____.	01
	1.3	$\int_0^1 \int_0^1 dx\, dy$ = _____.	01
	1.4	Change $\int\int\int dx\, dy\, dz$ to cylindrical polar coordinate system.	01
	1.5	The auxiliary equation of $\frac{d^3y}{dx^3} - \frac{3d^2y}{dx^2} + 2y = 0$ is _____.	01
	1.6	If the roots of Differential equations are $2 \pm 3i$, then the complementary function is _____.	01
	1.7	the tangent vector to the curve $\vec{r}(t) = (t^2 + 5)\hat{\imath} + (2t - 1)\hat{\jmath} + t\hat{k}$ at $t = 1$ is _____.	01
	1.8	If $\vec{A} = 2x^3\hat{\imath} - 3xyz\hat{\jmath} + x^2z\hat{k}$ then $div\vec{A}$ = _____.	01
	1.9	If the number of intervals is multiple of three, then _____ method is used to evaluate the definite integral.	01
	1.10	In Regular Falsi method, for determining root of $f(x) = 0$ over an interval (a, b), curve is replaced by _____.	01
	1.11	If $u^2 = x^2 + y^2$ and $V = x^2 - y^2$ then $J\left(\frac{u,v}{x,y}\right)$ = _____.	02
	1.12	Sketch the domain of the integral $\iint_R f(x,y)dxdy$ where R is the region bounded by $y = x^2$ and $y = x$.	02
	1.13	If $\vec{A} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ then $curl\ \vec{f}$ = _____.	02
	1.14	Reduce the equation $x^2y'' + xy' - 9y = x$ into a linear differential equation with constant coefficients.	02
	1.15	The Taylor's series solution of $\frac{dy}{dx} = 2y + 3e^x, y(x_0) = y_0$ upto 2nd degree terms is _____.	02

PART-B

2	a	Using Maclaurin's series, expand $\log(1 + x)$ in powers of x. Hence deduce series for $log\sqrt{\frac{1+x}{1-x}}$.	08
	b	If $u = x^2 + y^2 + z^2,\ v = xy + yz + zx,\ w = x + y + z$ find $J(\frac{u,v,w}{x,y,z})$.	08

3	a	Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy\, dx$ by changing the order of integration.	08
	b	Evaluate $\int_0^a \int_0^b \int_0^c x^2y^2z^2 dz\, dy\, dx$.	08
		OR	
4	a	Evaluate $\iint_R xy\, dA$ $R \rightarrow$Region bounded by the x-axis, ordinate at $x = 2a$ and $x^2 = 4ay$.	08
	b	Transform to polar coordinates and hence evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y\sqrt{x^2+y^2} dx\, dy$.	08
5	a	Solve $y''' - 2y'' - 5y' + 6y = e^{-x}$.	08
	b	Solve $x^2y'' + xy' + y = 2\, sin(log\, x)$.	08
		OR	
6	a	Solve $y'' + 8y' + 16y = \sin 4x$.	08
	b	Solve $(2x-1)^2y'' - 6(2x-1)y' + 16y = 8(2x-1)^2$.	08
7	a	Find the magnitude of velocity and accelerating of a particle which moves along he curve $\vec{r} = \cos 2t\, \hat{\imath} + \sin 2t\, \hat{\jmath} + t\hat{k}$ at any time $t = \frac{\pi}{8}$.	05
	b	Find $div\, \vec{F}\ at$ (1,1,1) where $\vec{F} = grad\ (x^3y + y^3z + z^3x)$.	05
	c	Prove that $\vec{F} = (3x^2 - 2yz)\hat{\imath} + (3y^2 - 2zx)\hat{\jmath} + (3z^2 - 2xy)\hat{k}$ is irrotational.	06
8	a	Find the real root of the equation $3x = \cos x + 1$ by Newton Raphson method. Perform 4 iterations.	08
	b	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using: i) Simpson's $3/8^{th}$ rule. ii) Weddle's rule.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution Affiliated to VTU

III Semester B. E. Fast Track Examinations July-18

Common to AS / CV / BT / CH / ME / IM

BRIDGE COURSE MATHEMATICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The expansion of $y = e^x$ in ascending powers of $'x'$ is _____.	02
	1.2	If $x = r.\cos(\theta)$ and $y = r.\sin(\theta)$, then the Jacobian $\frac{\partial(x,y)}{\partial(r,\theta)}$ = _____.	02
	1.3	Evaluate : $\int_0^2 \int_0^2 dx\,.dy$	02
	1.4	Sketch the domain of integration $\iint_R f(x,y)dA$, where 'R' is the region bounded by the ordinate $x = 2a$, parabola $x^2 = 4ay$ and $y = 0$.	02
	1.5	The complementary solution of the ODE $(D^2 - 1)y = 0$ is _____.	02
	1.6	If $D = \frac{d}{dx}$, then $\frac{1}{(D^2+1)}\cos(x)$ = _____.	02
	1.7	The vector function $\vec{f} = (2x + y)\hat{\imath} - (3y + 2z)\hat{\jmath} + (x + az)\hat{k}$ is solenoidal if a = _____.	02
	1.8	If $\emptyset = e^{r^2}$, where $r^2 = x^2 + y^2 + z^2$, then $grad(\emptyset)$ = _____.	02
	1.9	In the method of false position for determining root of $f(x) = 0$ over (a, b), curve is replaced by _____.	02
	1.10	Weddle's rule to determine $\int_a^b f(x)dx$ with seven ordinates is _____.	02

PART-B

2	a	The steady state two dimensional heat equation is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Show that the function $u = tan^{-1}\left(\frac{y}{x}\right), x \neq 0$ satisfies the above equation.	08
	b	If $u = f(X, Y, Z)$, where $X = \frac{x}{y}, Y = \frac{y}{z}$ and $Z = \frac{z}{x}$ show that $x.\frac{\partial u}{\partial x} + y.\frac{\partial u}{\partial y} + z.\frac{\partial u}{\partial z} = 0$.	08
3	a	Evaluate : $\int_{x=0}^{1} \int_{y=0}^{\sqrt{1+x^2}} \frac{dy\,.dx}{(1+x^2+y^2)}$.	08
	b	Evaluate: $\int_{x=0}^{a} \int_{y=0}^{b} \int_{z=0}^{c} (x^2 + y^2 + z^2)dz.dy.dx$	08
		OR	

4	a	By changing the order of integration, evaluate $\int_{x=0}^{\infty}\int_{y=x}^{\infty}\frac{e^{-y}}{y}dy.dx$.	08
	b	Evaluate: $\int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}}\int_{z=0}^{\sqrt{1-x^2-y^2}}\frac{dz.dy.dx}{\sqrt{1-x^2-y^2-z^2}}$.	08
5	a	Find the complementary solution of the ODE, $(D^3-6D^2+11D-6)y=0$.	08
	b	Solve: $y''-y'-2y=e^{3x}$	08
		OR	
6	a	Solve: $y''+3y'-4y=\sin(2x)$	08
	b	Solve: $x^2y''-2xy'+2y=\frac{1}{x^2}$	08
7	a	A particle moves along the curve: $\vec{r}=cos(wt)\,\hat{\imath}+sin(wt)\,\hat{\jmath}$, where '$w$' is a constant. Show that, i) The velocity $\vec{V}$ is perpendicular to $\vec{r}$ ii) The acceleration $\vec{a}$ is directed towards the origin and has a magnitude proportional to the distance from origin.	08
	b	Define divergence and curl of a vector point function $\vec{f}$. Hence find divergence and curl of the vector function, $\vec{f}=(3x^2-3yz)\hat{\imath}+(3y^2-3xz)\hat{\jmath}+(3z^2-3xy)\hat{k}$	08
8	a	Find a positive real root of the equation $3x=\cos(x)+1$, correct to four places of decimal using Newton – Raphson method.	08
	b	Using Range – Kutta method of order four, find $y\,at\,x=0.1, x=0.2$, $h=0.1$ from $\frac{dy}{dx}=x+y,\ y(0)=1$.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution Affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Common to AS / CV / BT / CH / ME / IM

BRIDGE COURSE MATHEMATICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The Maclaurins series expansion of $\sin x$ is _____.	01
	1.2	If $z = e^x \sin y + x^2y^3 + 2y^2 + 4$ then $\frac{\partial z}{\partial x}$ = _____.	01
	1.3	If $u = x^2 - 2y^2, v = 2x^2 - y^2$ then $\frac{\partial(u,v)}{\partial(x,y)}$ = _____.	02
	1.4	$\int_0^1 \int_0^1 dxdy$ = _____.	01
	1.5	Sketch the domain of the integral $\iint_R dxdy$ where R is the region bounded by the lines $x = 0, y = 0$ and $x + y = 1$.	02
	1.6	Change the $\iiint_V dxdydz$ to cylindrical polar coordinate system.	01
	1.7	Solve $y'' - y' - 6y = 0$.	02
	1.8	Reduce the equation $x^2y'' + xy' - 9y = \log x$ into a linear differential equation with constant coefficients.	02
	1.9	The unit tangent vector to the curve $\vec{r}(t) = t^3\,\hat{\imath} + 2t^2\hat{\jmath} + \hat{k}$ at $t = 1$ *is* _____.	02
	1.10	If $\vec{A} = 2x^2\hat{\imath} - 3yz\hat{\jmath} + xz^2\hat{k}$, then $div\,\vec{A}$ = _____.	01
	1.11	If $\vec{F}$ is a conservative force field them $curl\,\vec{F}$ = _____.	01
	1.12	In Regular-Falsi method for determining root of $f(x) = 0$, over an interval (a, b), curve is replaced by _____.	01
	1.13	The interval in which the real root of the equation $x^2 - 2x - 5 = 0$ lies is _____.	01
	1.14	Given x: 4.0, 4.2, 4.4, 4.6, 4.8, 5.0, 5.2 y: 1.3863, 1.4351, 1.4816, 1.5261, 1.5686, 1.6094, 1.6487 then $\int_4^{5.2} \log x\,dx$ by Simpson's $3/8^{th}$ rule is _____.	02

PART-B

2	a	Obtain Taylor series expansion of $\log_e x$ in powers of $(x - 1)$ upto the term containing $(x - 1)^4$.	05
	b	If $V = (x^2 + y^2 + z^2)^{\left(-\frac{1}{2}\right)}$, prove that $V_{xx} + V_{yy} + V_{zz} = 0$.	05

	c	If $u = x + 3y^2 - z^3, v = 4x^2yz, w = 2z^2 - xy$, evaluate $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at the point $(1,-1,0)$.	06
3	a	Evaluate $\int_0^2 \int_0^{x^2} x(x^2+y^2)dxdy$.	08
	b	Find the volume of the region bounded by $z = x^2 + y^2, z = 0, x = -a, x = a, y = -a$ & $y = a$.	08
		OR	
4	a	Evaluate $\iint_R xydxdy$ where R is the triangular region bounded by the axes of co-ordinates and the line $\frac{x}{a} + \frac{y}{b} = 1$.	08
	b	Transform to polar co-ordinates and hence evaluate $\int_{-a}^{a} \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dxdy$	08
5	a	Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4y = 2\sinh 2x$.	08
	b	Solve $(2x+1)^2 \frac{d^2y}{dx^2} + 2(2x+1)\frac{dy}{dx} + 16y = 4\sin[2\log(2x+1)]$.	08
		OR	
6	a	Solve $(D^4 + 2D^2 + 1)y = 0$.	08
	b	Solve $x^2y'' + xy' - 9y = \log x$.	08
7	a	Find the magnitude of the velocity and acceleration of a particle which moves along the curve $x = 2\sin 3t, y = 2\cos 3t, z = 8t$ at any time $t > 0$.	05
	b	If $\vec{A} = 3xyz^2\hat{\imath} + 2xy^3\hat{\jmath} - x^2yz\hat{k}$ and $\phi = 3x^2 - yz$, Find i) $\nabla \cdot \vec{A}$(ii) ii) $\vec{A} \cdot \nabla\phi$	05
	c	Prove that $\vec{F} = (6xy + z^3)\hat{\imath} + (3x^2 - z)\hat{\jmath} + (3xz^2 - y)\hat{k}$ is irrotational.	06
8	a	Using Newton – Raphson method, find correct to four decimal places, the root between 0 and 1 of the equation $x^3 - 6x + 4 = 0$.	08
	b	Dividing the range into 10 equal parts, find the approximate value of $\int_0^{\pi} \sin x\, dx$ by simpson's $1/3^{rd}$ rule.	08